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Report of

END-OF-STUDIES PROJECT

In partial fulfillment for:

Applied Bachelor's Degree in Information Systems Development

Development of an Internal Platform for Managing Requests, Projects, and Tasks at ArabSoft

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Dedication

To all those whose presence illuminated every step of this journey…

To my father, Adel,

You taught me that hard work and perseverance
are the keys to every success.
Your constant support and trust in me
have been my anchor throughout this journey.
This work is the result of your precious guidance.

To my mother, Amira,

An endless source of love and encouragement,
your caring presence has always carried me.
Every effort I made contains a part of you,
because you have been my greatest strength.

To my sister, Sarra,

For your joy, your support, and your generosity.
With one smile, you always gave me courage
in the most difficult moments.
Your presence is a precious gift.

To all my friends and colleagues, and to everyone who contributed,
directly or indirectly, to the completion of this work.

Abderrahmen Ben Hattab

Acknowledgements

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Table of Contents

List of Figures

List of Tables

General Introduction

Digital transformation has become a major strategic driver for companies seeking to improve organizational efficiency and ensure smooth management of internal processes. The centralization of administrative requests, project tracking, and internal communication represents significant challenges when operations are handled manually or in a fragmented manner. The growing complexity of internal workflows, together with the need for effective coordination between departments, makes the adoption of modern and structured digital tools essential.

In this context, my work focuses on the **development of an internal platform for managing requests, projects, and tasks at ArabSoft**. The objective is to design a practical and reliable solution that centralizes administrative requests, monitors project progress, and improves communication between employees, team leaders, and HR managers. By providing a global and real-time view of organizational activities, this platform aims to strengthen collaboration, improve process efficiency, and support better operational decision-making.

The ambition of this project is to provide a tool that addresses current organizational needs while remaining adaptable to future evolution. Whether for handling internal requests, tracking project progress, assigning tasks, or improving information flow between stakeholders, the platform is designed as a complete and intuitive solution that integrates smoothly into existing work practices.

This report presents the different phases of project realization. From requirements analysis to deployment, including design, development, and testing phases, I explain how this platform was built to meet user expectations and deliver concrete added value to the company. Particular attention is given to the methodological approach, technical choices, and implementation strategy adopted throughout the project.

Through this work, I aim to contribute to the modernization of internal management practices and to the continuous improvement of organizational performance. This project represents an opportunity to rethink how administrative and project activities are managed by making them more structured, collaborative, and efficient.

Chapter 1

Project Context Presentation

I. Host organization presentation

1. General presentation of ArabSoft

ArabSoft is a company specialized in software engineering and digital solutions for organizations. It operates in several areas such as business process management, enterprise application development, systems integration, and digital transformation support. Through its experience in information systems, the company helps institutions modernize operations and improve service quality.

ArabSoft combines **technological expertise** and **business understanding** to deliver reliable and scalable solutions adapted to client needs. With a structured project approach and multidisciplinary teams, the company supports organizations in optimizing internal workflows, improving traceability, and increasing overall operational efficiency.

Figure 1: ArabSoft logo

2. ArabSoft in Tunisia

In Tunisia, **ArabSoft** supports both public and private organizations in the digitalization of their internal processes. Its teams work on the design, development, deployment, and maintenance of business-oriented platforms, with a focus on security, scalability, and long-term maintainability. Key missions include process automation, workflow management, data centralization, and decision support through structured reporting.

II. Project presentation

1. Current-state study

At present, internal process management at ArabSoft relies on a combination of office tools and partially connected workflows. Daily practices include Excel files, email exchanges, messaging channels, and manual follow-up sheets to handle administrative requests and monitor project execution.

Although these tools support day-to-day operations, information remains distributed across multiple sources and teams. This fragmentation limits process visibility, slows coordination, and increases the risk of inconsistencies between request status, project progress, and assigned tasks.

The current operational flow mainly covers:

- **Administrative request handling**, to receive, validate, and process internal service requests.
- **Project and task follow-up**, to plan activities, assign responsibilities, and track execution progress.
- **Operational coordination**, to synchronize information between employees, team leaders, and HR managers and ensure timely communication.

2. Current-state critique

Despite the use of multiple tools, several difficulties remain and reduce operational efficiency. Main issues include:

- **Data fragmentation:** Request, project, and task information is spread across multiple files and channels, which makes a coherent global view difficult.
- **Delayed follow-up:** Manual tracking slows the handling of requests and makes it harder to monitor priorities, deadlines, and responsibilities.

- **Inefficient coordination:** Employees, team leaders, and HR managers do not share a single operational workspace, which increases duplication and communication gaps.
- **Limited access to updated information:** Stakeholders do not always have real-time access to the latest status, which reduces responsiveness.

These limitations show that the current tools and processes no longer fully meet ArabSoft's needs for traceability, responsiveness, and coordination in the management of internal requests, projects, and tasks.

3. Proposed solution

To address these challenges, I propose the development of an **internal platform for managing requests, projects, and tasks at ArabSoft**. The platform is designed to centralize operational data, structure workflow execution, and improve coordination across departments through a practical and intuitive environment. Key features include:

- **Data centralization:** All requests, projects, and tasks are stored in a unified database accessible in real time.
- **Workflow management:** Structured request lifecycle with clear statuses, validation steps, and processing history.
- **Project and task tracking:** Planning, assignment, prioritization, and progress monitoring through dedicated views.
- **Role-based access:** Secure access control according to user profiles and organizational responsibilities for employees, team leaders, and HR managers.
- **Notifications and integrations:** Alerts and automated exchanges with external services when needed to support authentication, messaging, and reporting.
- **Dashboards and reporting:** Operational indicators for monitoring performance, identifying bottlenecks, and supporting decision-making.

This solution provides a global real-time view of internal activities and improves efficiency, responsiveness, and traceability across the organization.

Modular, secure, and scalable, the platform is designed to adapt to ArabSoft's evolving needs while enabling progressive adoption and continuous process improvement.

III. Adopted methodologies and formalism

1. Modeling and design methodology

For project modeling and design, I chose **UML (Unified Modeling Language)**. UML is a standardized visual language used to represent both system structure and behavior.

Using UML in this project allows me to:

- **Visualize requirements:** Use-case diagrams capture expected functionalities and actor-system interactions.
- **Structure the system:** Class and sequence diagrams define module architecture and data/interaction flows.
- **Facilitate communication:** UML provides a common language for developers, project managers, and clients.

UML was selected for its flexibility, standardization, and ability to fit complex project needs.

Figure 2: UML logo

2. Working methodology

For project management, I adopted the **Agile** methodology, specifically the **Scrum** framework. This iterative and incremental approach supports requirement changes and continuous user feedback.

2.1. Methodological choice (Scrum)

Scrum was chosen for several reasons:

- **Flexibility:** Scrum adapts quickly to change, which is critical when requirements evolve.
- **Transparency:** Regular meetings such as daily stand-ups and sprint reviews ensure clear communication.
- **Continuous delivery:** Sprint cycles provide continuous validation of implemented features.

2.2. Why Scrum?

Scrum is well suited to this project because it supports multidisciplinary teams and collaboration. It also helps to:

- **Maximize value:** Focus on high-priority features aligned with project objectives.
- **Improve quality:** Frequent feedback and regular testing enable rapid issue detection and correction.

2.3. Scrum team roles

In this project, Scrum roles are distributed as follows:

- **Product Owner:** Defines priorities and manages the backlog to ensure delivered features meet user needs. In this project, the Product Owner is **Mr. Mouadh Zidi**, my company supervisor.
- **Scrum Master:** Ensures Scrum principles are followed and helps remove obstacles. In this project, the Scrum Master is **Ms. Anissa CHALOUAH**, my ISET supervisor.
- **Development Team:** Responsible for design, development, and testing of platform features. In this project, the development team is composed of **Abderrahmen Ben Hattab** only.

This role distribution enables efficient collaboration and clear responsibilities, contributing directly to project success.

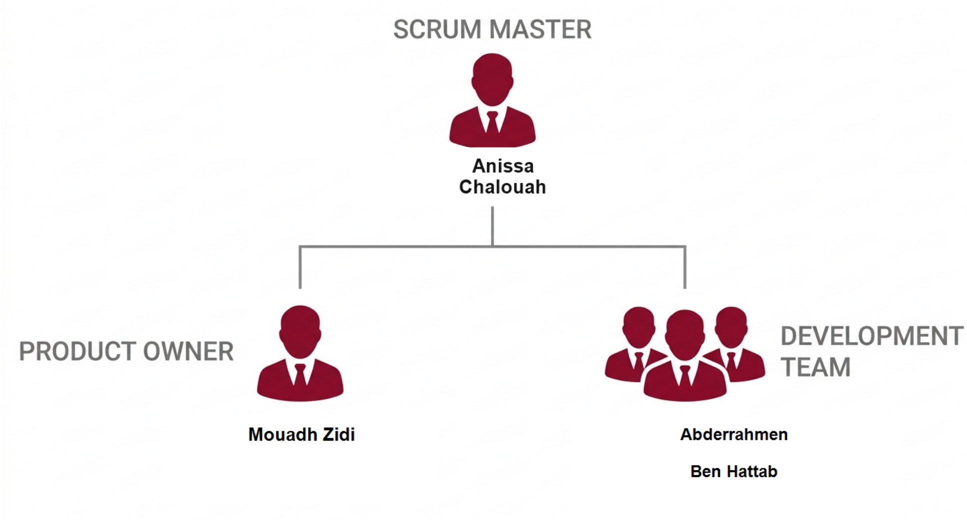


Figure 3: Scrum team roles

IV. Development environment

To complete this project, an appropriate development environment was set up. It includes hardware resources, software tools, and modern technologies to ensure effective design, development, and deployment.

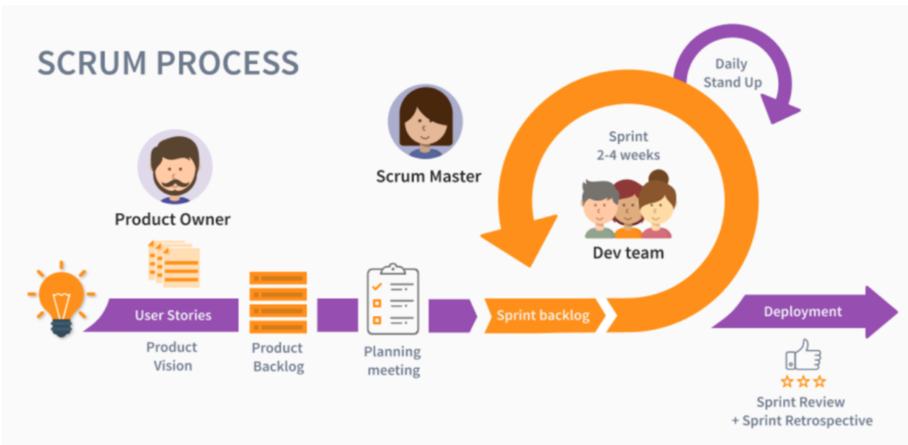
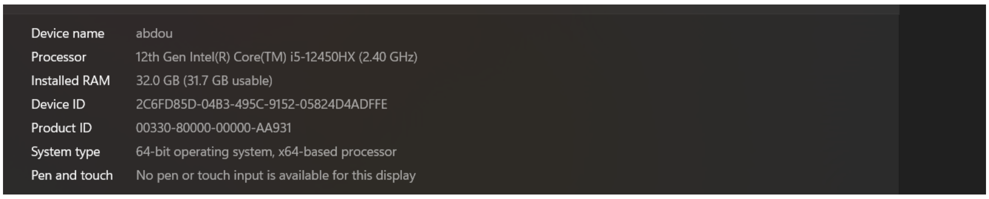


Figure 4: SCRUM method

1. Hardware environment

This project was developed on a laptop with the following specifications:



Device name	abdou
Processor	12th Gen Intel(R) Core(TM) i5-12450HX (2.40 GHz)
Installed RAM	32.0 GB (31.7 GB usable)
Device ID	2C6FD85D-04B3-495C-9152-05824D4ADFFE
Product ID	00330-80000-00000-AA931
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Figure 5: PC specifications used for development

2. Software environment

For developing the internal request, project, and task management platform, several software tools were used. Each tool was selected for its specific features and its fit with project requirements.

- **Visual Studio Code :**

Main code editor used for **frontend** development with **React** and **Tailwind CSS**. Its lightweight design and extension ecosystem provide a productive environment for UI implementation.

- **IntelliJ IDEA :**

Integrated development environment used for **backend** development with **Spring Boot** and **PostgreSQL**. IntelliJ IDEA provides advanced autocompletion, debugging, refactoring, and native integration with Maven and Spring.

- **Postman :**

Used to test and validate REST APIs developed with Spring Boot through an intuitive interface and strong debugging and documentation features.

- **Draw.io :**

  Used to design UML diagrams such as use-case, class, and sequence diagrams, as well as functional mockups, thanks to its ease of use and free access.

- **Git and GitHub :**

  Used for version control and source-code management, with reliable workflows and easy integration in Visual Studio Code and IntelliJ IDEA.

- **Jira :**

  Used for task management and Scrum sprint tracking. It supports user-story planning, sprint monitoring, and progress reporting.

- **LaTeX :**

  LaTeX was selected for report writing to ensure a professional layout, accurate references, and typographic consistency throughout the document.

- **Figma :**

  Used to create wireframes for the user interface. It enables quick visual prototyping and ergonomic validation before development.

3. Technologies used

The following technologies were chosen for their fit with project requirements and their ability to deliver a modern, high-performance solution.

- **React :**

  Main frontend library used in the View layer of the platform interface. React provides reusable components, efficient rendering, and flexible state management for scalable user interfaces.

- **Tailwind CSS :**

  Utility-first CSS framework used for interface styling and layout. Tailwind CSS enables fast, consistent, and responsive UI development while supporting design customization.

- **Spring Boot :**

  Java framework used for **backend** development based on a **layered architecture**. Spring Boot simplifies secure and high-performance REST API development through auto-configuration and a rich ecosystem.

- **PostgreSQL :**

  Relational database management system used for platform data storage and management. PostgreSQL was selected for reliability, performance, and strong support for complex queries and data integrity.

- **Keycloak :**

  Identity and access management solution used to handle secure authentication, role-based authorization, and token-based security for the platform.

- **Docker :**

Containerization platform used to package and run application services consistently across development and deployment environments.

- **Kafka :**

Distributed event-streaming platform used for asynchronous communication, especially for notifications and background event processing in the system.

V. Application architecture

1. Global architecture

The application follows a web-based architecture composed of two complementary parts: a **React frontend structured with MVVM** and a **layered** backend. This organization fits an internal platform used by employees, team leaders, and HR managers to manage requests, projects, tasks, notifications, and dashboards. It ensures clear separation of responsibilities, smooth integration through REST APIs, and maintainable system evolution. External services such as Keycloak, Kafka, JasperReports, and email notifications are integrated when needed to support authentication, event handling, reporting, and communication.

2. Frontend part: MVVM architecture with React

The frontend part is implemented with **React** and **Tailwind CSS** and organized according to the **MVVM** pattern. This choice is appropriate for a role-aware internal platform where employees, team leaders, and HR managers access different views over the same operational data. React provides the View layer, view models coordinate presentation logic and state, and service classes handle communication with the Spring Boot REST API. This organization ensures a clear separation between interface rendering, presentation behavior, and service communication, which improves maintainability and functional consistency.

- **View layer:** Contains React pages and reusable UI components used to display request forms, task boards, project dashboards, notification panels, and role-specific interfaces.

- **ViewModel layer:** Handles presentation logic, state coordination, form behavior, interaction flows, and the transformation of backend data into structures adapted to the interface.
- **Model / Service layer:** Groups frontend data models, DTO-shaped structures when needed, and service classes or functions responsible for calling the backend API and managing request and response data.

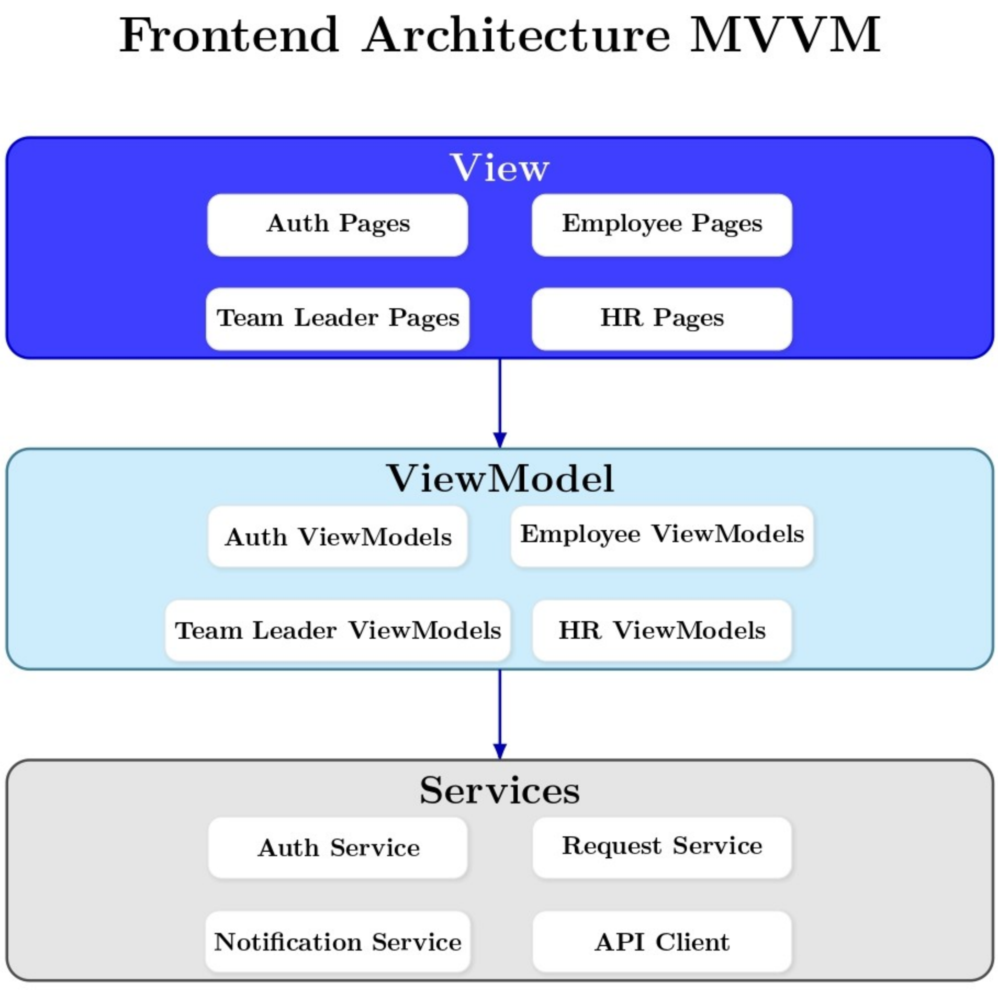


Figure 6: MVVM frontend architecture with React

3. Backend part: layered architecture with Spring Boot

The backend is based on a **layered architecture**, implemented with **Spring Boot** and **PostgreSQL**. This architecture structures the backend into clear functional layers, improving maintainability, testability, and scalability.

- **Controller layer:** Spring MVC REST controllers expose API endpoints and handle HTTP requests from the React frontend.
- **Service layer:** Contains business logic, validation rules, and orchestration of application use cases.
- **Repository layer:** Spring Data JPA repositories handle persistence and data access in **PostgreSQL**.

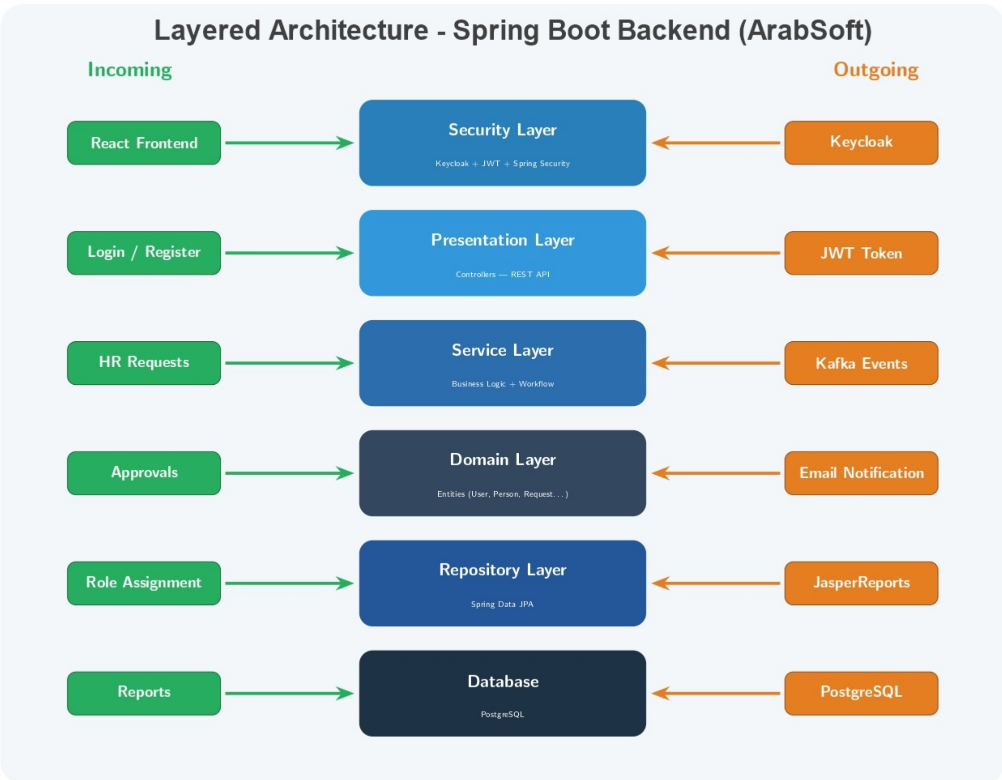


Figure 7: Layered architecture with Spring Boot

4. Why these architectures?

These architectures were selected for several reasons:

- **Separation of concerns:** MVVM separates the interface, presentation logic, and service access in the frontend, while the backend keeps controllers, services, and repositories clearly isolated.
- **Maintainability and testability:** View models can be tested and evolved independently from React pages, and backend service logic remains independent from persistence concerns.
- **Reusability and consistency:** A structured frontend makes it easier to reuse UI components and shared presentation logic across requests, projects, tasks, dashboards, and notifications.
- **Scalability and integration:** React communicates efficiently with the Spring Boot REST API, which supports progressive extension of the platform without compromising organization.

5. Global architecture illustration

The figure below presents the global application architecture in a cleaner and more report-ready form. It shows the React frontend structured with MVVM, the Spring Boot backend organized in layers, PostgreSQL as the persistent storage, and the external services integrated to support authentication, asynchronous communication, reporting, and notifications. The diagram also highlights the role-based usage of the platform by employees, team leaders, and HR managers.

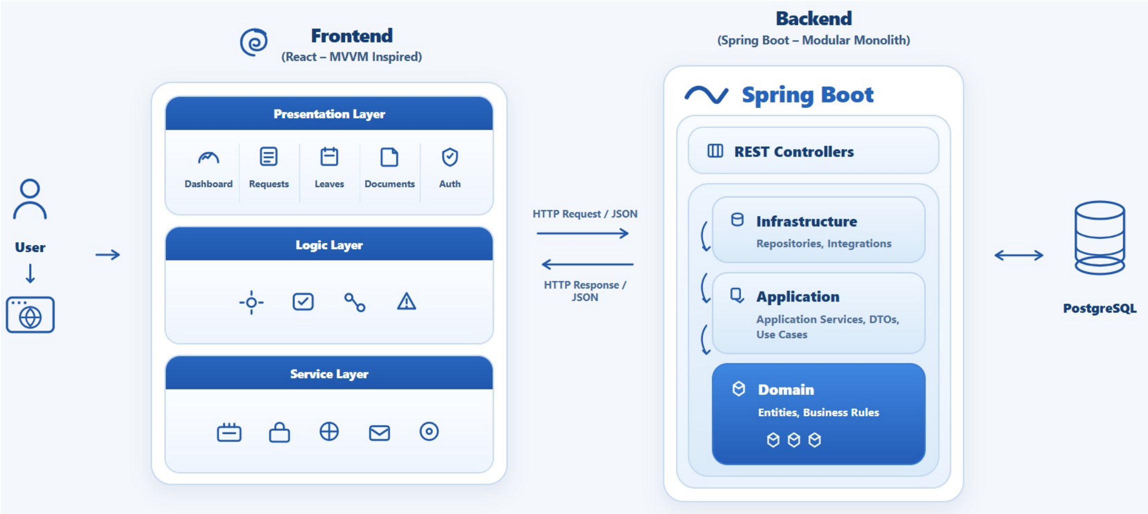


Figure 8: Global application architecture

Conclusion

This chapter presented the general project context: the host organization ArabSoft, the current internal management challenges related to requests, projects, and tasks, the proposed solution, and the adopted methodologies and development environment. These elements provide the foundation for the rest of the work. The next chapter focuses on product backlog planning and functional requirement identification.

Chapter 2

Product Backlog Planning

Introduction

This chapter defines the product backlog planning for the proposed internal platform at ArabSoft. It identifies the actors, structures the functional and non-functional requirements, and formalizes user stories and backlog priorities that will guide sprint planning and implementation.

I. Actor Identification

1. Employee

The employee submits administrative requests, tracks processing status, accesses assigned tasks, and receives notifications related to approvals, rejections, and project activities.

2. Team Leader

The team leader supervises team activities, validates requests requiring managerial approval, assigns and follows tasks, and monitors project execution indicators.

3. HR Manager

The HR manager manages employee records and roles, performs final validation of administrative requests, and monitors compliance through dashboards and reports.

4. New User

A new user is registered in the platform but remains in a restricted state until HR validation and role assignment are completed.

II. Functional Requirements

1. Authentication and access management

The system shall support registration, secure authentication, logout, account activation by HR, and role-based access control according to user profile.

2. Administrative request management

The platform shall manage submission, validation, rejection, and history tracking for leave requests, document requests, loan requests, and authorization requests.

3. Team and employee management

The system shall provide profile management, team assignment, leader designation, and maintenance of employee administrative information.

4. Project and task management

The platform shall support project creation, task assignment, workload organization, progress tracking, and status visibility for all authorized stakeholders.

5. Notification and reporting

The system shall send operational notifications and provide reports and dashboards to support monitoring, traceability, and decision-making.

6. AI assistance and decision support

The platform shall provide AI-assisted support to help employees prepare administrative requests, assist team leaders in reviewing requests and workload constraints, and support HR managers with recommendation-oriented decision assistance while preserving human validation, traceability, and auditability.

III. Non-Functional Requirements

1. Security

The application must guarantee secure authentication, protection of sensitive data, strict role-based permissions, and complete audit traceability.

2. Performance

The platform must maintain acceptable response times under expected internal usage and support concurrent request processing without significant delays.

3. Usability

Interfaces must be clear and intuitive to reduce training effort and facilitate adoption by employees, team leaders, and HR managers.

4. Maintainability

The system architecture must allow efficient correction, enhancement, and extension through modular and well-documented components.

5. Scalability

The platform should support future growth in users, workflows, and modules without requiring major redesign.

IV. User Stories

1. Employee User Stories

Employee User Stories

As an employee, I want to sign in securely so that I can access the platform according to my role.	As an employee, I want to sign out securely so that I can protect my account after use.	As an employee, I want to manage my profile information so that my personal data stays accurate.
As an employee, I want to change my password securely so that I can keep my account protected.	As an employee, I want to access my personal dashboard so that I can quickly view the most relevant information.	As an employee, I want to submit and cancel a leave request before decision so that my leave workflow stays accurate.
As an employee, I want to submit a document request so that I can obtain official HR documents.	As an employee, I want to submit a loan request so that I can ask for financial support through the platform.	As an employee, I want to submit an authorization request so that I can formalize exceptional permissions.

As an employee, I want to track request status and history so that I can follow leave and administrative workflows.

As an employee, I want to view my assigned tasks and project calendar so that I can organize my work efficiently.

As an employee, I want to filter tasks by project so that I can focus on the work context concerned.

As an employee, I want to receive notifications and emails about role, request, leave, and task updates so that I stay informed.

2. Team Leader User Stories

Team Leader User Stories

As a team leader, I want to sign in securely so that I can access the platform according to my role.

As a team leader, I want to sign out securely so that I can protect my account after use.

As a team leader, I want to manage my profile information so that my personal data stays accurate.

As a team leader, I want to change my password securely so that I can keep my account protected.

As a team leader, I want to access my dashboard so that I can quickly monitor team activity.

As a team leader, I want to review team leave requests so that I can process them according to team constraints.

As a team leader, I want to approve or reject leave requests with a reason so that decisions remain clear and traceable.

As a team leader, I want to monitor team workload so that I can balance responsibilities effectively.

As a team leader, I want to assign tasks to team members so that project work is distributed properly.

As a team leader, I want to follow task progress and receive completion notifications so that I can monitor delivery.

As a team leader, I want to view the team leave calendar within my scope so that I can anticipate availability issues.

As a team leader, I want to use employee-side leave and request actions for my own needs so that personal workflows remain consistent.

3. HR Manager User Stories

HR Manager User Stories

As an HR manager, I want to sign in securely so that I can access the platform according to my role.

As an HR manager, I want to sign out securely so that I can protect my account after use.

As an HR manager, I want to manage my profile information so that my personal data stays accurate.

As an HR manager, I want to change my password securely so that I can keep my account protected.

As an HR manager, I want to access my dashboard so that I can monitor operational activity.

As an HR manager, I want to assign roles and trigger user notification so that access changes are communicated clearly.

As an HR manager, I want to activate or deactivate accounts so that access remains controlled.

As an HR manager, I want to manage employee administrative information so that records stay complete and accurate.

As an HR manager, I want to make final decisions on leave and administrative requests so that workflows are completed properly.

As an HR manager, I want to manage teams and team leaders so that the organizational structure remains consistent.

As an HR manager, I want broad leave calendar visibility and filters so that I can supervise teams, employees, dates, and statuses.

As an HR manager, I want to review pending user accounts so that I can complete their access setup.

As an HR manager, I want to upload official request files and notify users so that final HR outputs are communicated.

As an HR manager, I want to trace request history and cancellations before decision so that workflow follow-up remains clear.

V. Product Backlog

The product backlog is organized into six functional sprints. Each sprint groups a coherent set of user stories and technical tasks according to business value, implementation dependencies, and the progressive delivery of the platform. The backlog was tracked through Jira and structured around access management, leave workflows, administrative requests, HR operations, collaboration features, reporting, notifications, and AI-assisted support.

Table 1: Product Backlog

Sprint	Num	User Story / Task	Description	Complexity
Sprint 1 Access and Account Management	**US 01**	**User registration**	As a new user, I want to create an account so that I can request access to the platform.	Medium
	US 02	**Secure authentication**	As a user, I want to sign in securely so that I can access features according to my role.	Medium
	US 03	**Pending access state**	As a new user, I want to see a pending-access page until HR assigns my role.	Low
	US 04	**Password reset**	As a user, I want to reset my password so that I can recover access to my account.	Medium
	US 05	**Profile update**	As a user, I want to update my profile information so that my personal data remains accurate.	Low
	US 06	**Role assignment**	As an HR manager, I want to assign roles so that users receive the correct access rights.	Medium
	US 07	**Account control**	As an HR manager, I want to activate or deactivate accounts so that platform access remains controlled.	Medium
	US 08	**Pending user review**	As an HR manager, I want to review pending user accounts so that I can complete their access setup.	Medium
	TS 01	**Security configuration**	Configure Keycloak, PostgreSQL, and Spring Security for secure role-based access.	High
	TS 02	**Role notification**	Send notification and email after role assignment.	Low

Table 1: Product Backlog (continued)

Sprint	Num	User Story / Task	Description	Complexity
Sprint 2 Leave Management	**US 09**	**Leave submission**	As an employee, I want to submit a leave request so that I can ask for time off.	Medium
	US 10	**Leave balance and history**	As an employee, I want to view my leave balance and leave history.	Medium
	US 11	**Working-day validation**	As an employee, I want the system to validate leave dates using working days and public holidays.	High
	US 12	**Overlap prevention**	As an employee, I want the system to prevent overlapping leave requests.	High
	US 13	**Edit and cancellation**	As an employee, I want to edit or cancel a leave request before final decision.	Medium
	US 14	**Team leader decision**	As a team leader, I want to approve or reject team leave requests.	Medium
	US 15	**HR final decision**	As an HR manager, I want to make the final decision on leave requests.	High
	US 16	**Leave calendar**	As a user, I want to view leave requests in a calendar.	Medium
	US 17	**Leave filters**	As an HR manager, I want to filter leave records by employee, status, and date.	Low
	TS 03	**Leave notifications**	Send leave decision notifications and emails.	Low
	TS 04	**Leave reports**	Generate leave approval reports.	Medium

Table 1: Product Backlog (continued)

Sprint	Num	User Story / Task	Description	Complexity
Sprint 3 Administrative Requests	**US 18**	**Document request**	As an employee, I want to submit a document request so that I can obtain official HR documents.	Medium
	US 19	**Document processing**	As an HR manager, I want to process document requests and upload final files.	Medium
	US 20	**Loan request**	As an employee, I want to submit a loan request so that I can request financial support.	Medium
	US 21	**Loan eligibility**	As an HR manager, I want to check loan eligibility before making a decision.	High
	US 22	**Loan meeting**	As an HR manager, I want to schedule a loan meeting with valid time slots.	High
	US 23	**Authorization request**	As an employee, I want to submit an authorization request for exceptional permission.	Medium
	US 24	**HR request decision**	As an HR manager, I want to approve or reject administrative requests.	High
	US 25	**Request tracking**	As a user, I want to track the status and history of my administrative requests.	Low
	US 26	**Request filters**	As an HR manager, I want to search and filter administrative requests.	Low
	TS 05	**Request notifications**	Send request status notifications and emails.	Low
	TS 06	**Final files**	Manage final request files and downloads.	Medium

Table 1: Product Backlog (continued)

Sprint	Num	User Story / Task	Description	Complexity
Sprint 4 HR Management and Calendar	**US 27**	**Employee records**	As an HR manager, I want to manage employee records.	Medium
	US 28	**Department management**	As an HR manager, I want to manage departments.	Medium
	US 29	**Job title management**	As an HR manager, I want to manage job titles.	Medium
	US 30	**Team management**	As an HR manager, I want to create teams and assign team leaders.	High
	US 31	**Employee movement**	As an HR manager, I want to move employees between teams.	Medium
	US 32	**User search and filters**	As an HR manager, I want to search and filter users.	Low
	US 33	**HR dashboard**	As an HR manager, I want to view HR dashboard indicators.	Medium
	US 34	**HR calendar**	As an HR manager, I want to supervise leave planning through the HR calendar.	Medium
	US 35	**Team calendar**	As a team leader, I want to view my team calendar to understand availability.	Medium
	TS 07	**Calendar filters**	Add calendar filters by employee, status, and date.	Low
	TS 08	**HR approval views**	Improve HR request and approval views.	Medium

Table 1: Product Backlog (continued)

Sprint	Num	User Story / Task	Description	Complexity
Sprint 5 Projects, Tasks, Reports and Notifications	**US 36**	**Project management**	As a team leader, I want to create and manage projects.	Medium
	US 37	**Task assignment**	As a team leader, I want to create and assign tasks to team members.	Medium
	US 38	**Task status update**	As an assignee, I want to update task status so that progress remains visible.	Low
	US 39	**Task filters**	As a user, I want to filter tasks by project and status.	Low
	US 40	**Availability check**	As a team leader, I want to check availability before assigning work.	High
	US 41	**Notifications**	As a user, I want to receive in-app notifications about important updates.	Medium
	US 42	**PDF reports**	As a user, I want to download generated PDF reports.	Medium
	US 43	**QR verification**	As a public verifier, I want to verify official documents using a QR code.	Medium
	US 44	**Reports and statistics**	As an HR manager, I want to monitor reports and statistics.	Medium
	TS 09	**Kafka events**	Configure Kafka event publishing and consumers.	High
	TS 10	**Outbox monitoring**	Monitor failed outbox events and replay them.	High

Table 1: Product Backlog (continued)

Sprint	Num	User Story / Task	Description	Complexity
Sprint 6 AI, Deployment and Stabilization	**US 45**	**Platform questions**	As a user, I want to ask platform questions using the assistant.	Medium
	US 46	**Role-aware answers**	As a user, I want the assistant to answer according to my role.	High
	US 47	**Leave draft assistance**	As an employee, I want the assistant to help draft a leave request.	High
	US 48	**Draft validation**	As an employee, I want the system to validate the leave draft before submission.	High
	US 49	**Explicit confirmation**	As an employee, I want leave submission to require explicit confirmation.	Medium
	US 50	**Unsafe-action refusal**	As a user, I want the assistant to refuse unsafe actions.	Medium
	TS 11	**React assistant integration**	Integrate the React chatbot with the Spring Boot assistant gateway.	High
	TS 12	**FastAPI integration**	Connect the Spring Boot assistant gateway with the FastAPI AI service.	High
	TS 13	**Demo deployment preparation**	Prepare local demonstration deployment configuration.	Medium
	TS 14	**Final build checks**	Run final frontend, backend, and AI service build checks.	Medium
	TS 15	**Final stabilization**	Stabilize the platform and correct final integration issues.	High

VI. Release Organization

The six planned sprints are grouped into three major releases to ensure progressive and coherent delivery of the platform.

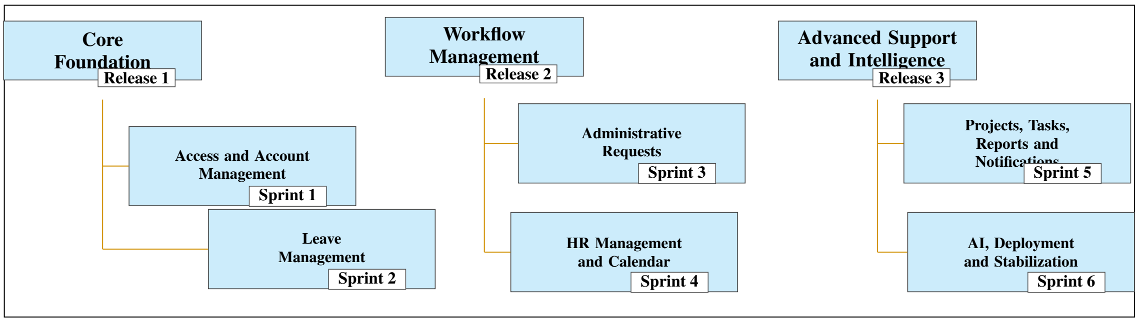


Figure 9: Organization of Releases

Conclusion

This chapter established the backlog-planning foundation of the project through actor identification, requirement structuring, user stories, and prioritized backlog definition. It provides the planning baseline for sprint execution and the progressive realization of the platform at ArabSoft.

Chapter 3

Release 1 – Foundations

Introduction

This chapter focuses on Release 1, which forms the functional and technical foundation of the ArabSoft internal platform. Through its use case diagrams, system sequence diagrams, state diagrams, class diagrams, and interface illustrations, this chapter presents how the first core functionalities of the platform are structured. It highlights the implementation logic of secure access management and the first complete administrative workflow. More specifically, Release 1 covers user registration, authentication, role assignment, account control, and leave request management. Together, these elements establish the first operational base of the platform for employees, team leaders, and HR managers.

I. Release Organization

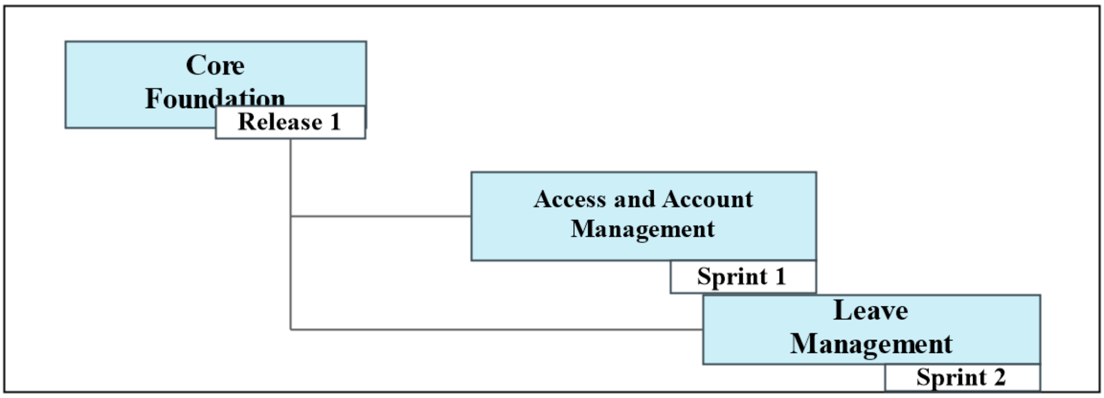


Figure 10: Release 1 Organization

II. Sprint 1: Access and Account Management

1. Sprint Goal

This first sprint aims to establish the technical and functional access foundation of the platform. It implements the core access features: user registration, secure authentication, password reset, HR user account setup, role assignment with notification and email, account activation and deactivation, profile update, and access control. It also introduces the restricted

NEW_USER state to control onboarding before HR validation. The sprint is planned as a two-week iteration. The objective is to obtain an operational base for the next functional developments of the platform.

2. Visual Sprint Journal

The visual sprint journal will document the selected user stories of Sprint 1 and their completion status in Jira. The corrected Sprint 1 journal screenshot will be inserted after validation.

3. Sprint Backlog

The table below presents the backlog of the first sprint, planned as a two-week iteration.

Sprint	Stories	Tasks	Duration
Sprint 1	**Configure security and infrastructure**	Configure Keycloak realm, client, and roles	2d
		Set up PostgreSQL, Docker, and Kafka	
		Connect Spring Boot to PostgreSQL and Keycloak	
	Registration and access control	Implement user registration	3d
		Enforce the restricted **NEW_USER** state	
		Implement secure login	
		Implement frontend logout	
	Password recovery and profile management	Implement password reset	2d
		Implement profile update	
	HR account governance	Implement HR role assignment	2d
		Trigger role assignment notification and email	
		Implement account activation and deactivation	
	HR account setup validation	Validate pending users and complete their access setup	1d

Table 2: Sprint Backlog – Sprint 1

4. Sprint Implementation

Requirements Analysis

Sprint 1 covers the access and account management foundation, including registration, authentication, password recovery, profile management, HR account setup, and account status control.

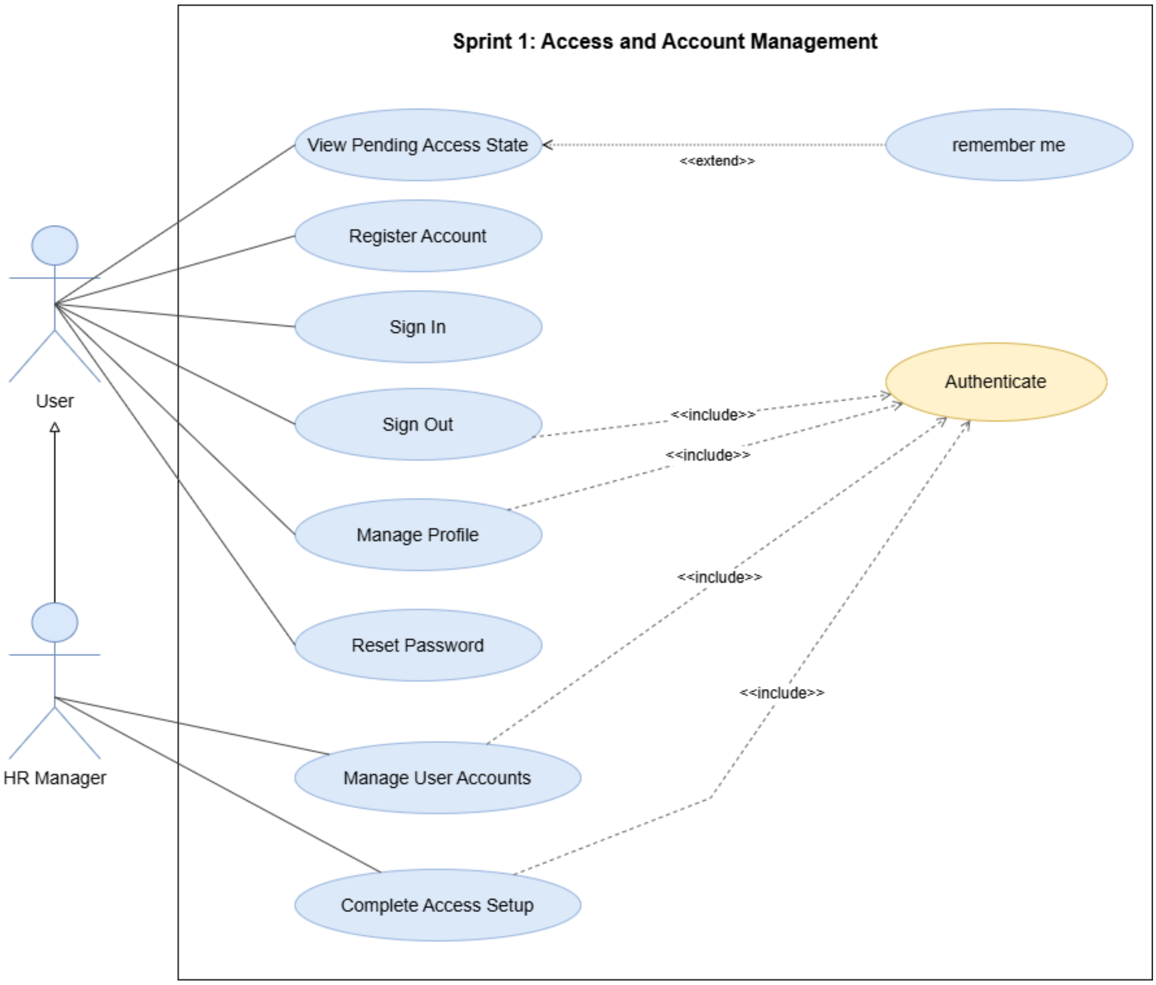


Figure 11: Use Case Diagram – Sprint 1 Access and Account Management

System Sequence Diagrams

These diagrams summarize the main user-system exchanges for Sprint 1 access scenarios.

User Registration

This diagram shows how a visitor registers and remains pending until HR validation.

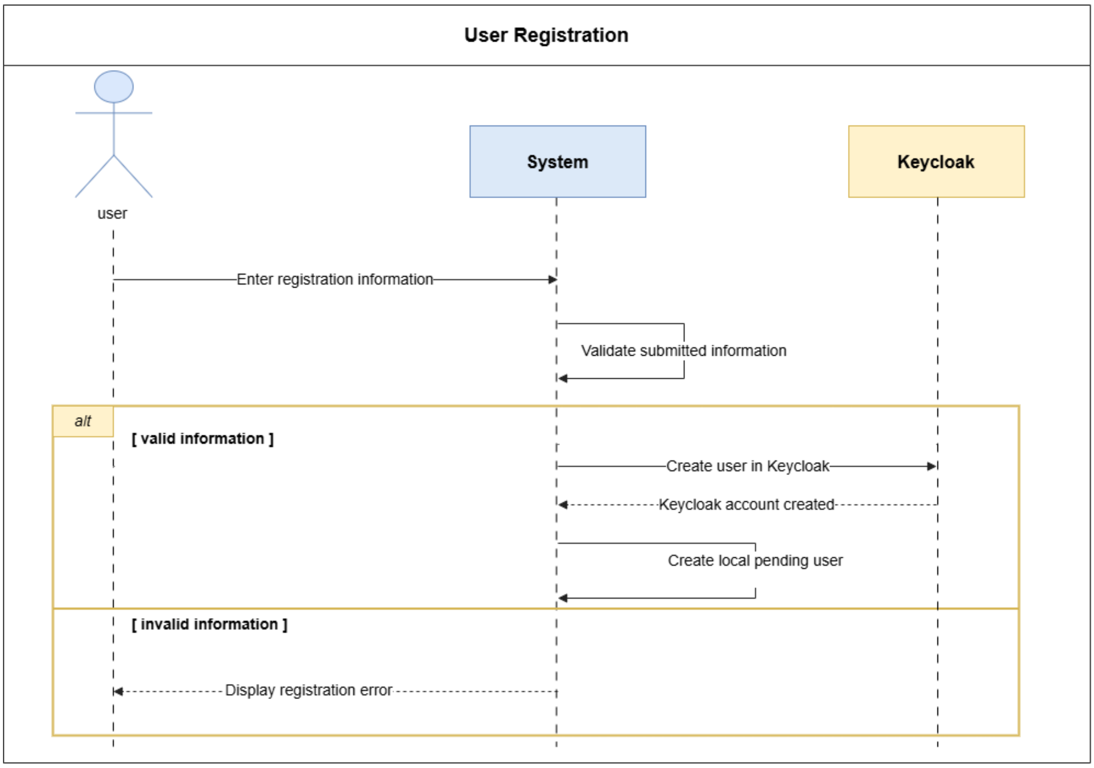


Figure 12: System Sequence Diagram – User Registration

Secure Login

This diagram shows credential validation through Keycloak and role-based redirection.

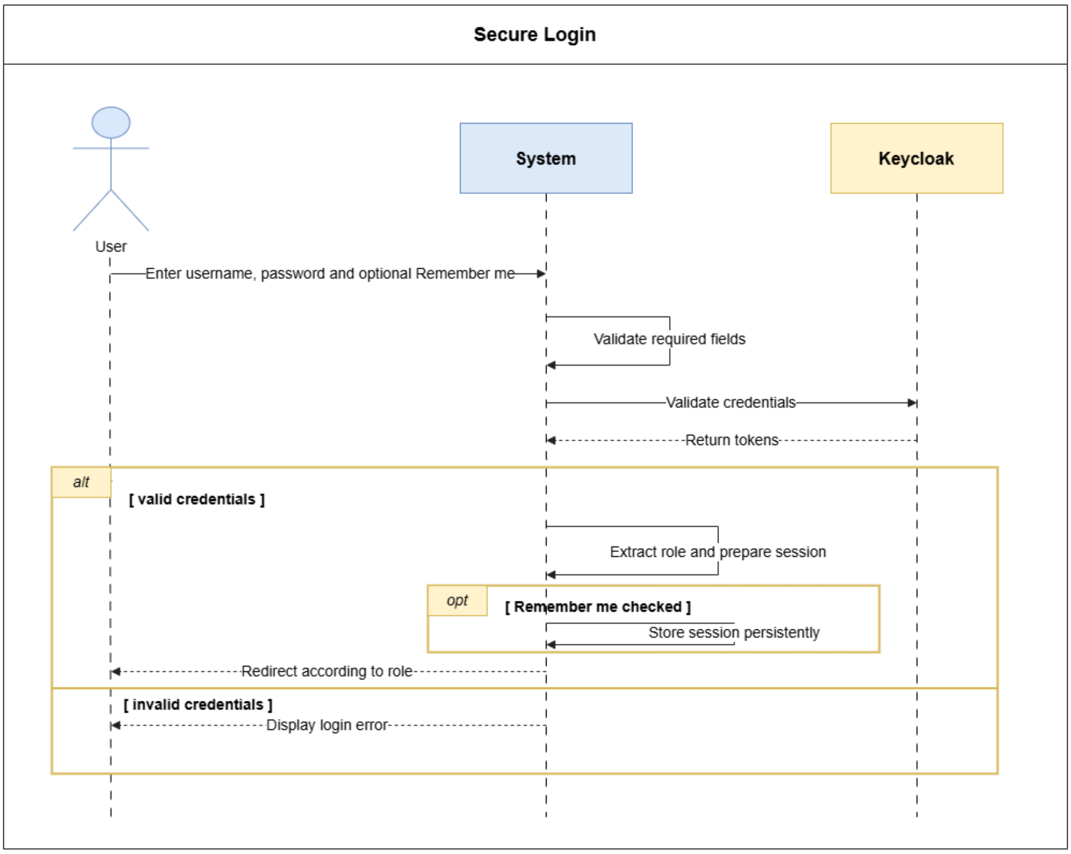


Figure 13: System Sequence Diagram – Secure Login

Logout

This diagram shows session cleanup, optional Keycloak logout, and redirection to the login page.

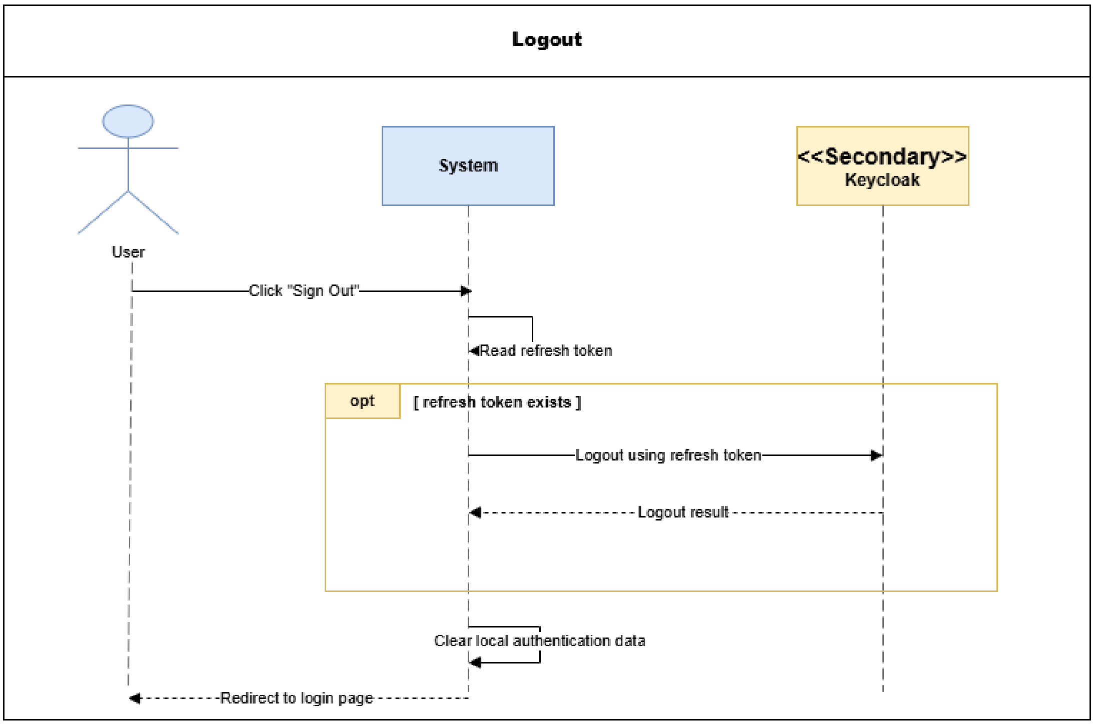


Figure 14: System Sequence Diagram – Logout

Use Case Diagram – Personal Account Management

This diagram groups the personal account actions available to an authenticated user.

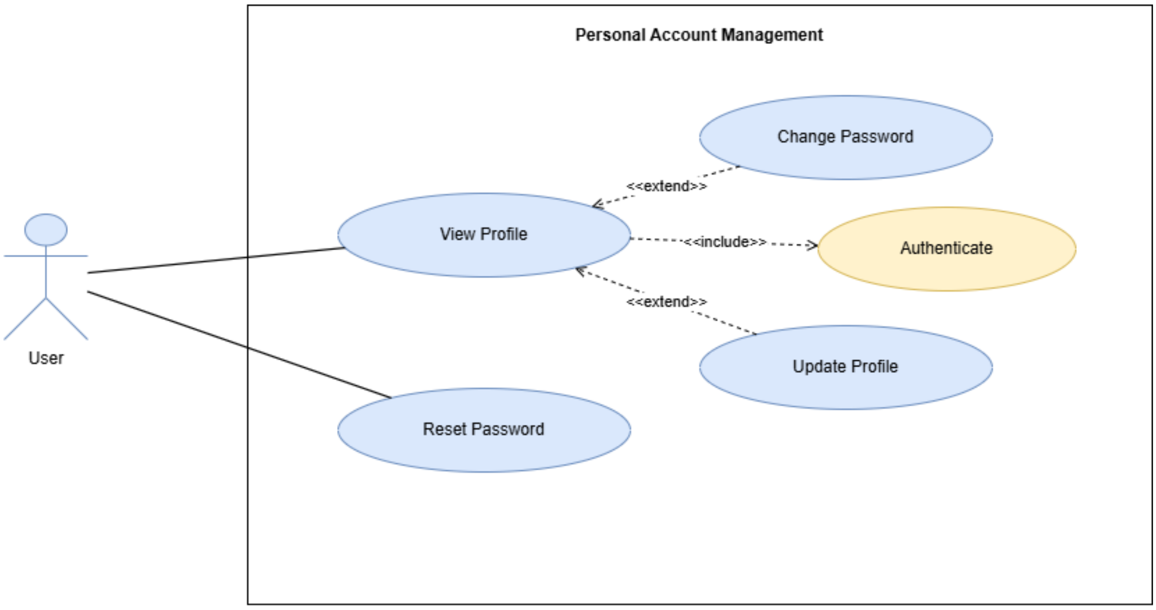


Figure 15: Use Case Diagram – Personal Account Management

Password Reset

This diagram shows how a user receives a reset code and defines a new password.

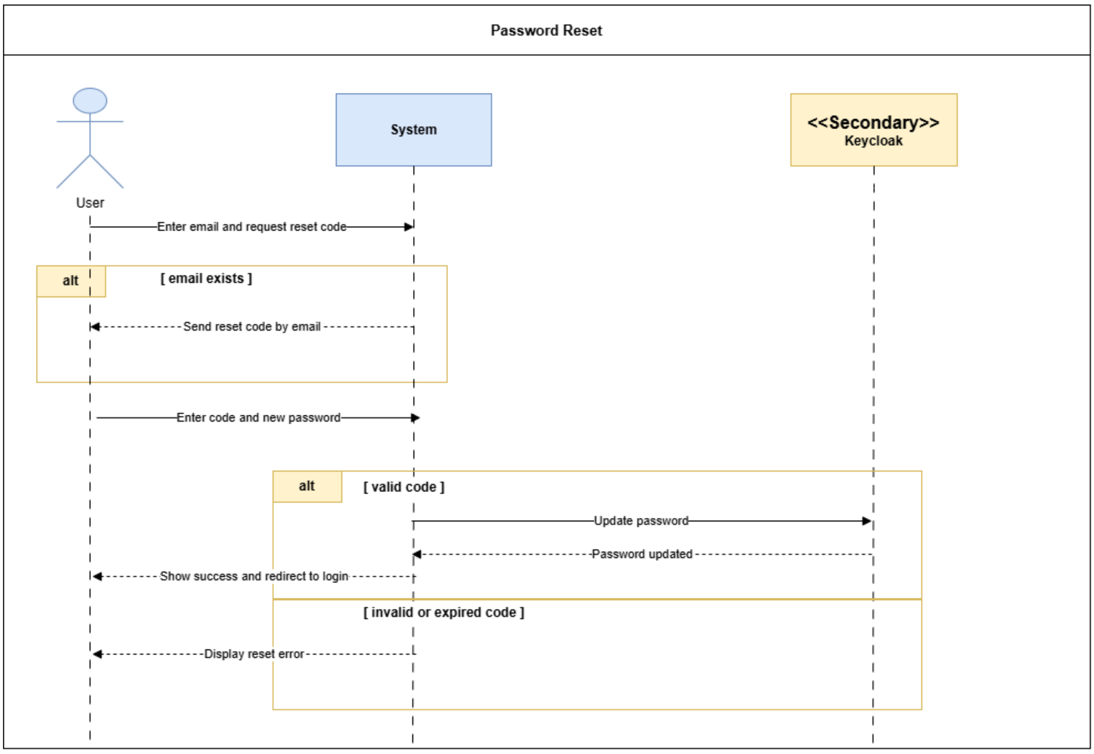


Figure 16: System Sequence Diagram – Password Reset

Profile Update

This diagram shows how an authenticated user updates allowed profile information.

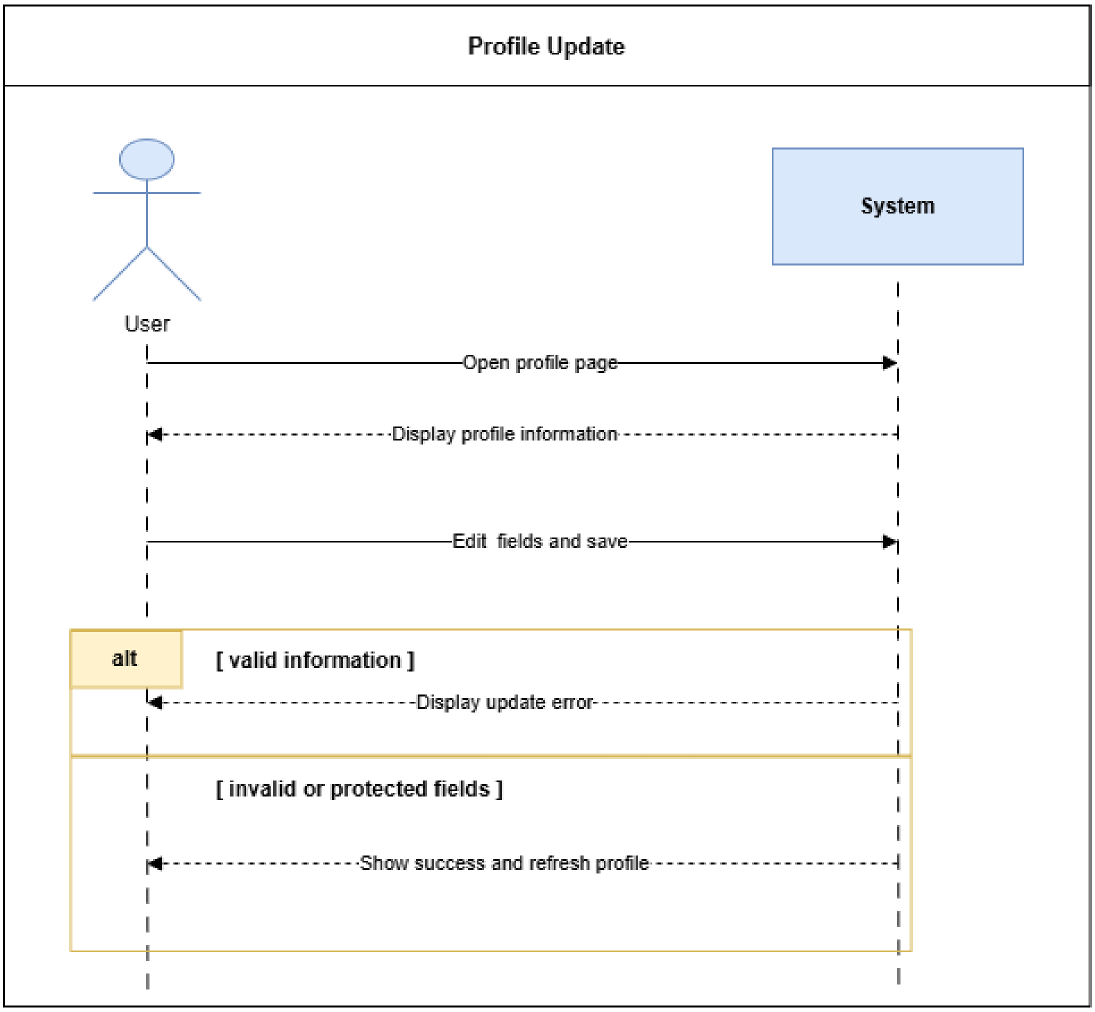


Figure 17: System Sequence Diagram – Profile Update

Authenticated Password Change

This diagram shows password change using the current password or OTP verification.

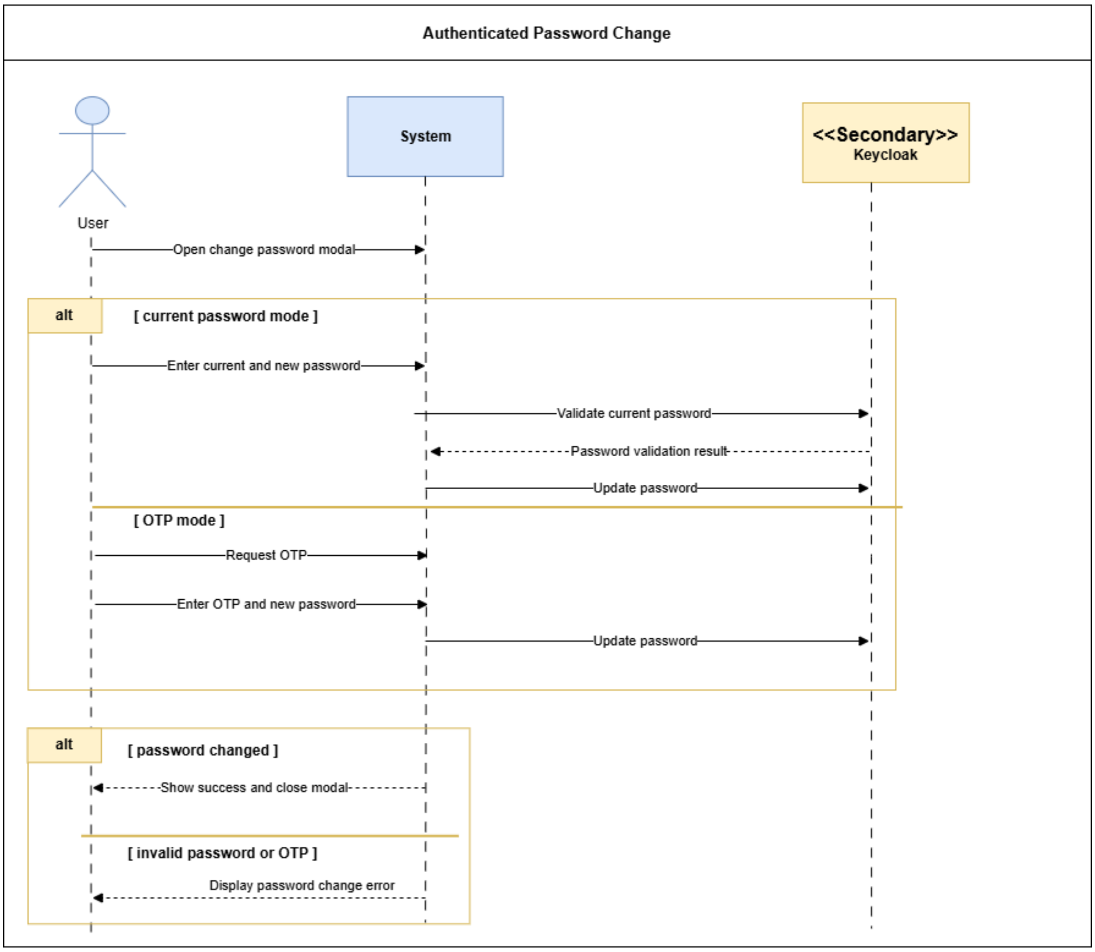


Figure 18: System Sequence Diagram – Authenticated Password Change

Use Case Diagram – HR User Account Management

This diagram groups the HR actions used to manage user accounts.

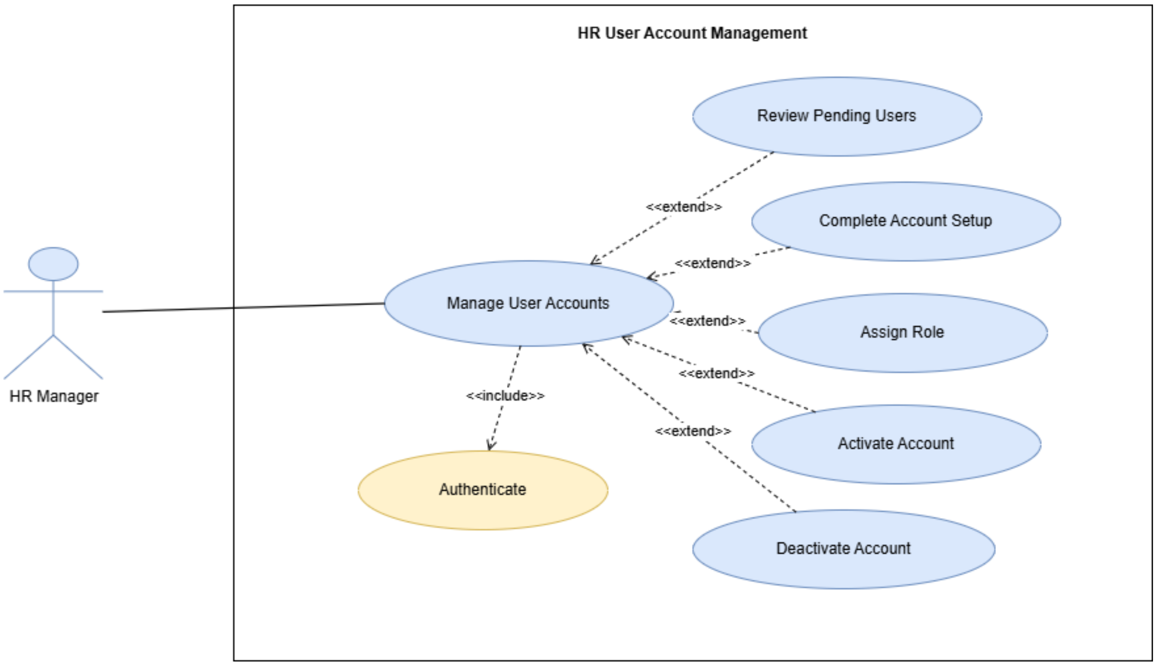


Figure 19: Use Case Diagram – HR User Account Management

HR User Account Setup

This diagram shows how HR completes a pending account setup and activates access.

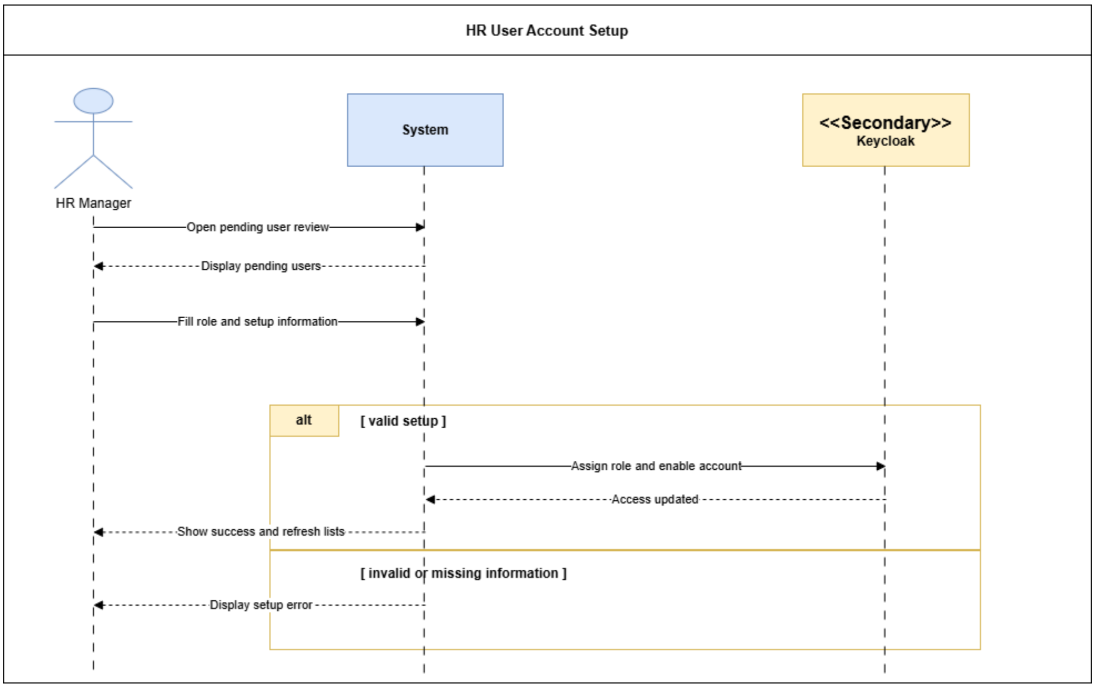


Figure 20: System Sequence Diagram – HR User Account Setup

Account Activation and Deactivation

This diagram shows HR-controlled account activation and deactivation.

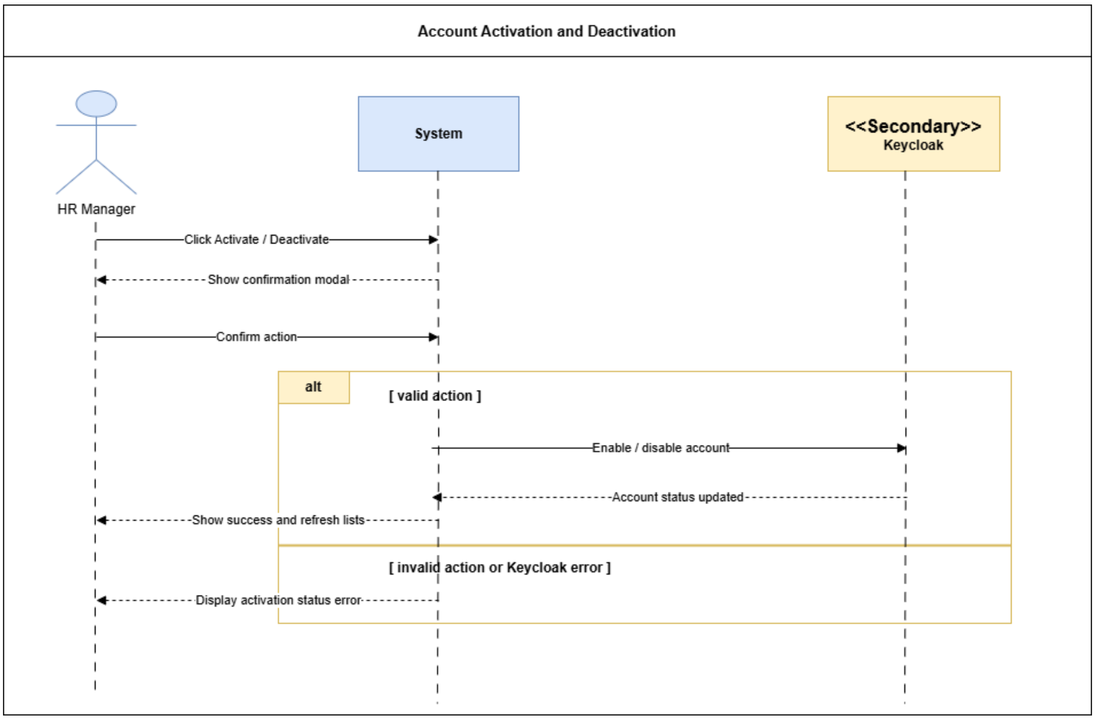


Figure 21: System Sequence Diagram – Account Activation and Deactivation

Analysis

This section summarizes the account lifecycle and the main business concepts used in Sprint 1.

Domain Model – Sprint 1 Access and Account Management

This diagram shows the main business concepts used for Sprint 1 account and access management.

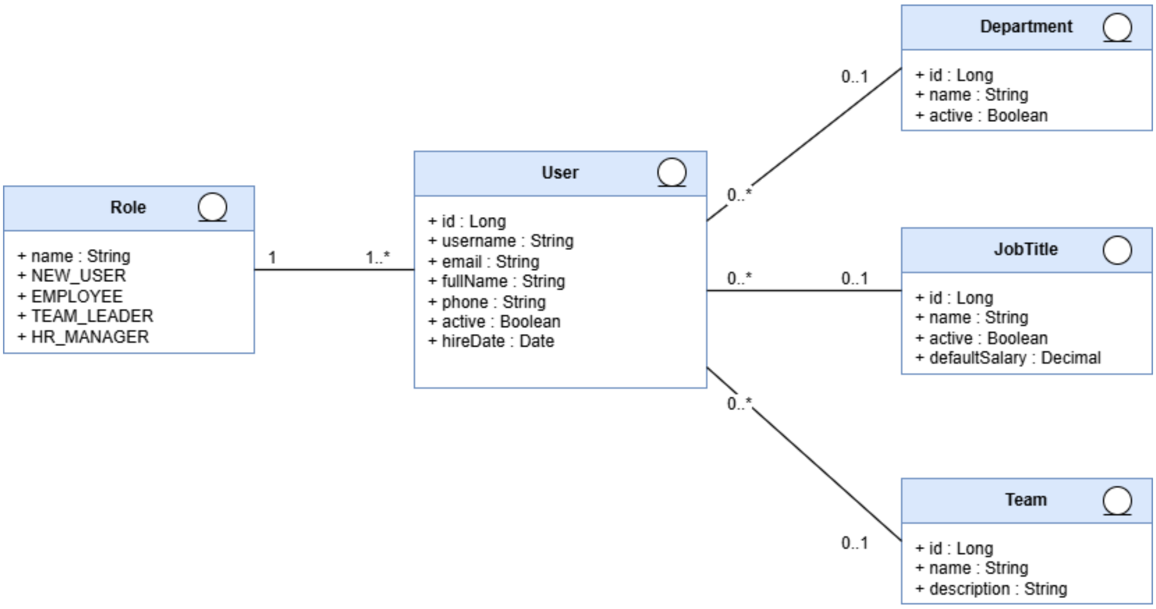


Figure 22: Domain Model – Sprint 1 Access and Account Management

Participant Class Diagrams

These diagrams identify the main frontend and backend participants involved in Sprint 1 scenarios.

Participant Class Diagram – Authentication and Access Control

This diagram shows the participants involved in authentication and role-based access control.

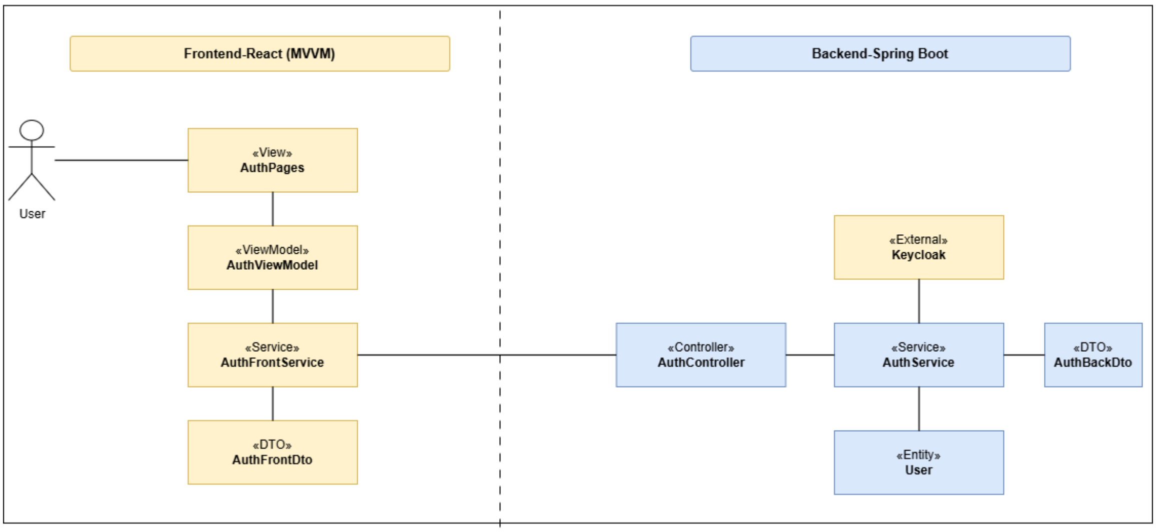


Figure 23: Participant Class Diagram – Authentication and Access Control

Participant Class Diagram – Password Reset and Password Change

This diagram shows the participants involved in password recovery and authenticated password change.

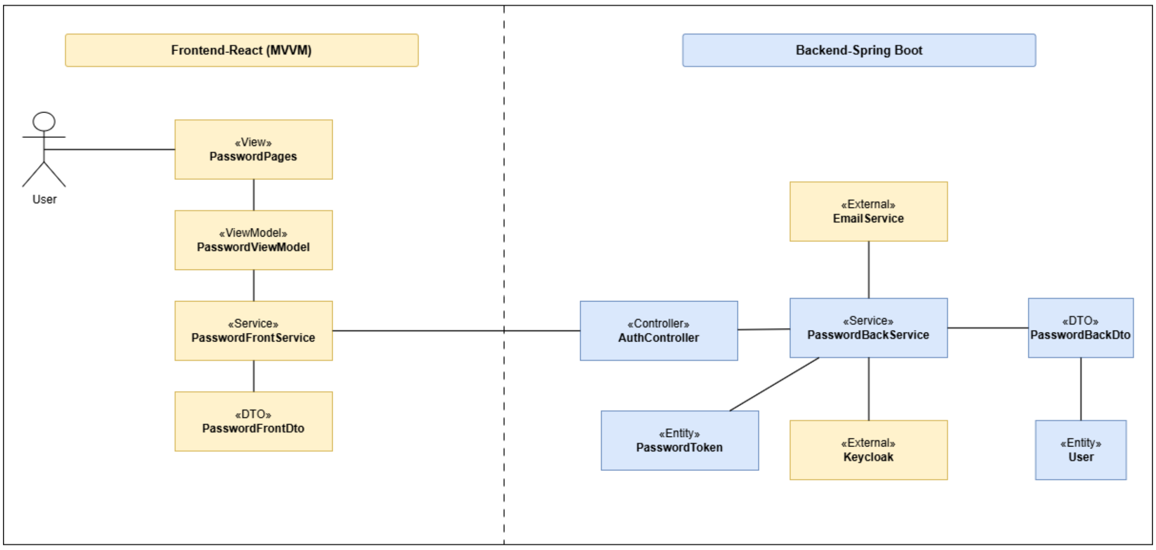


Figure 24: Participant Class Diagram – Password Reset and Password Change

Participant Class Diagram – HR Account Management

This diagram shows the participants involved in HR account setup, role assignment, and account status control.

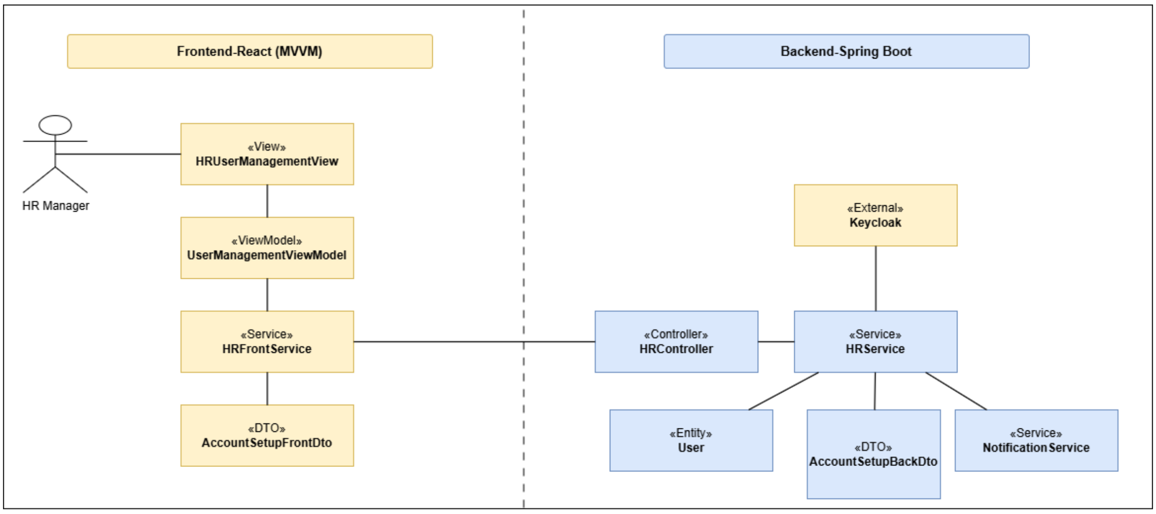


Figure 25: Participant Class Diagram – HR Account Management

State Diagrams

This diagram presents the lifecycle of a user account during the access-management workflow.

State Diagram – User Account Lifecycle

This diagram shows the transition from a pending account to an active or deactivated account.

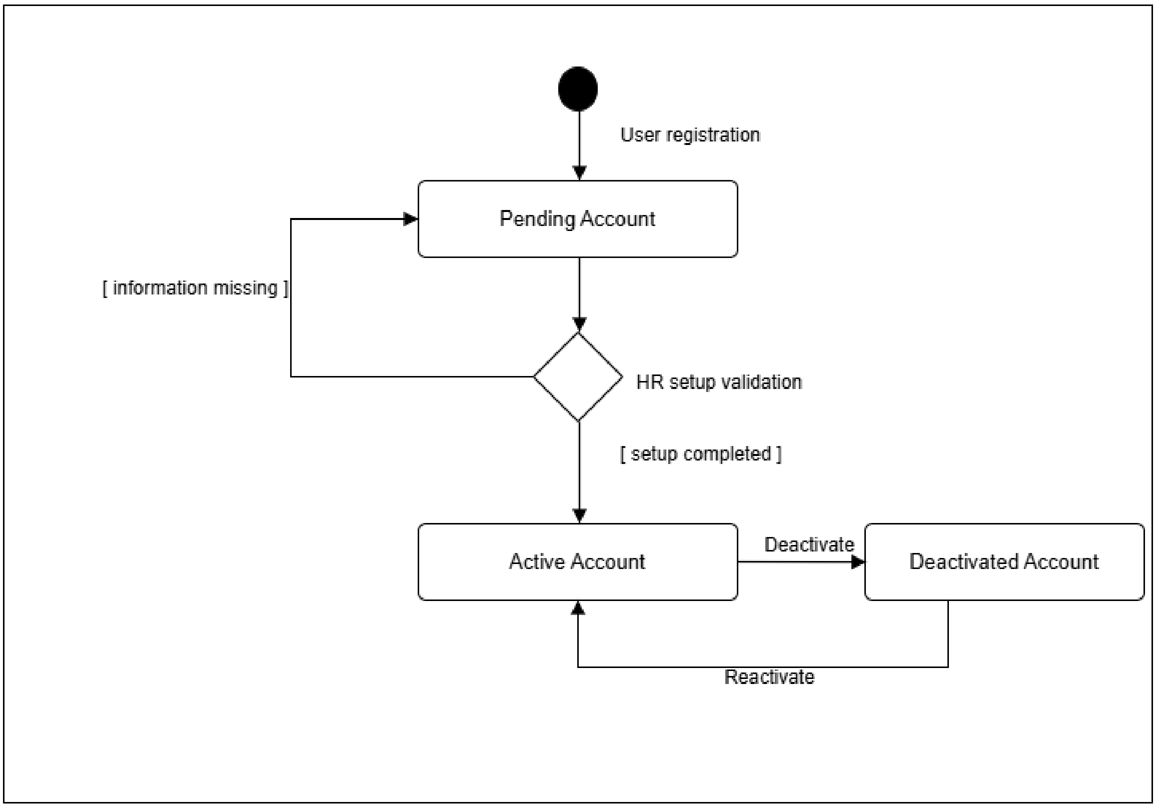


Figure 26: State Diagram – User Account Lifecycle

Design Class Diagrams

These diagrams present the main classes supporting Sprint 1 access-management features.

Design Class Diagram – Registration and Pending Access

This diagram illustrates the main classes used to register a new user and keep the account pending until HR validation.

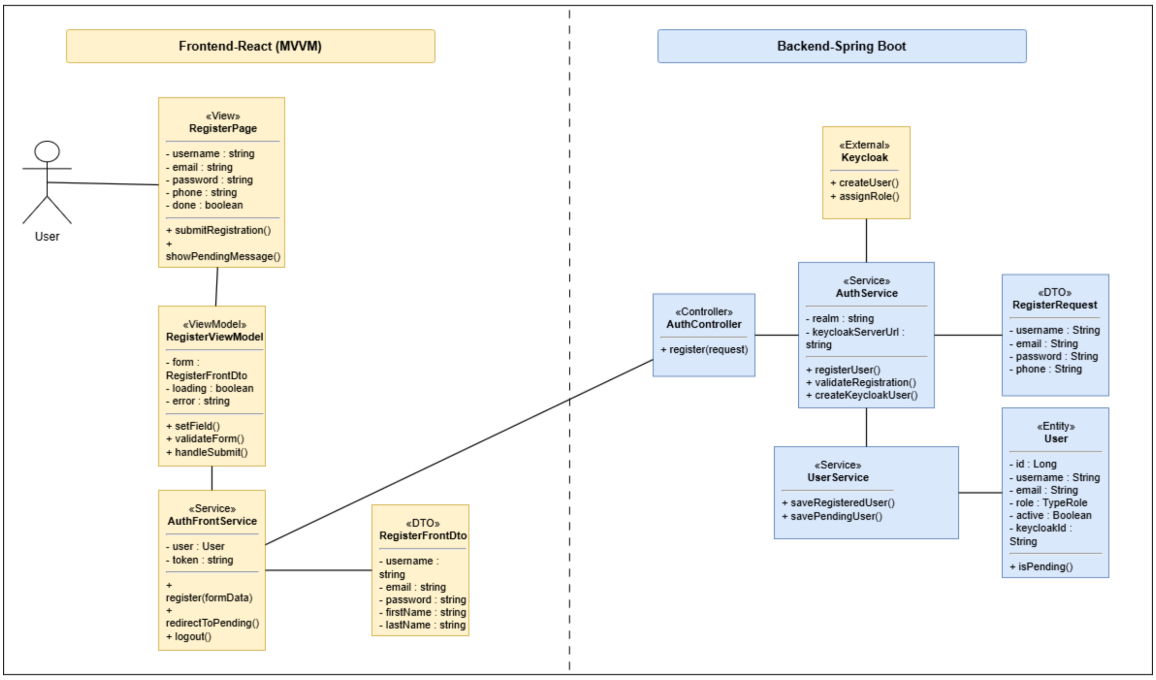


Figure 27: Design Class Diagram – Registration and Pending Access

Design Class Diagram – Security, Keycloak, and Access Control

This diagram shows the classes involved in secure authentication, Keycloak integration, and role-based access control.

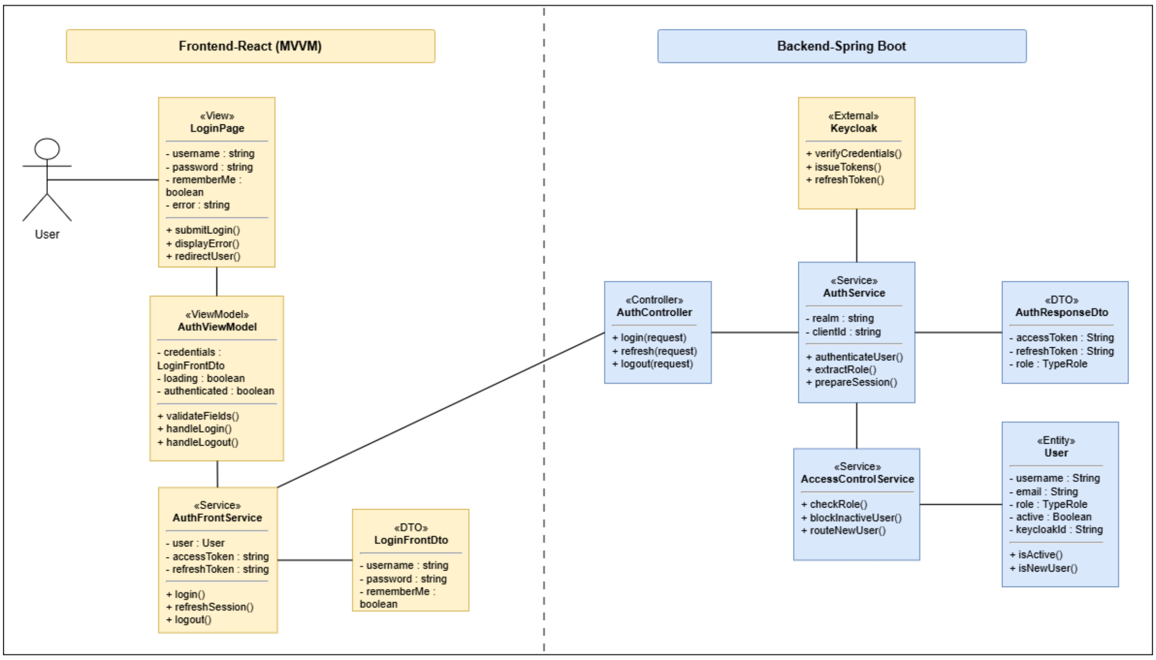


Figure 28: Design Class Diagram – Security, Keycloak, and Access Control

Design Class Diagram – HR Account Setup and Notification

This diagram presents the classes used by HR to complete account setup, update access rights, and trigger notifications.

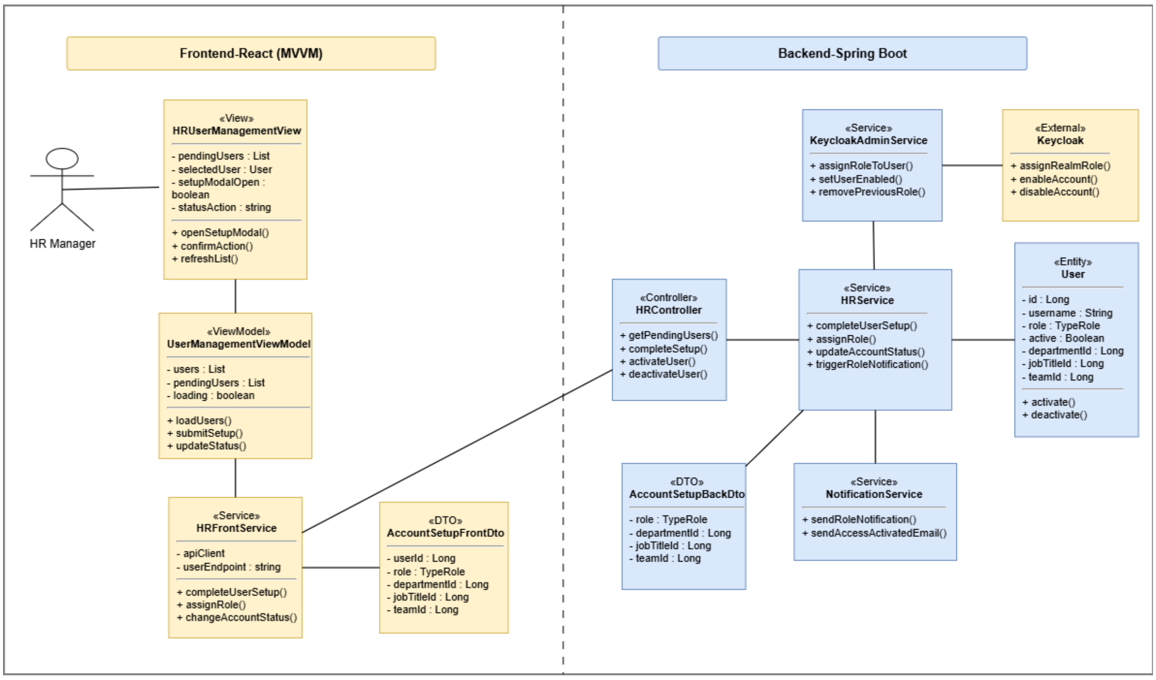


Figure 29: Design Class Diagram – HR Account Setup and Notification

Conception Sequence Diagrams

These diagrams detail the internal frontend and backend interactions behind the main Sprint 1 scenarios.

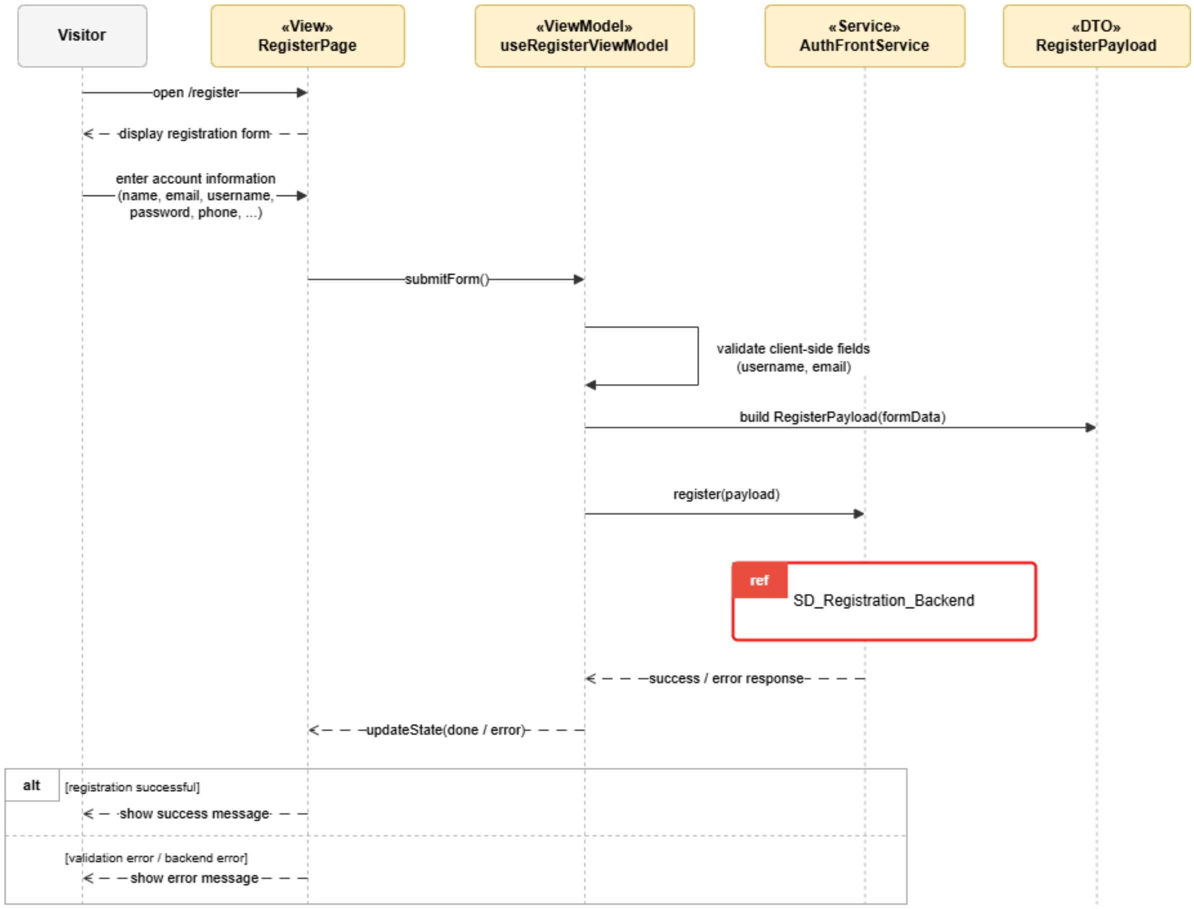


Figure 30: Conception Sequence Diagram – User Registration Frontend

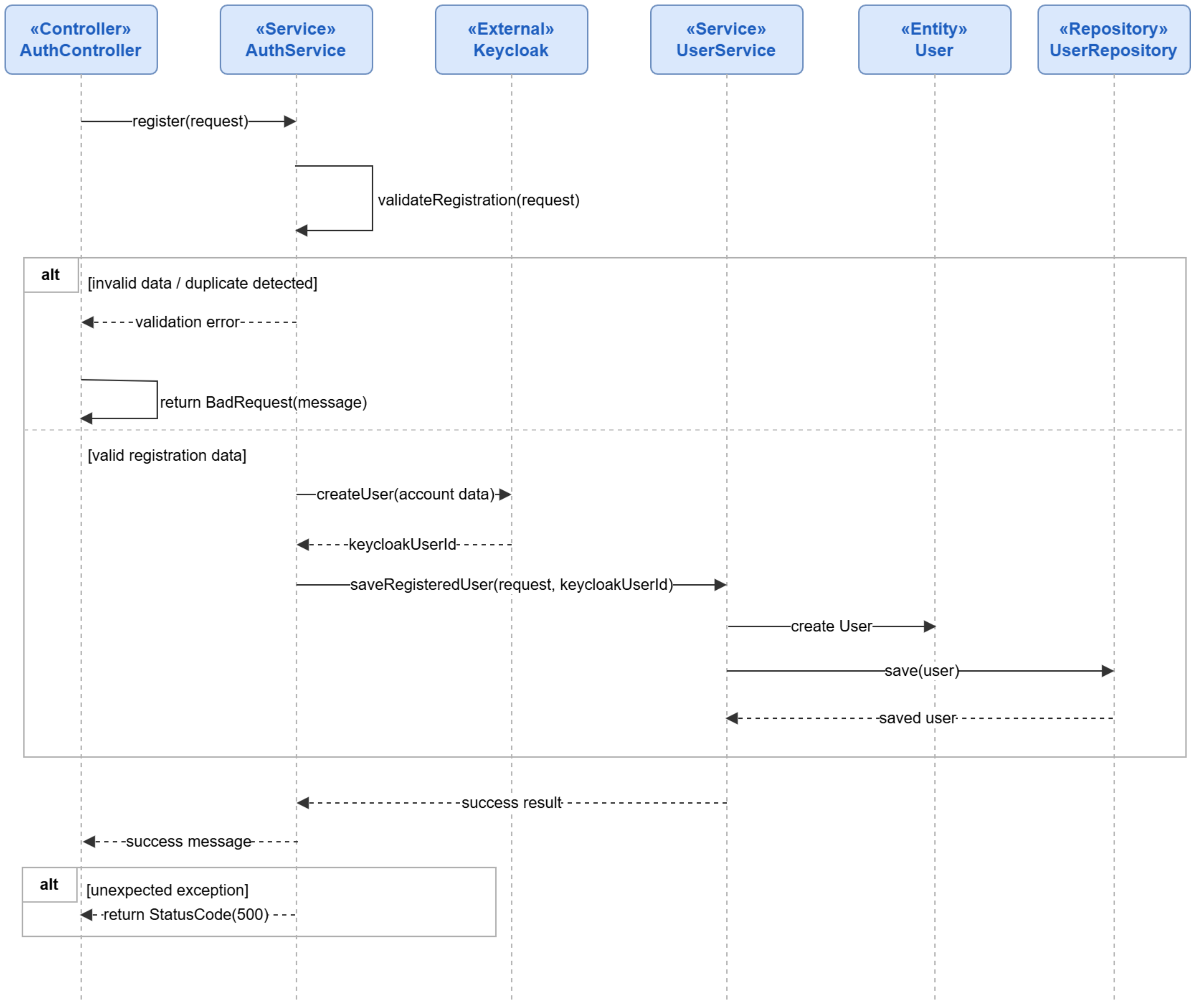


Figure 31: Conception Sequence Diagram – User Registration Backend

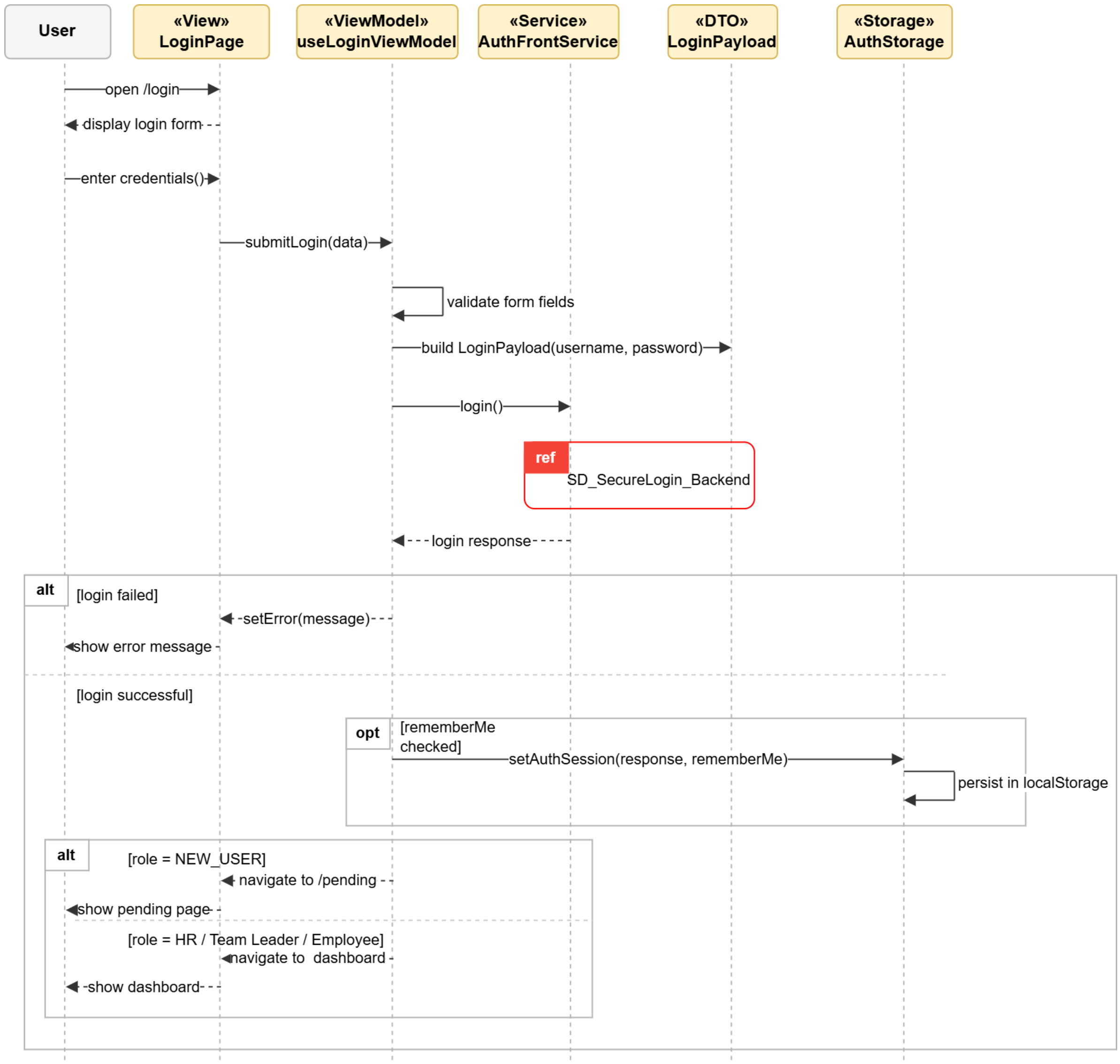


Figure 32: Conception Sequence Diagram – Secure Login Frontend

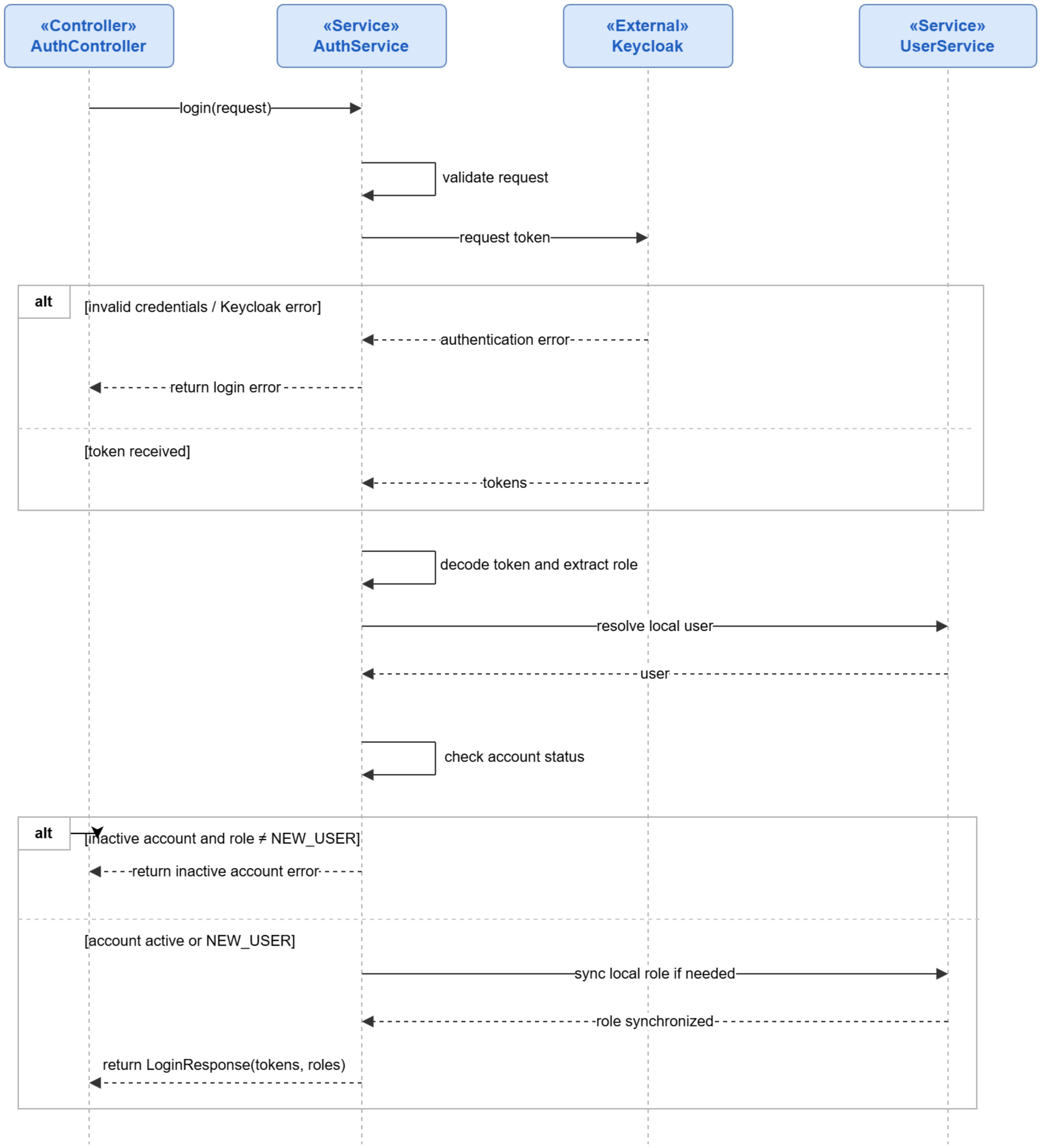


Figure 33: Conception Sequence Diagram – Secure Login Backend

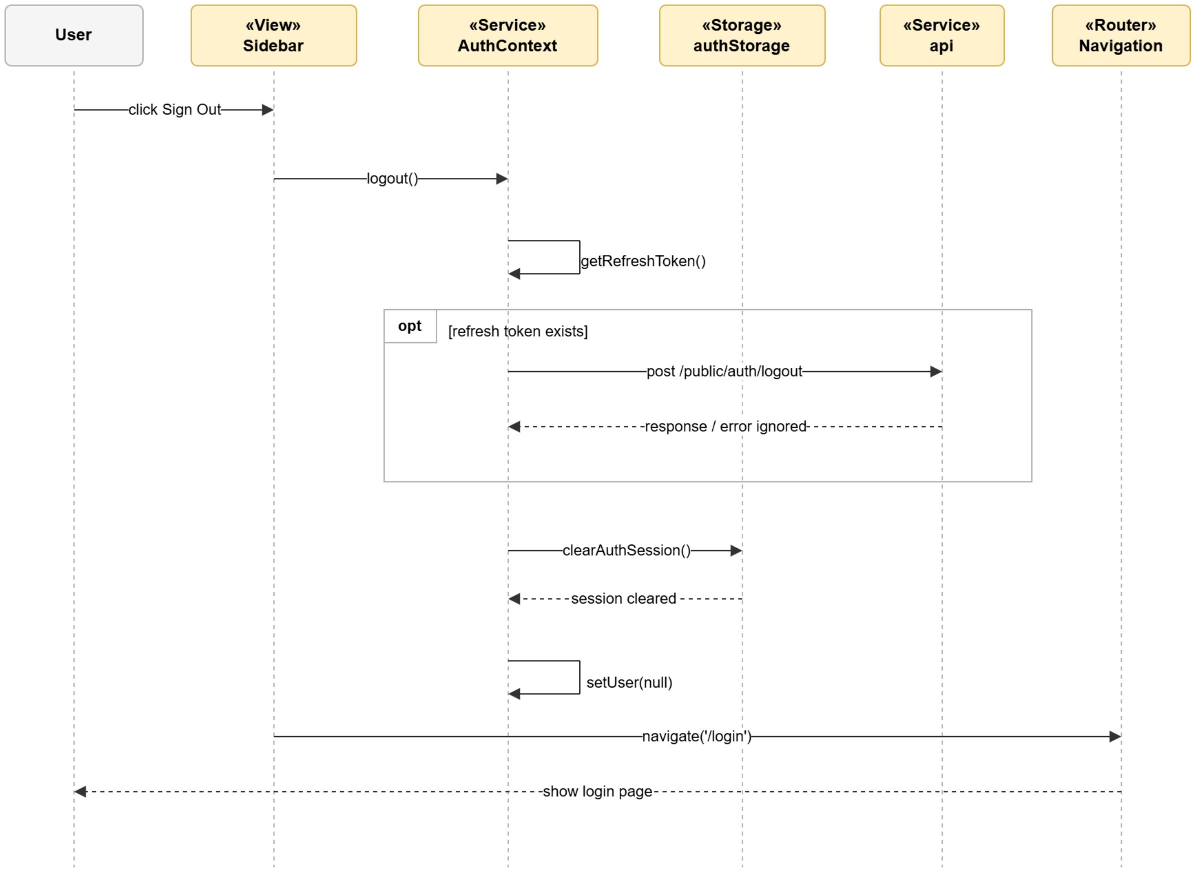


Figure 34: Conception Sequence Diagram – Logout Frontend

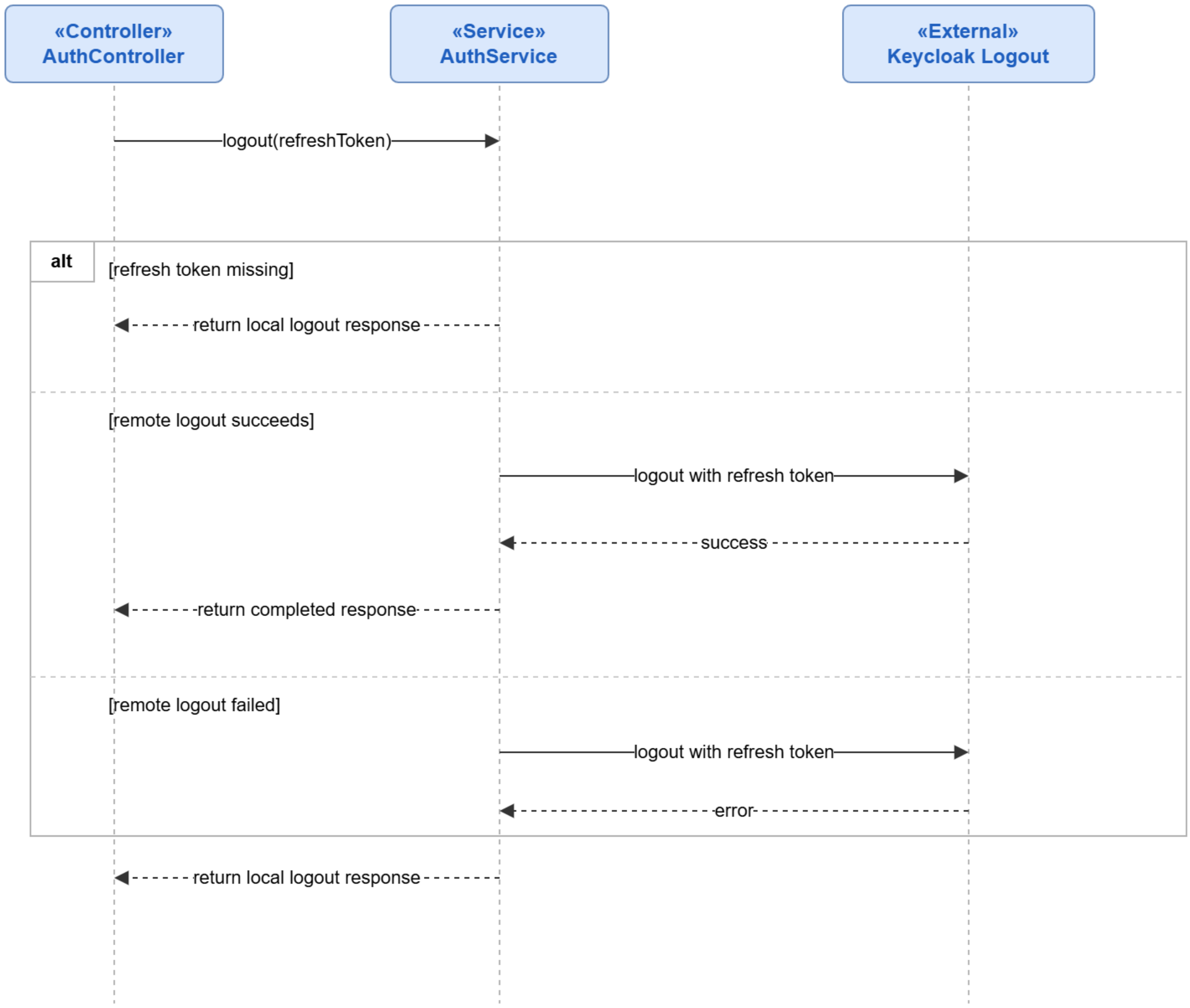


Figure 35: Conception Sequence Diagram – Logout Backend

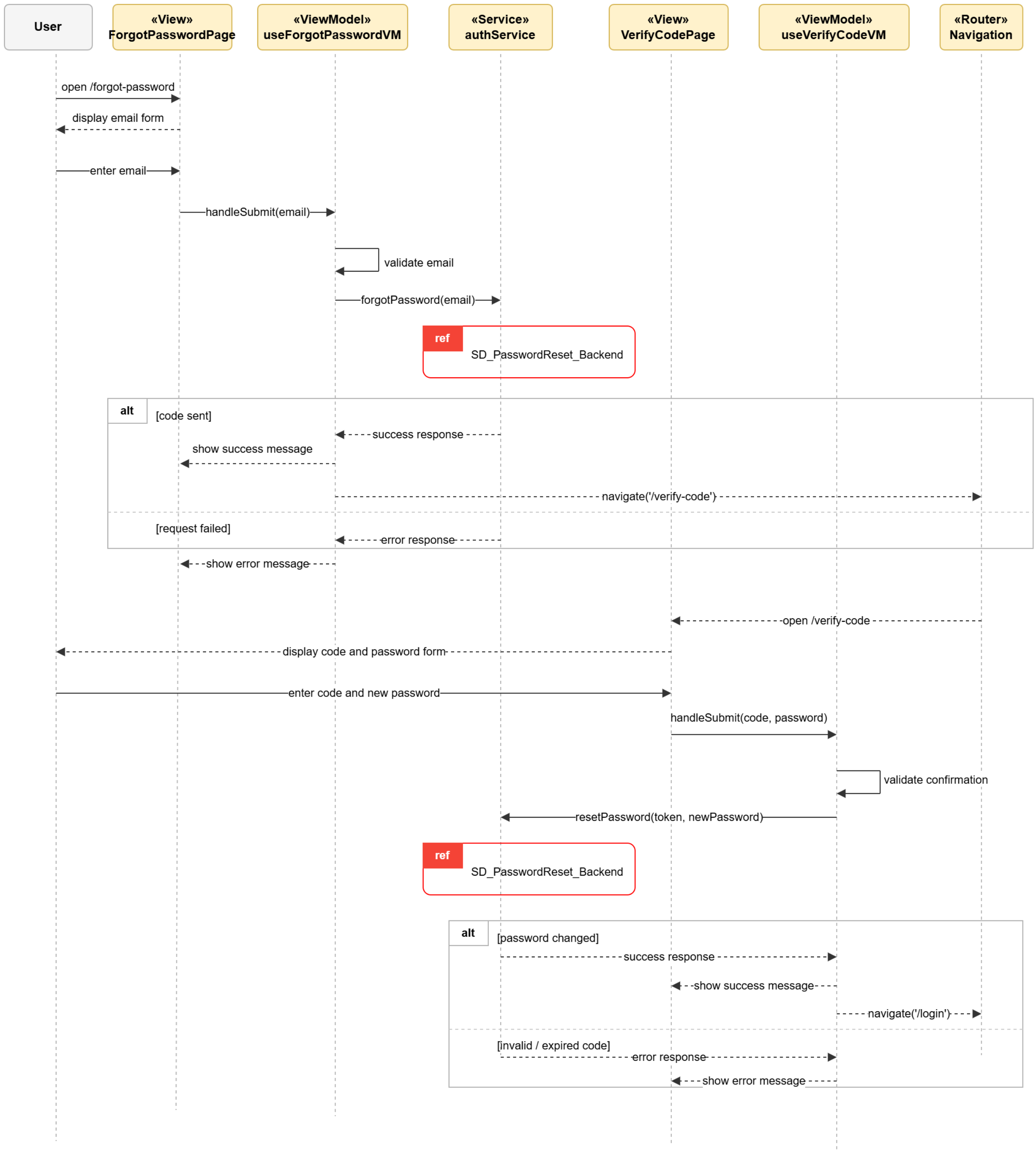


Figure 36: Conception Sequence Diagram – Password Reset Frontend

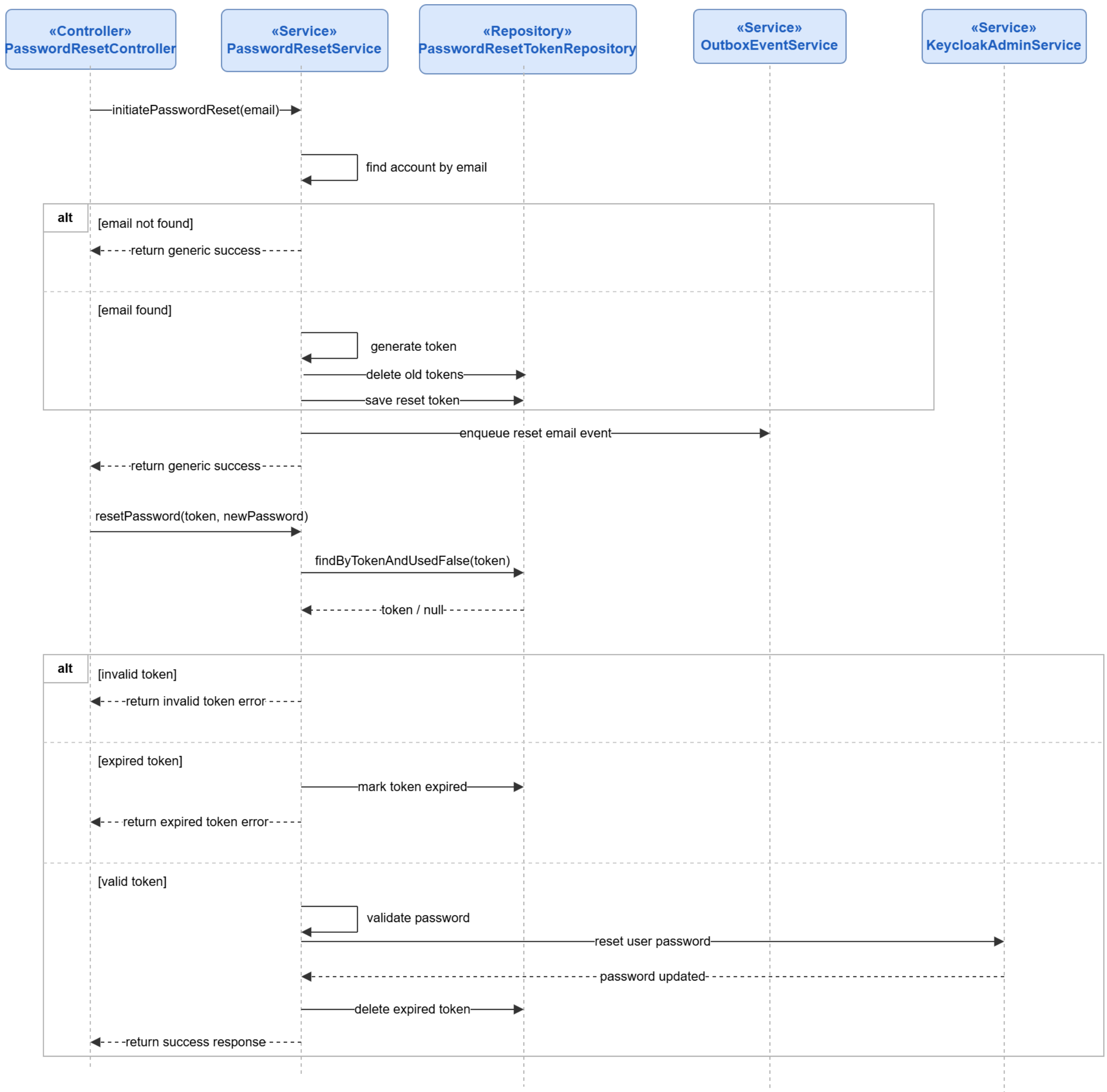


Figure 37: Conception Sequence Diagram – Password Reset Backend

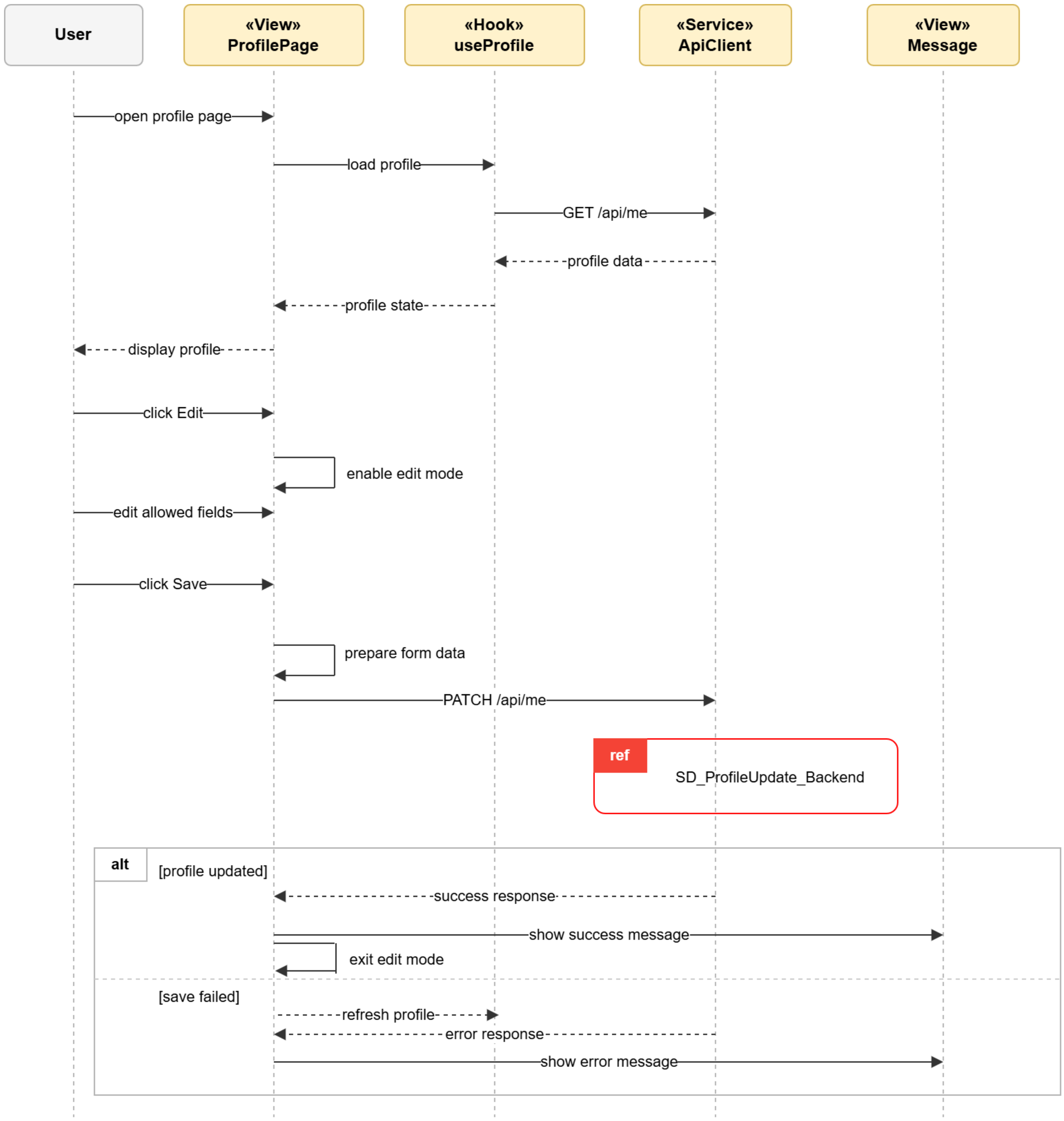


Figure 38: Conception Sequence Diagram – Profile Update Frontend

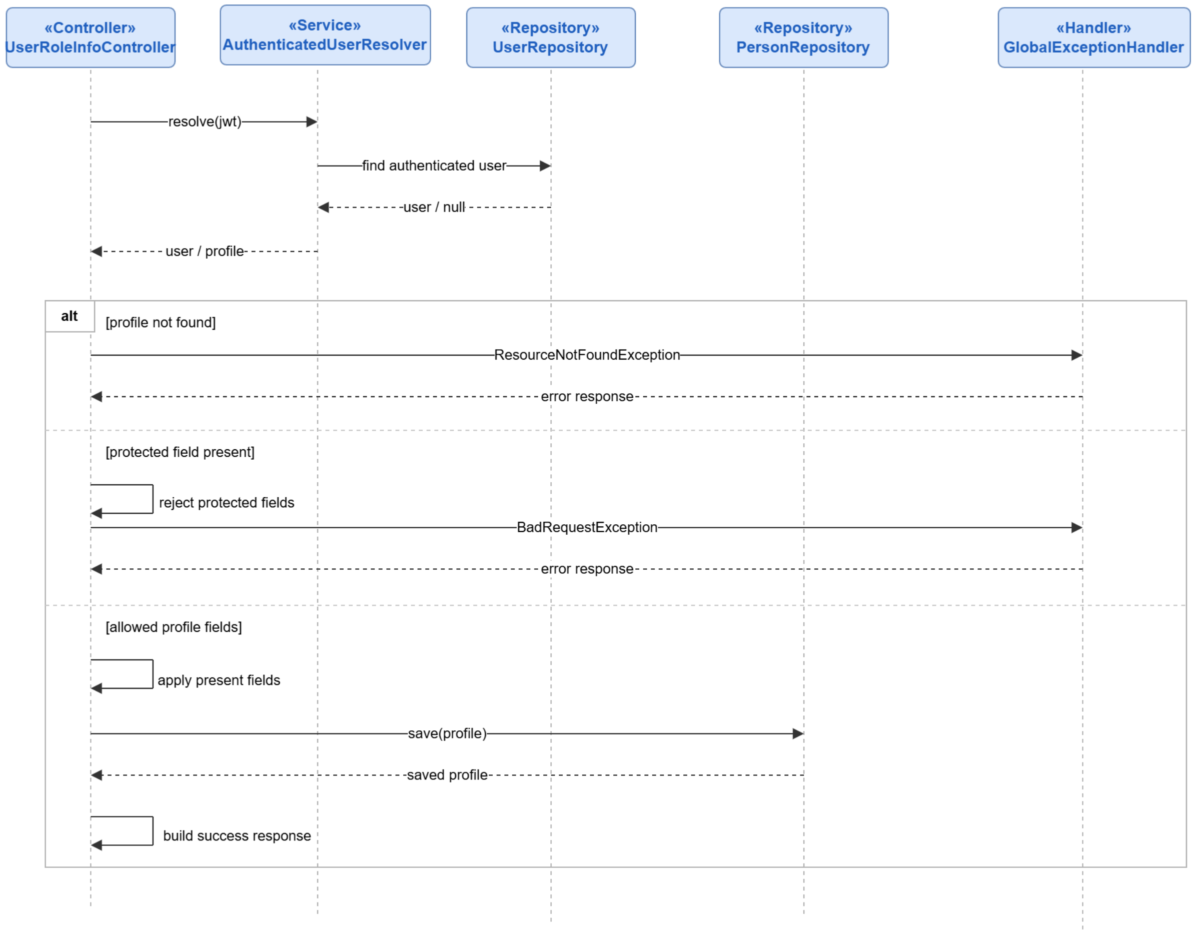


Figure 39: Conception Sequence Diagram – Profile Update Backend

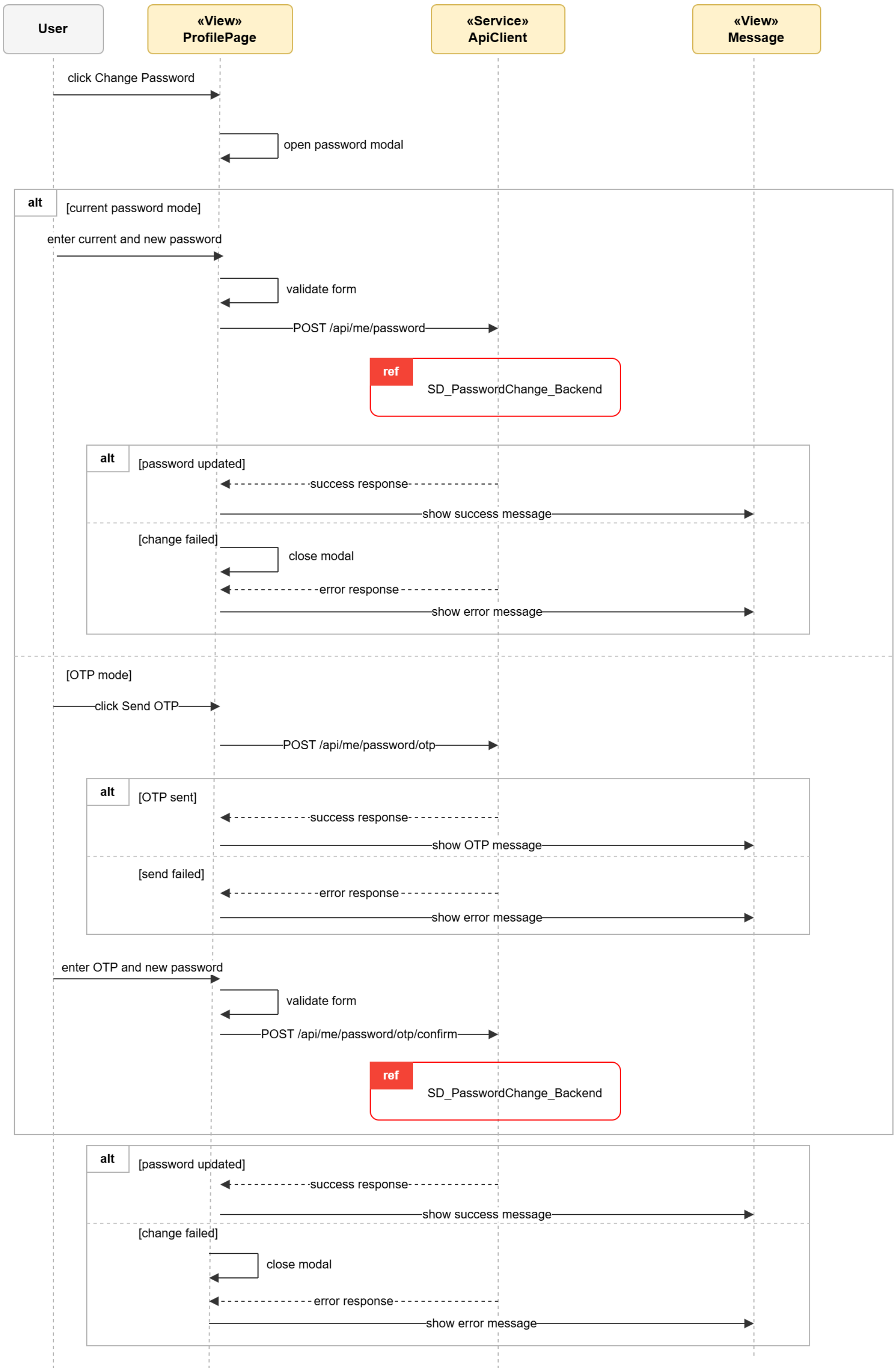


Figure 40: Conception Sequence Diagram – Authenticated Password Change Frontend

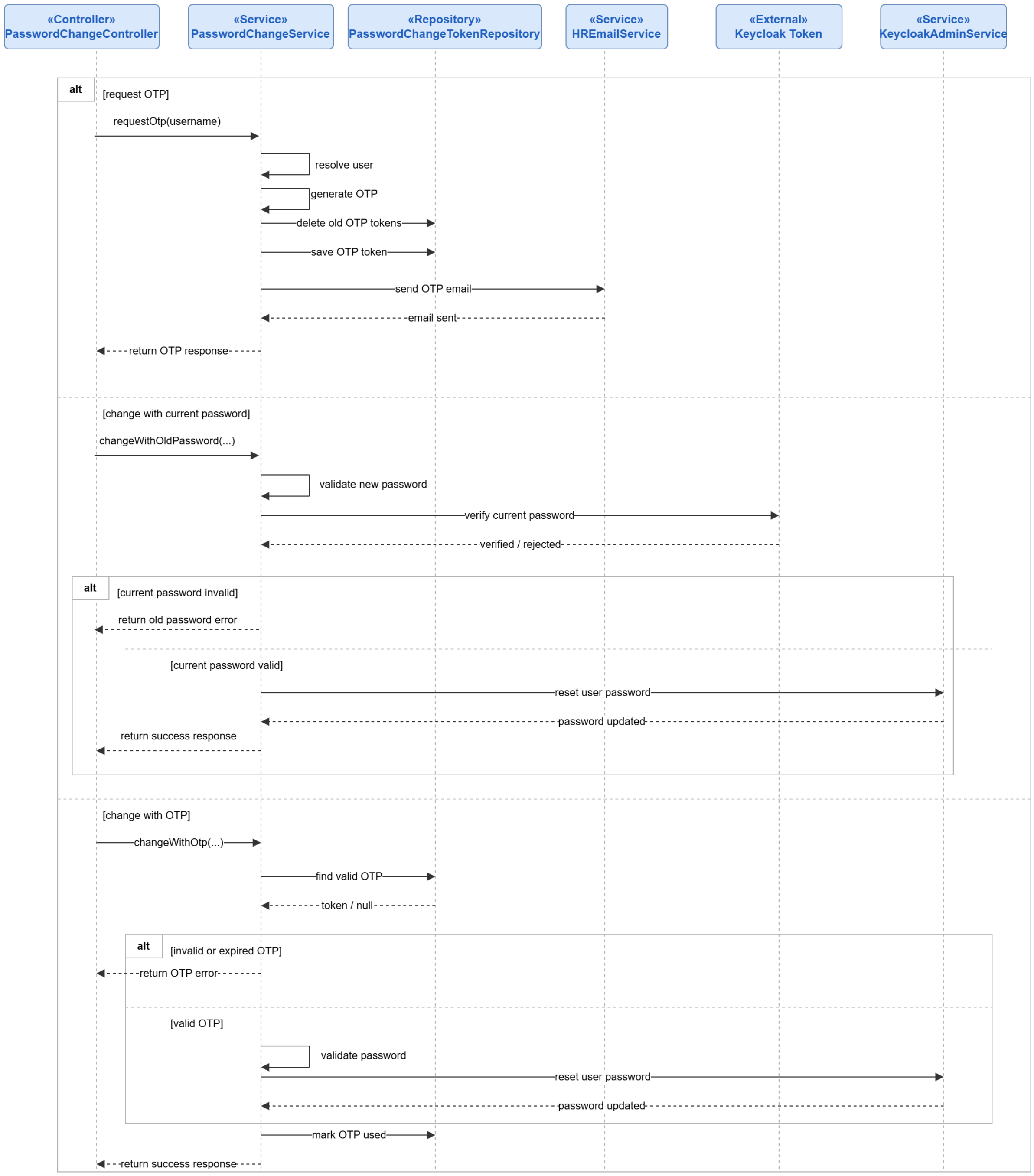


Figure 41: Conception Sequence Diagram – Authenticated Password Change Backend

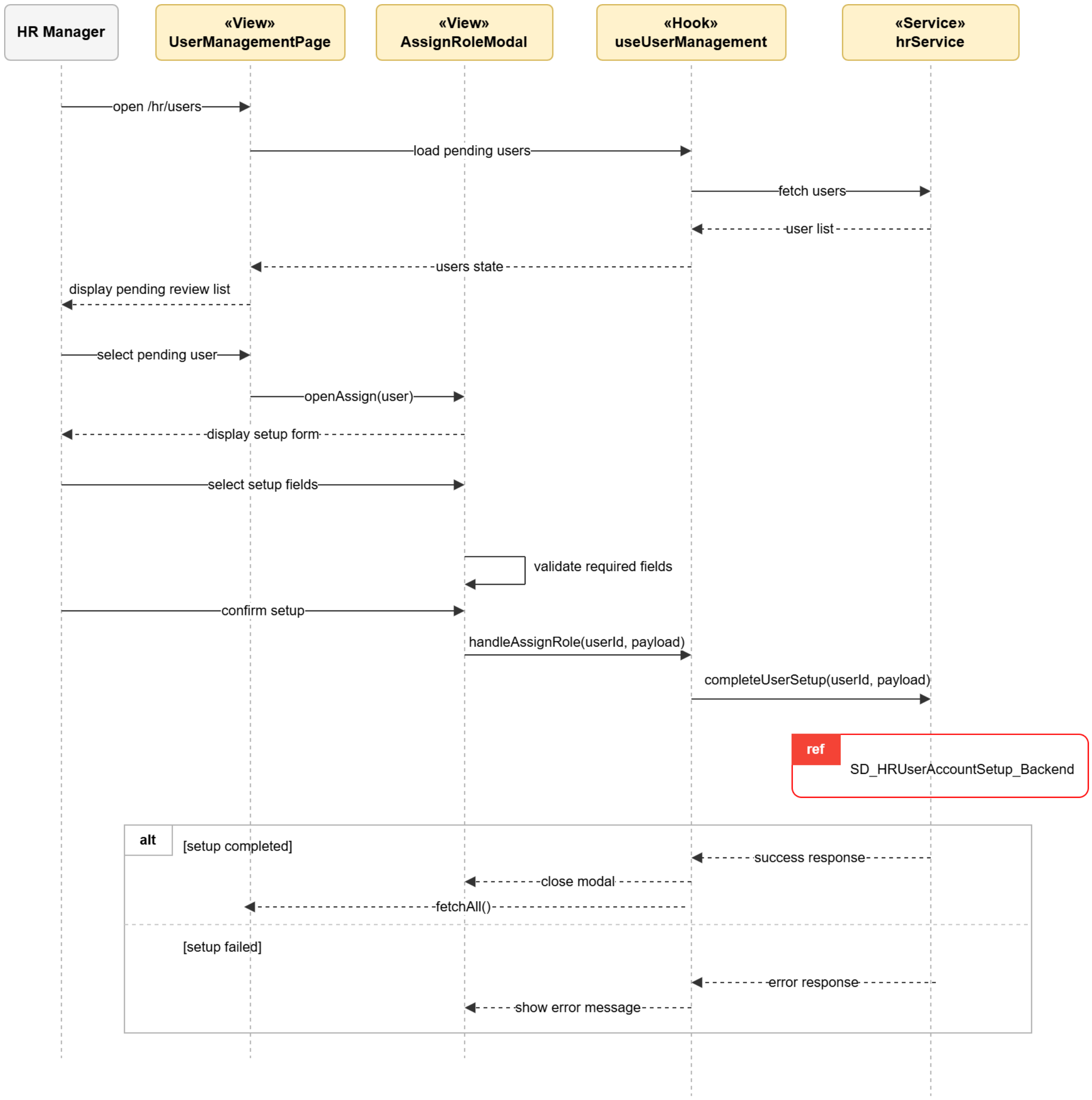


Figure 42: Conception Sequence Diagram – HR User Account Setup Frontend

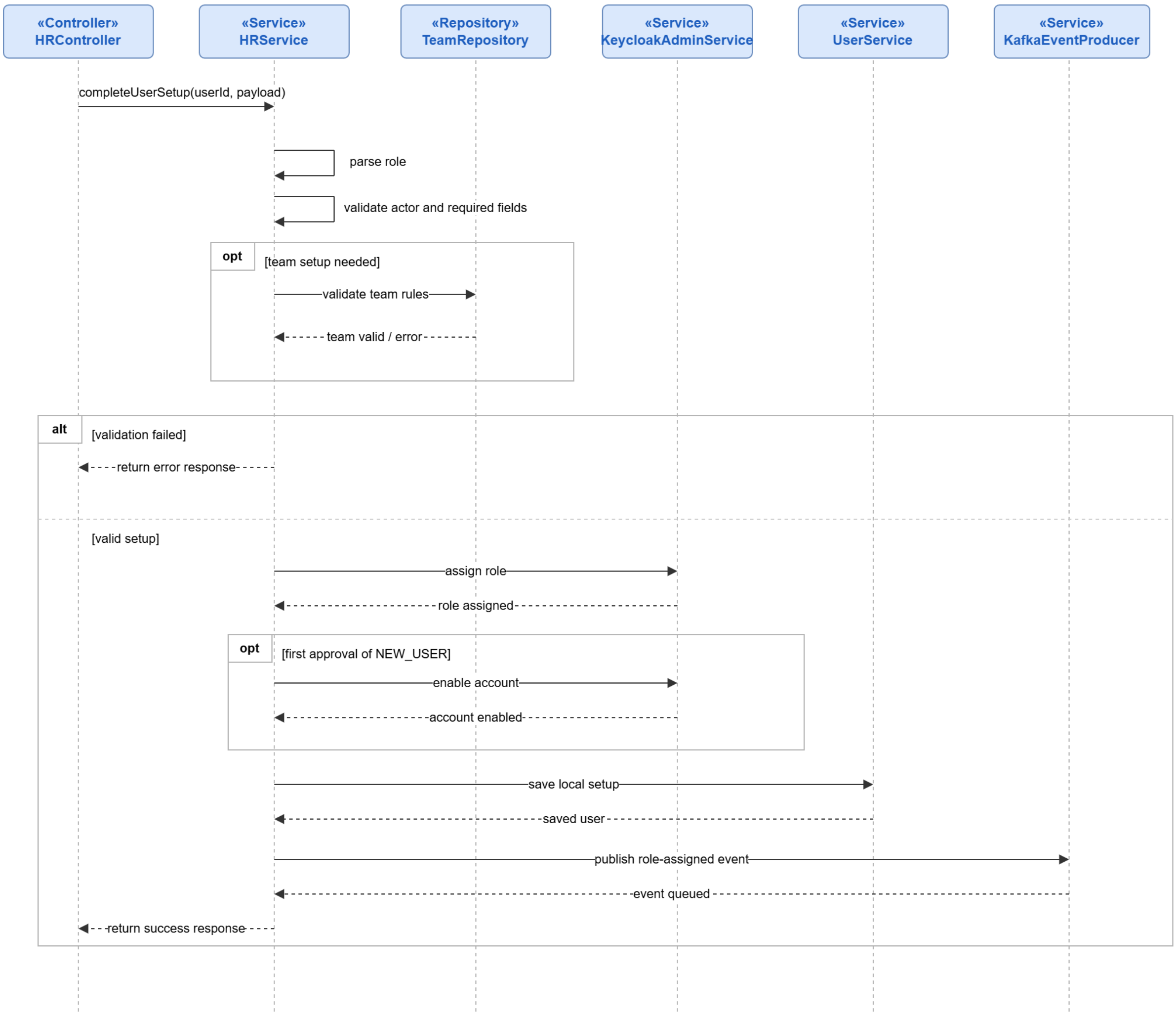


Figure 43: Conception Sequence Diagram – HR User Account Setup Backend

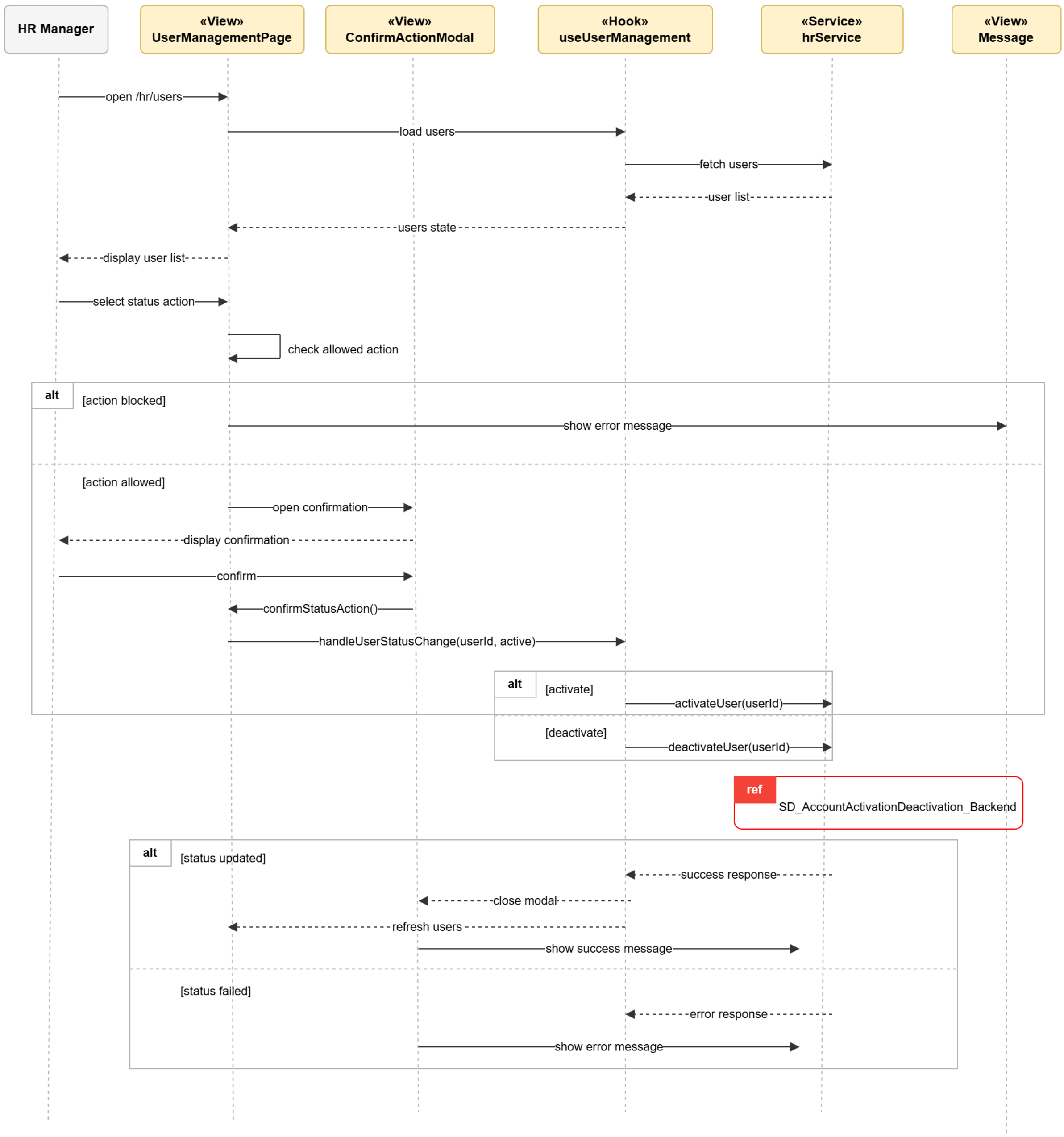


Figure 44: Conception Sequence Diagram – Account Activation and Deactivation Frontend

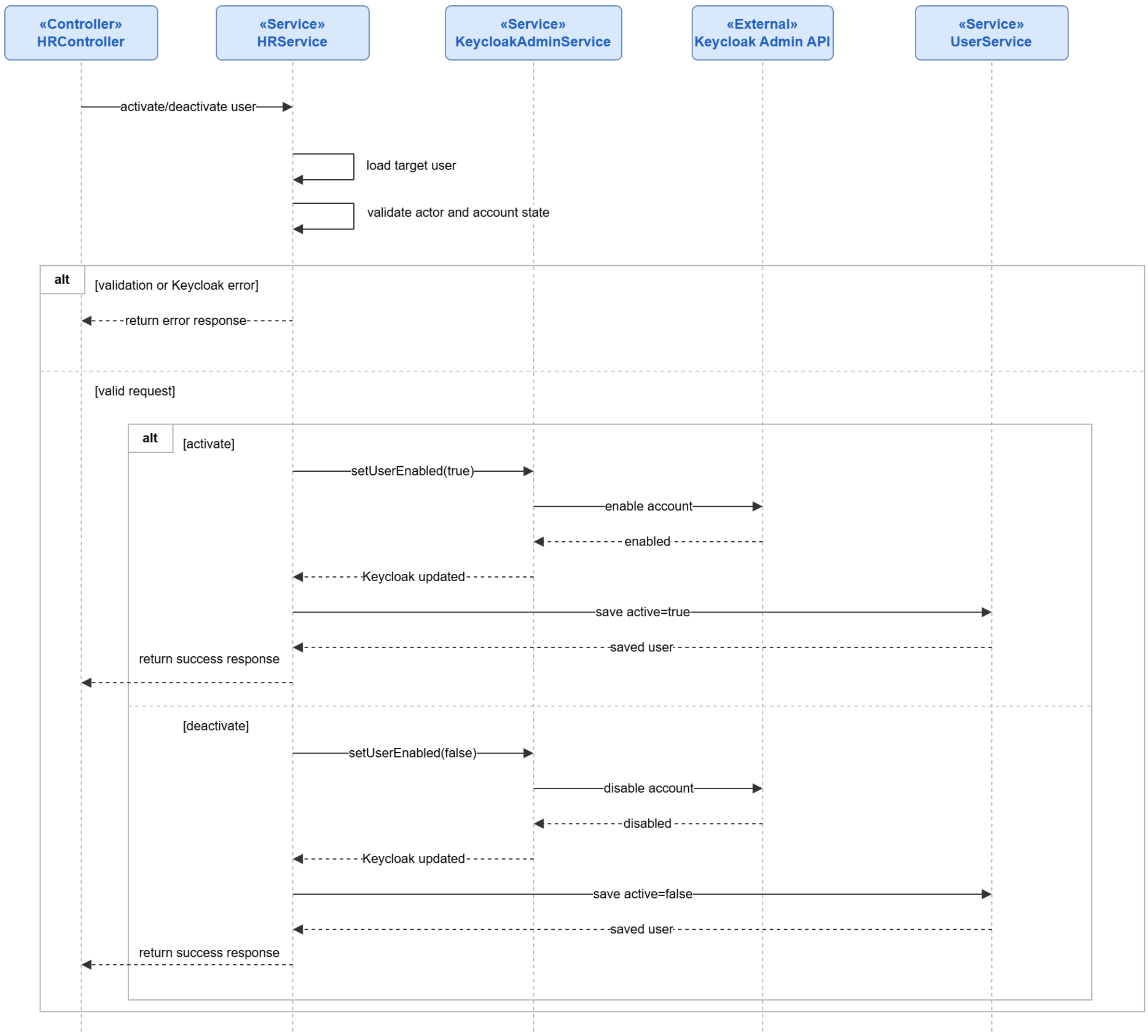


Figure 45: Conception Sequence Diagram – Account Activation and Deactivation Backend

Realization

This section will present the main user interfaces implemented during Sprint 1. The final screenshots will be inserted after the frontend redesign is completed.

Login Interface and Validation Feedback

`sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/login_validation_ui.png`

Registration and Pending Access Interface

sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/

registration_pending_access_ui.png

Password Reset Interface

sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/

password_reset_ui.png

Profile Update Interface

sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/

profile_update_ui.png

Authenticated Password Change Interface

sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/

authenticated_password_change_ui.png

HR User Account Setup Interface

sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/

hr_user_account_setup_ui.png

Account Activation and Deactivation Interface

sections/chapitre3/img/sprint1_access_account_management/09_ui_screenshots/

account_activation_deactivation_ui.png

5. Sprint Review

At the end of Sprint 1, the main access and account-management features were reviewed and validated. The sprint provided a secure foundation for the platform by implementing registration, authentication, logout, password recovery, profile update, HR account setup, role assignment, and account activation or deactivation. It also introduced the restricted **NEW_USER** state, which ensures that newly registered users cannot access the platform before HR completes their account setup.

6. Sprint Retrospective

What went well	What needs improvement
The sprint delivered a stable access-management base covering registration, authentication, account setup, and role-based access.	**Keycloak, local user records, roles, and account status had to remain carefully synchronized.**
The main access scenarios were clearly modeled through use-case, sequence, state, and class diagrams.	**Future modules should keep frontend feedback and backend validation aligned from the beginning.**

Table 3: Sprint Retrospective – Sprint 1

III. Sprint 2: Leave Management

1. Sprint Goal

This second sprint aims to implement the leave-management workflow of the ArabSoft platform after the access foundation established in Sprint 1. It allows employees to submit leave requests, consult their leave balance and history, update or cancel requests while they are still pending team leader review, and follow the progress of their leave workflow. The sprint also introduces working-day validation, weekend and public-holiday exclusion, overlap prevention, team leader approval or rejection, HR final decision after team leader approval, calendar visualization, filters, workflow notifications, and approved leave PDF download. The objective is to deliver the first complete administrative workflow involving employees, team leaders, and HR managers.

2. Visual Sprint Journal

The visual sprint journal will document the selected user stories of Sprint 2 and their completion status in Jira. The final Sprint 2 journal screenshot will be inserted after validation.

Jira Sprint Journal – Sprint 2 Leave Management

`sections/chapitre3/img/sprint2_leave_management/01_jira/sprint2_jira_journal.png`

3. Sprint Backlog

The table below presents the backlog of the second sprint, planned as a two-week iteration dedicated to leave management.

Sprint	Stories	Tasks	Duration
Sprint 2	**Leave request submission and balance consultation**	Implement leave request creation	2d
		Display leave balance and leave history	
		Connect employee leave actions with backend APIs	
	Working-day, holiday, and overlap validation	Validate leave dates using working days	2d
		Exclude weekends and public holidays from deducted days	
		Prevent overlapping leave requests	
		Display clear validation feedback to the user	
	Leave update and cancellation	Allow request update while pending team leader review	1d
		Allow request cancellation while pending team leader review	
	Team leader leave decision	Implement team leader approval	2d
		Implement team leader rejection with reason	
	HR final leave decision	Implement HR final approval	2d
		Implement HR final rejection and decision tracking	
	Leave calendar, filters, notifications, and approved PDF	Display leave requests in role-aware calendar views	1d
		Add leave filters by employee, status, date, department, and team	
		Send workflow events and leave notifications	
		Enable approved leave PDF download	

Table 4: Sprint Backlog – Sprint 2

4. Sprint Implementation

Requirements Analysis

Sprint 2 covers the leave-management workflow, including employee leave request management, team leader review, HR final decision after team leader approval, leave calendar consultation, filtering, workflow notifications, and approved leave PDF download. The following use-case diagram first presents the global scope of the sprint.

Use Case Diagrams

These diagrams summarize the main user-system exchanges for Sprint 2 leave-management scenarios. The section also introduces focused use-case views before the employee request-management scenarios and the approval workflow scenarios.

Use Case Diagram – Sprint 2 Leave Management

This diagram presents the general leave-management use cases of Sprint 2. It shows the main actions performed by the employee, the team leader, and the HR manager within the leave workflow.

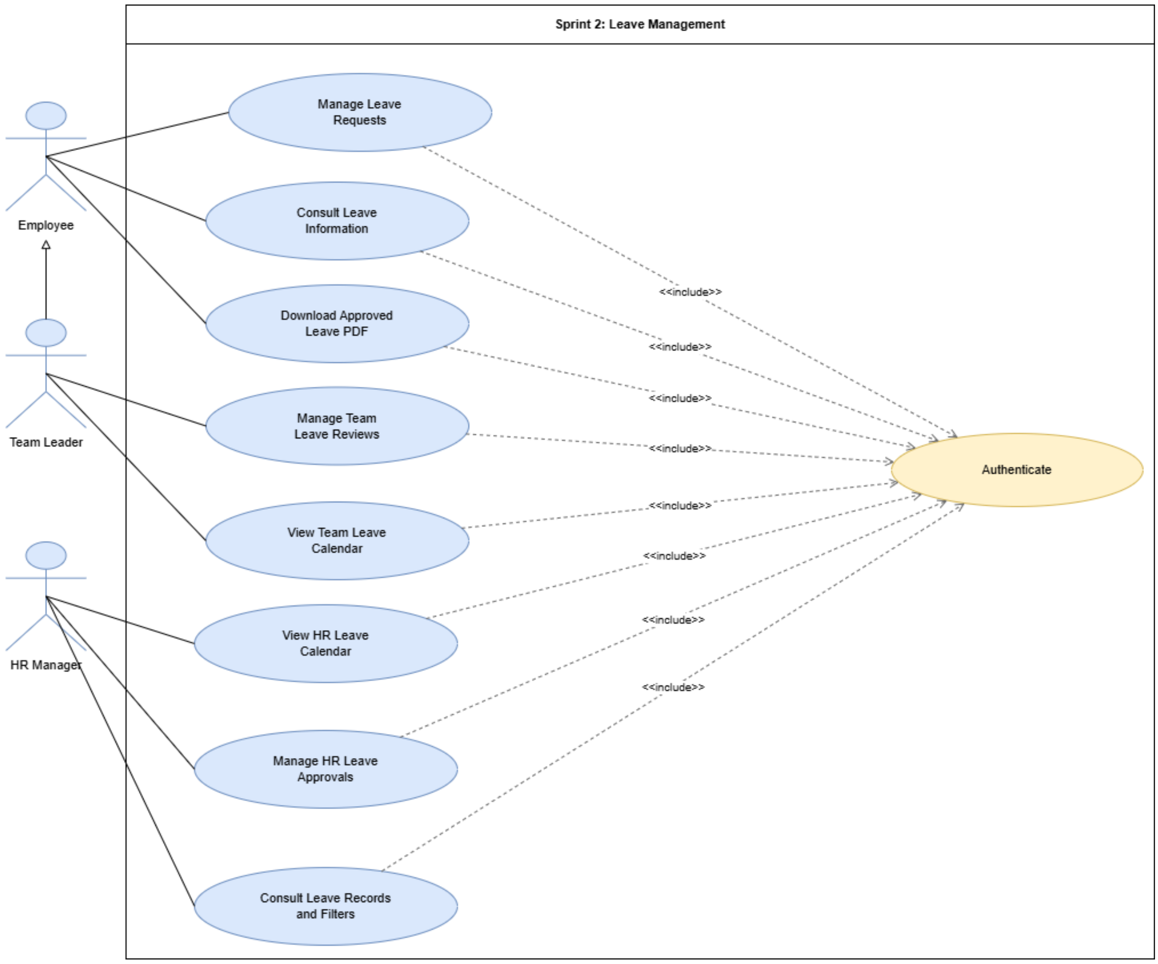


Figure 46: Use Case Diagram – Sprint 2 Leave Management

System Sequence Diagrams

These diagrams summarize the main user-system exchanges for Sprint 2 leave-management scenarios. The section also introduces focused use-case views before the employee request-management scenarios and the approval workflow scenarios.

Use Case Diagram – Employee Leave Request Management

This diagram groups the leave actions available to an employee, including request submission, balance consultation, history tracking, update or cancellation while pending team leader review, and approved leave PDF download.

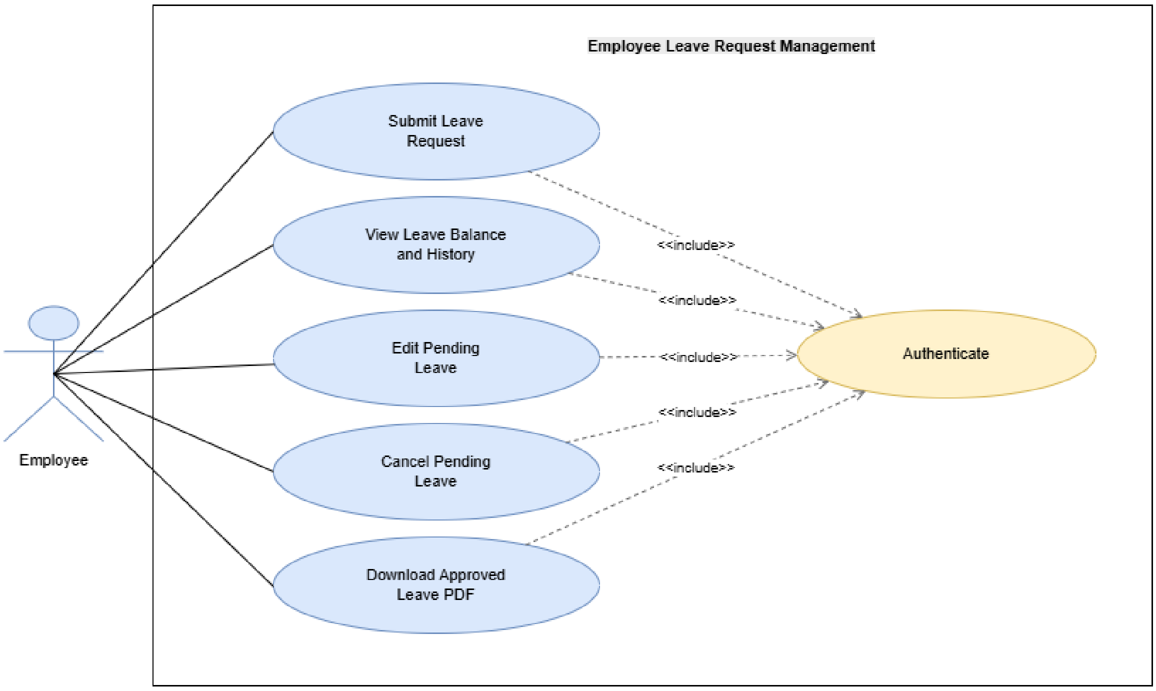


Figure 47: Use Case Diagram – Employee Leave Request Management

Submit Leave Request

This diagram shows how an employee submits a leave request and receives validation feedback from the system.

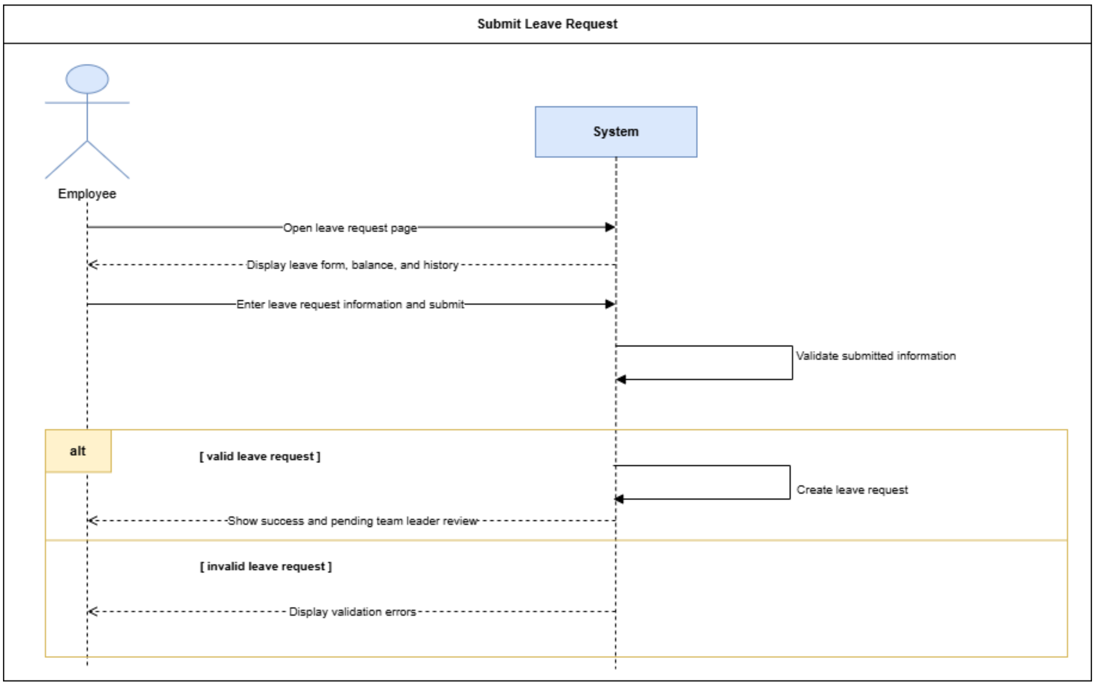


Figure 48: System Sequence Diagram – Submit Leave Request

Edit or Cancel Pending Leave

This diagram shows how an employee updates or cancels a leave request while it is still pending team leader review.

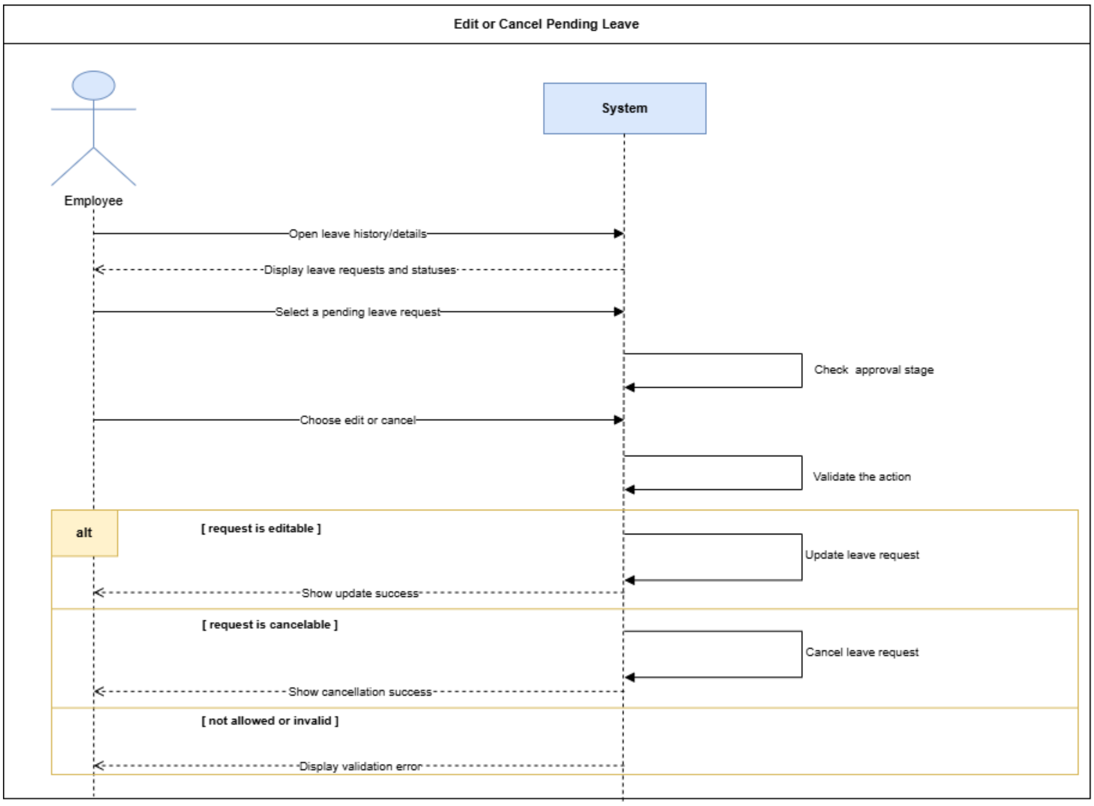


Figure 49: System Sequence Diagram – Edit or Cancel Pending Leave

Use Case Diagram – Leave Approval Workflow

This diagram presents the approval responsibilities of the team leader and HR manager, including team review, approval, rejection, HR final decision, leave records, and calendar consultation.

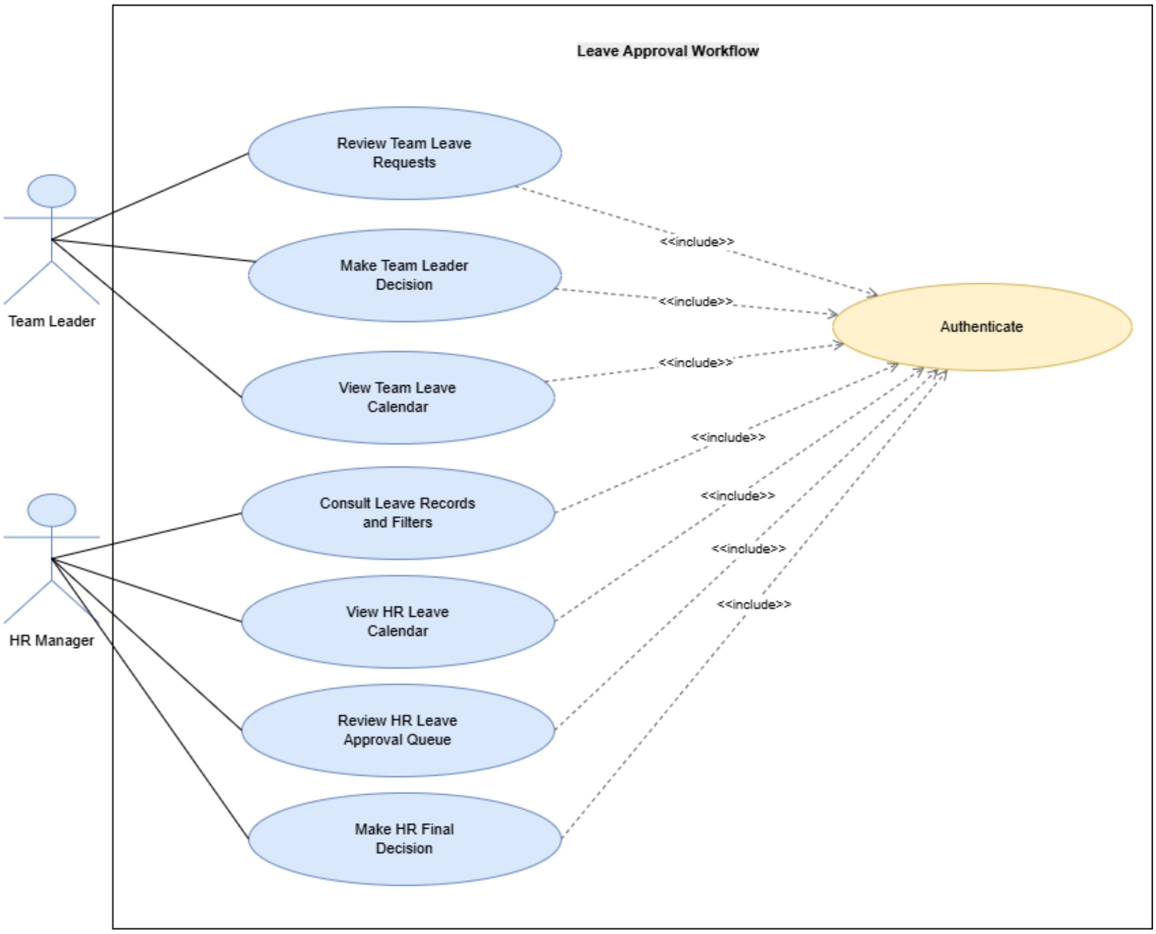


Figure 50: Use Case Diagram – Leave Approval Workflow

Team Leader Leave Decision

This diagram shows how a team leader approves or rejects a team member's leave request with a traceable decision.

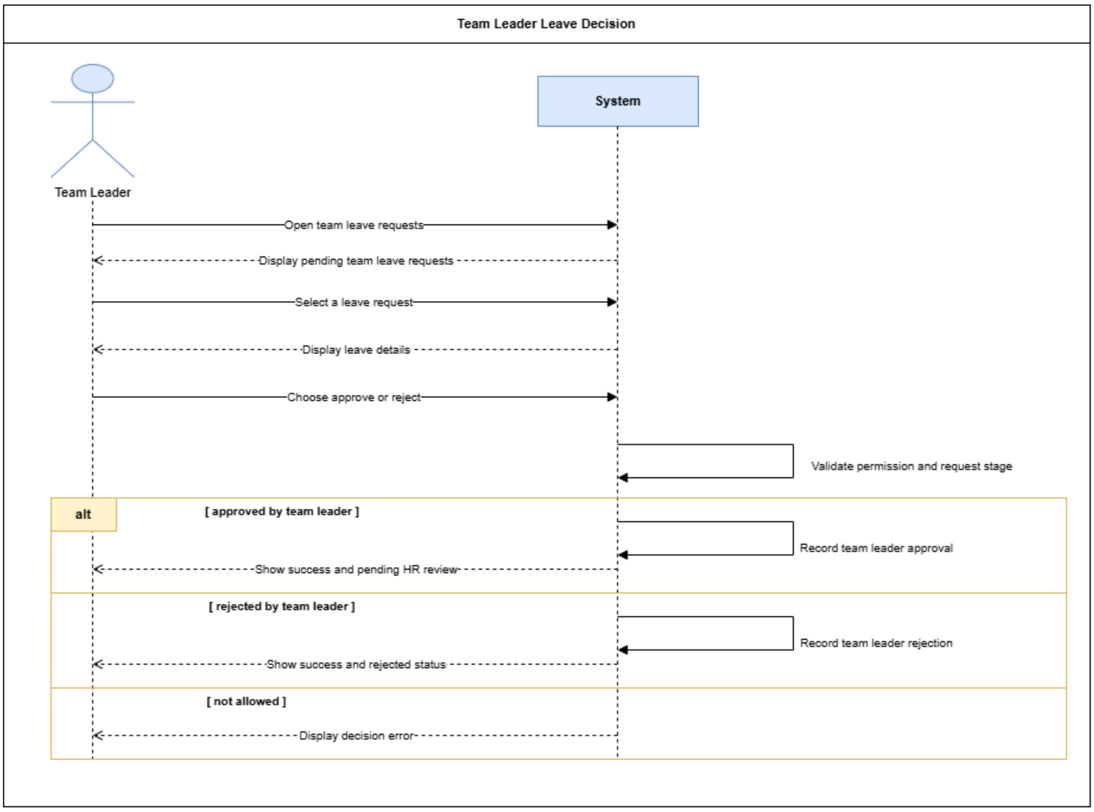


Figure 51: System Sequence Diagram – Team Leader Leave Decision

HR Final Leave Decision

This diagram shows how HR makes the final decision after team leader approval and updates the leave workflow accordingly.

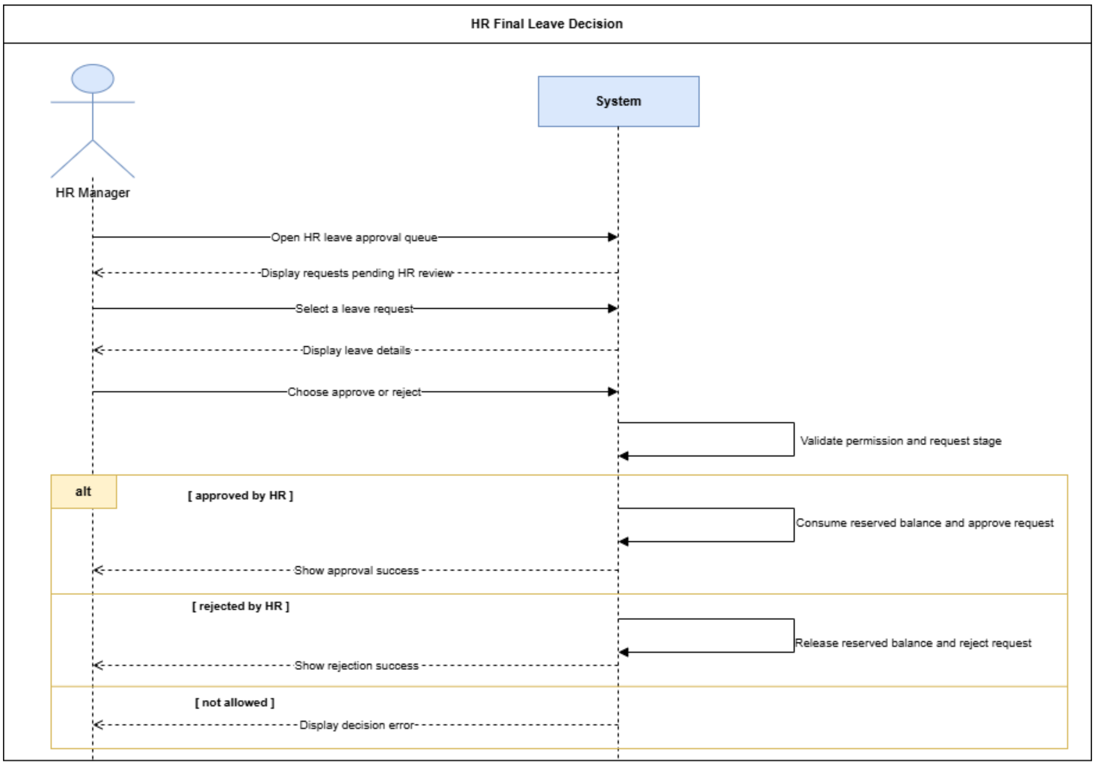


Figure 52: System Sequence Diagram – HR Final Leave Decision

Analysis

This section summarizes the main business concepts and lifecycle rules involved in the leave-management workflow.

Domain Model – Sprint 2 Leave Management

This diagram shows the main business concepts used for leave request management, including users, teams, leave requests, leave types, balances, public holidays, request history, and approval decisions.

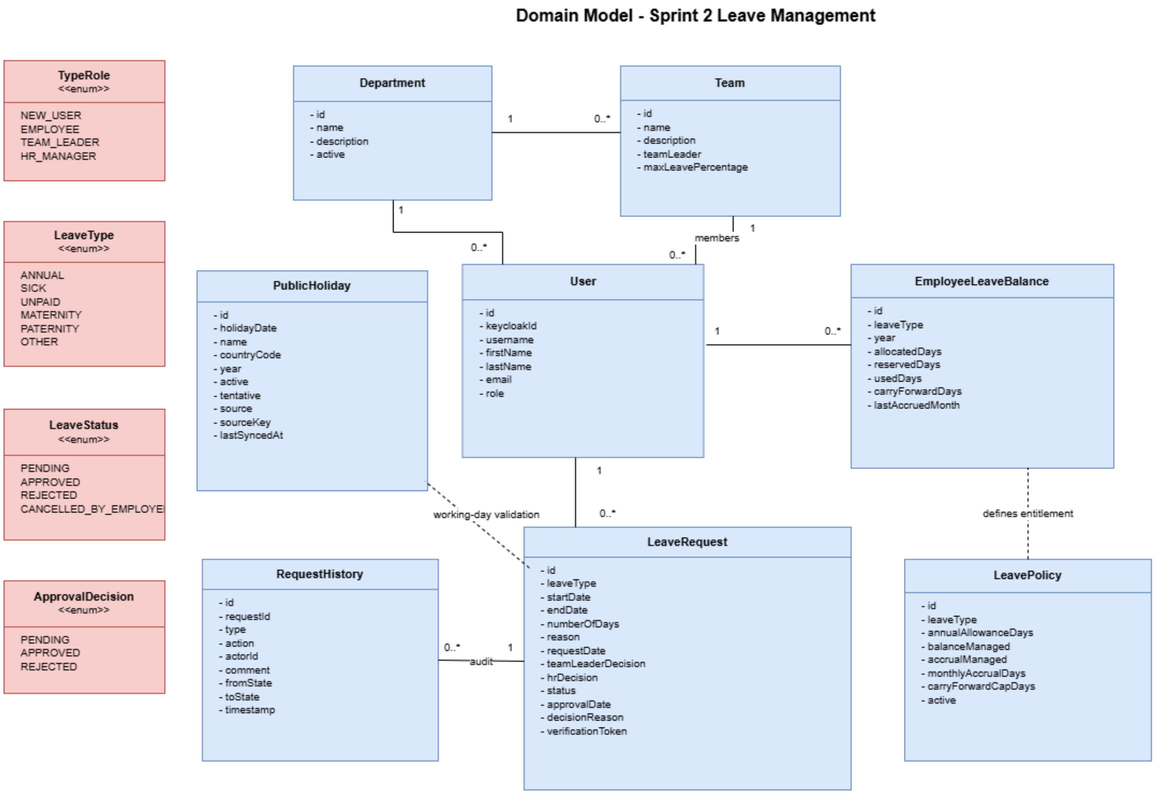


Figure 53: Domain Model – Sprint 2 Leave Management

Participant Class Diagrams

These diagrams identify the main frontend and backend participants involved in the leave-management workflow, following the main responsibility stages of the sprint: employee leave request management, team leader review, and HR final decision.

Participant Class Diagram – Employee Leave Request Management

This diagram identifies the main frontend and backend participants involved in employee leave submission, validation, balance consultation, and pending request update or cancellation.

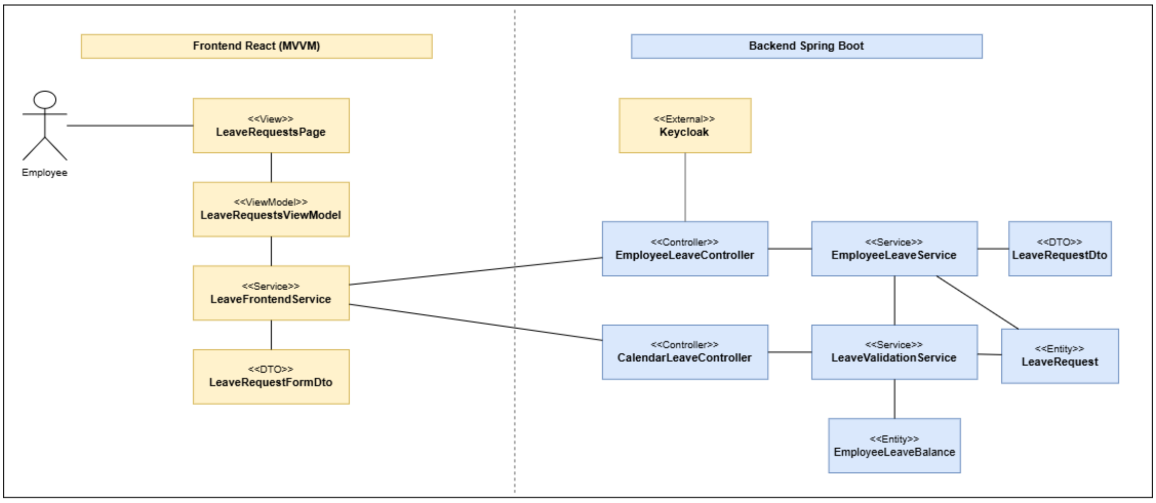


Figure 54: Participant Class Diagram – Employee Leave Request Management

Participant Class Diagram – Team Leader Leave Review

This diagram presents the participants involved in team leader leave review, including team request consultation, approval or rejection, and decision traceability.

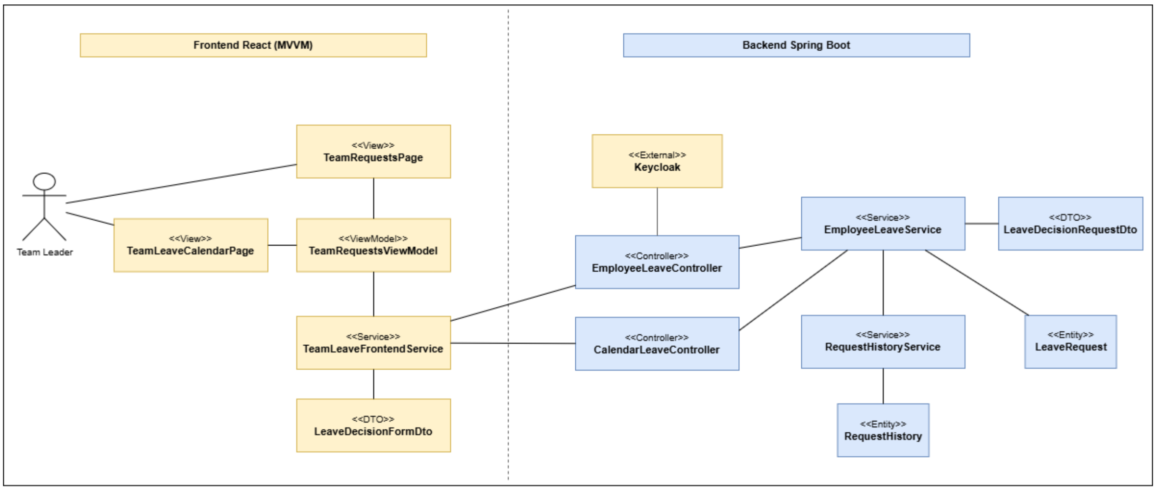


Figure 55: Participant Class Diagram – Team Leader Leave Review

Participant Class Diagram – HR Final Leave Decision

This diagram shows the participants involved in the HR final decision workflow, including approval queue consultation, final approval or rejection, and balance update.

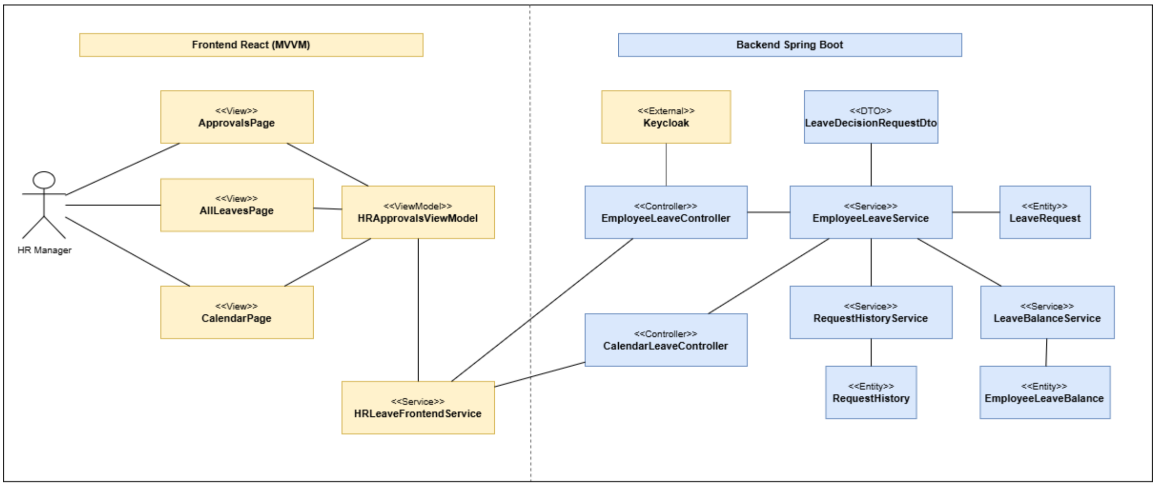


Figure 56: Participant Class Diagram – HR Final Leave Decision

State Diagrams

This section presents the lifecycle of a leave request using the real workflow states and stages of Sprint 2.

State Diagram – Leave Request Lifecycle

This diagram shows the leave request lifecycle from submission to team leader review, HR final decision, approval, rejection, or employee cancellation.

State Diagram – Leave Request Lifecycle

Reserved image placeholder

`sections/chapitre3/img/sprint2_leave_management/04_state_diagrams/`

Figure 57: State Diagram – Leave Request Lifecycle

Design Class Diagrams

These diagrams present the main classes supporting Sprint 2 leave-management features.

Design Class Diagram – Leave Request Creation and Validation

This diagram illustrates the main classes used to create leave requests and apply validation rules before persistence.

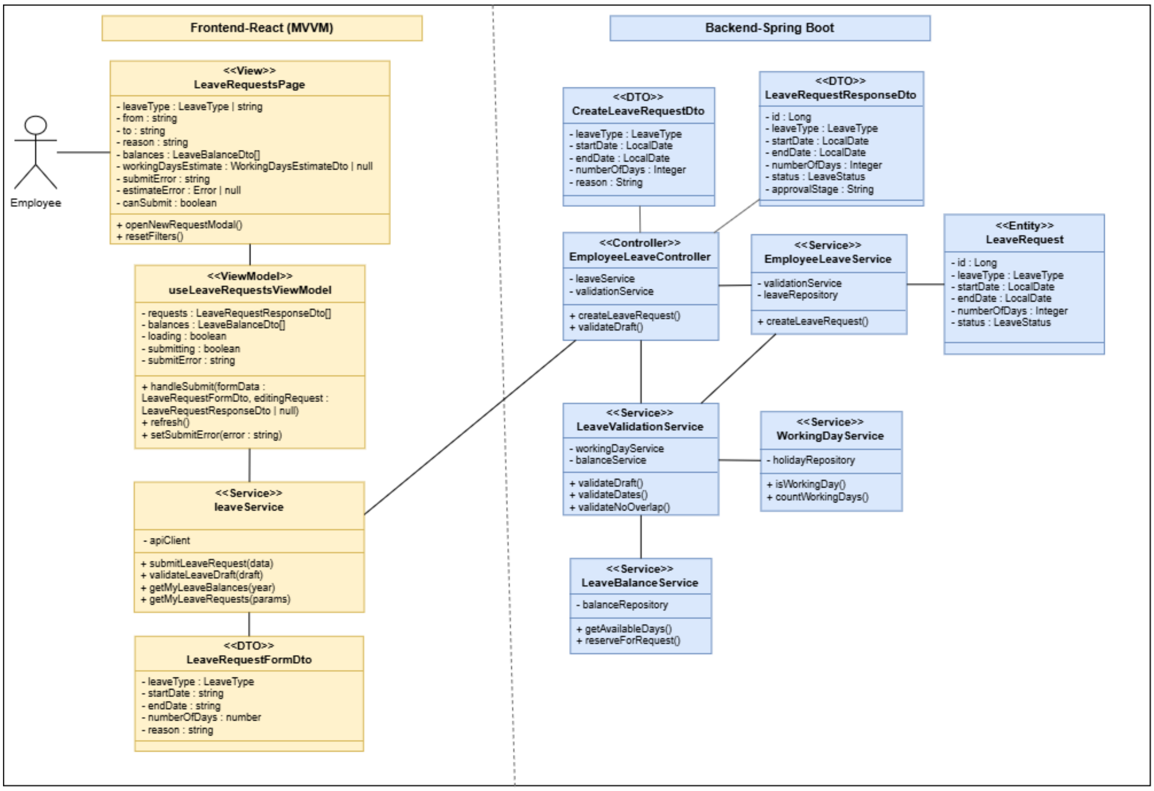


Figure 58: Design Class Diagram – Leave Request Creation and Validation

Design Class Diagram – Team Leader Leave Review

This diagram presents the main frontend and backend classes used for team leader leave consultation, approval, rejection, history recording, and balance release when needed.

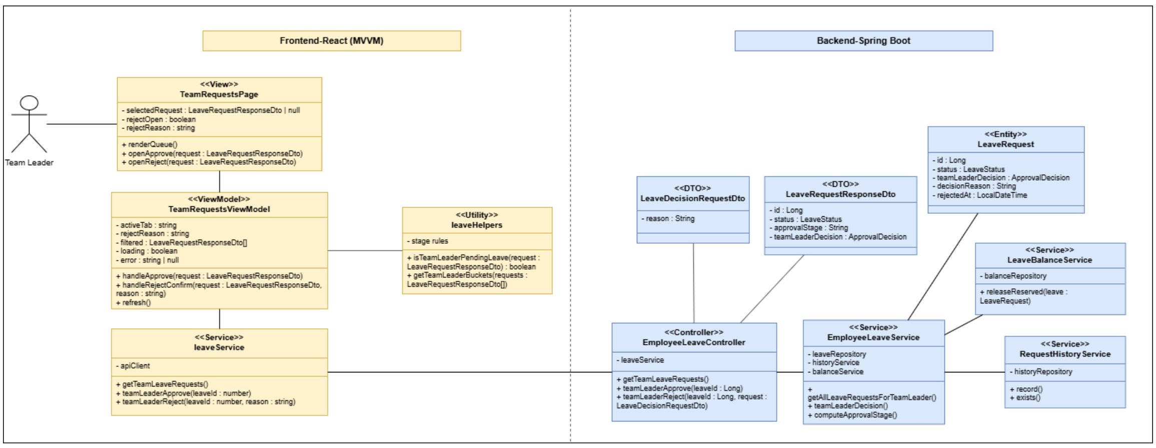


Figure 59: Design Class Diagram – Team Leader Leave Review

Design Class Diagram – HR Final Leave Decision

This diagram shows the classes used by HR to make the final leave decision, update the request status, record history, and consume or release reserved balance.

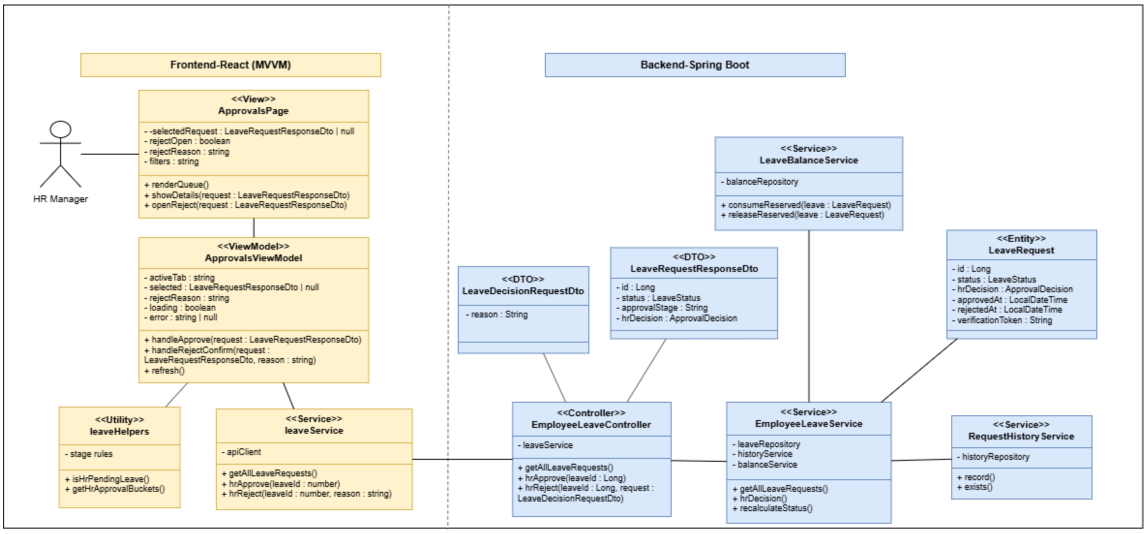


Figure 60: Design Class Diagram – HR Final Leave Decision

Design Class Diagram – Working Day, Holiday, and Balance Management

This diagram presents the classes used to calculate deducted working days, exclude weekends and public holidays, manage leave policy, and check available balance.

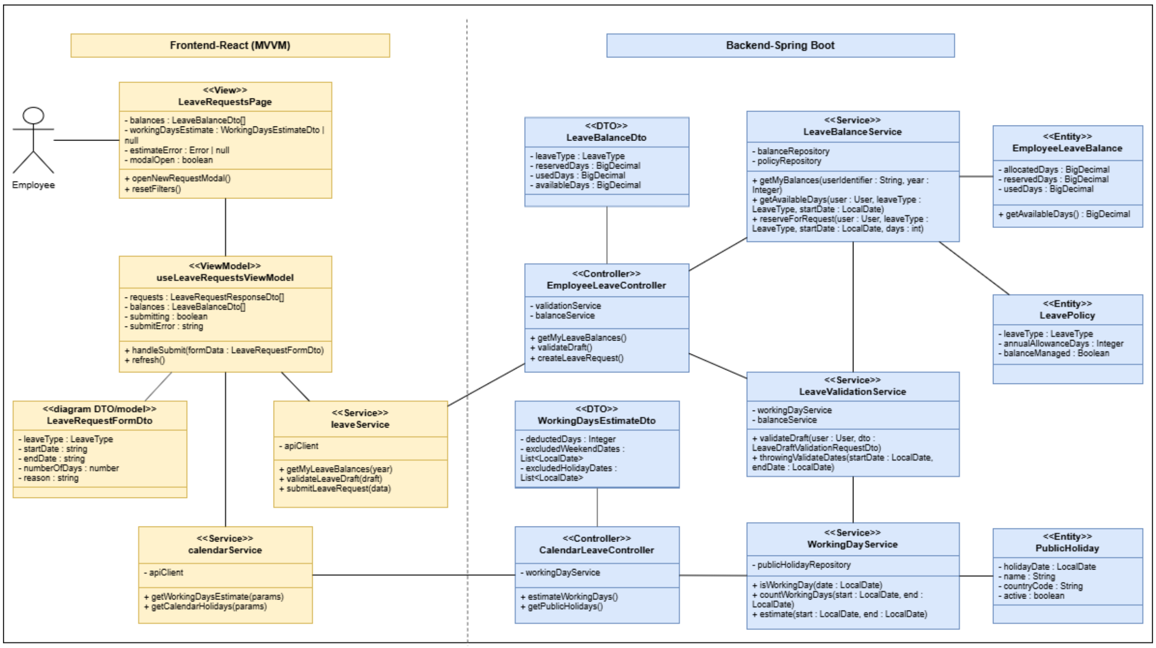


Figure 61: Design Class Diagram – Working Day, Holiday, and Balance Management

Conception Sequence Diagrams

These diagrams detail the internal frontend and backend interactions behind the main Sprint 2 scenarios: leave submission, pending leave update or cancellation, team leader review, and HR final review.

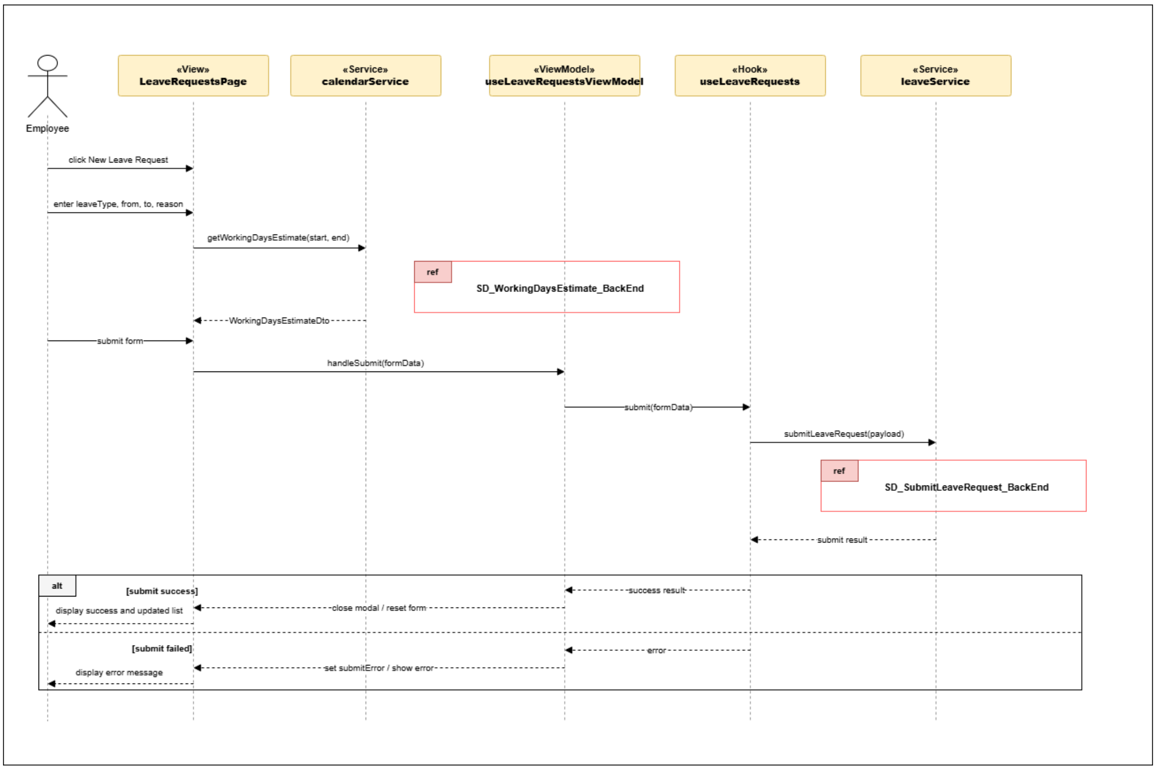


Figure 62: Conception Sequence Diagram – Submit Leave Request Frontend

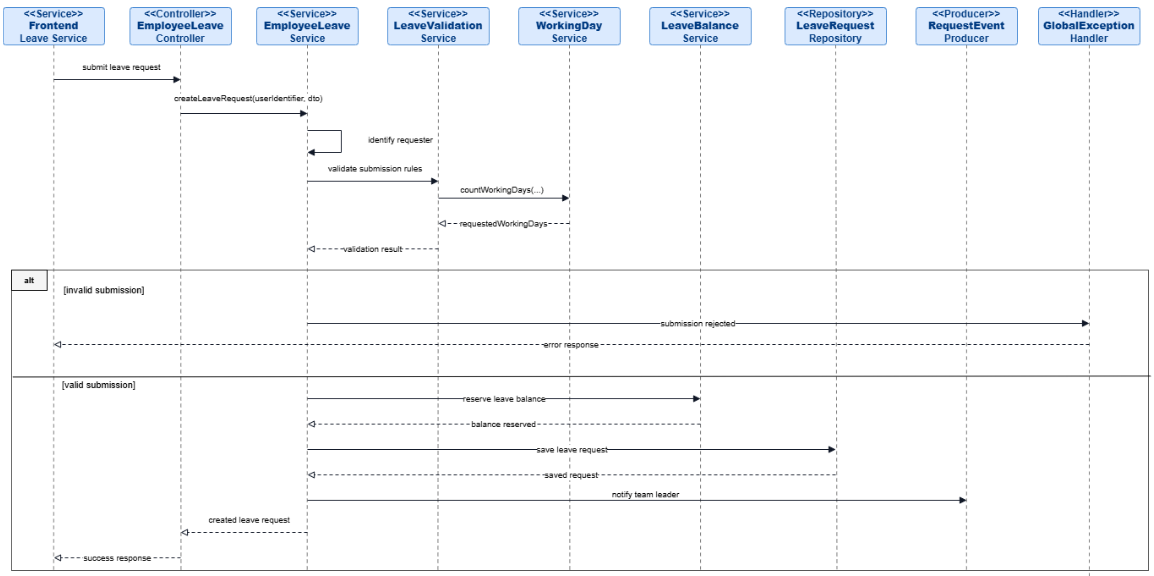


Figure 63: Conception Sequence Diagram – Submit Leave Request Backend

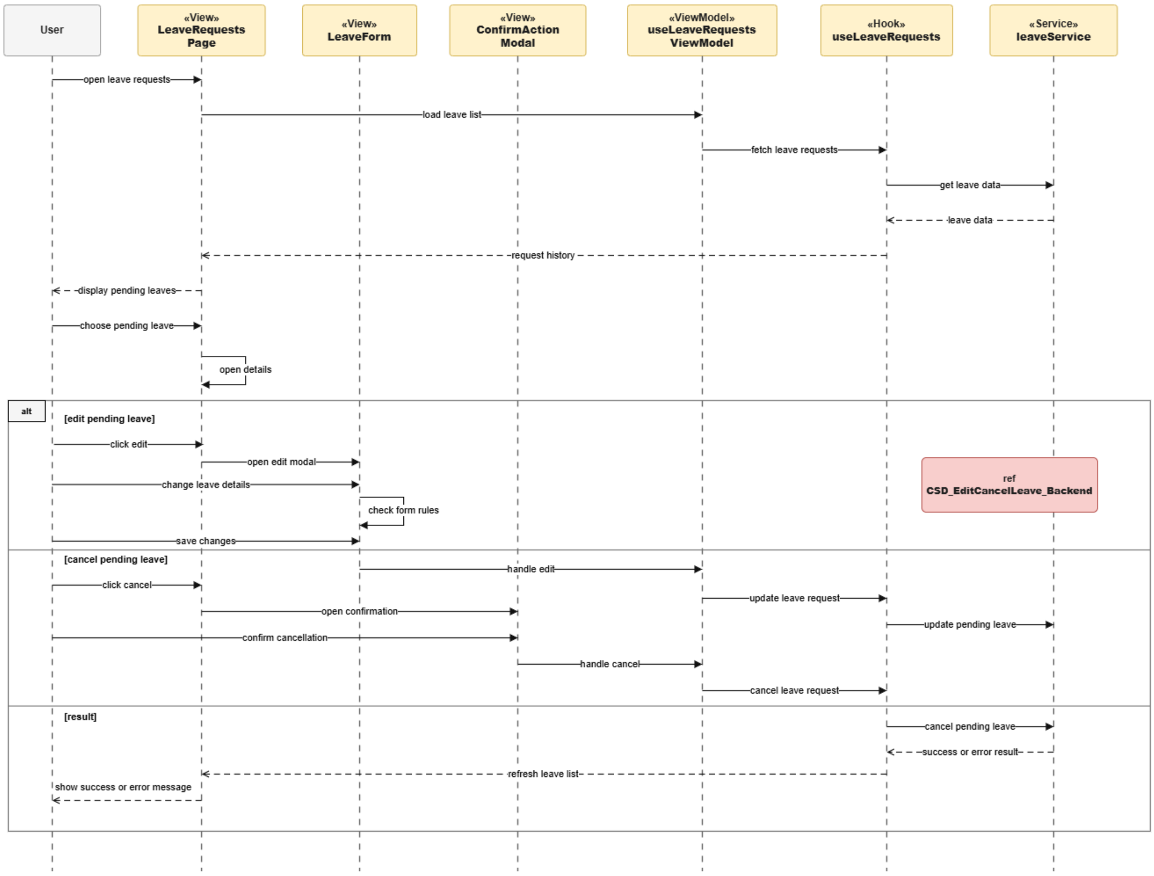


Figure 64: Conception Sequence Diagram – Edit or Cancel Pending Leave Frontend

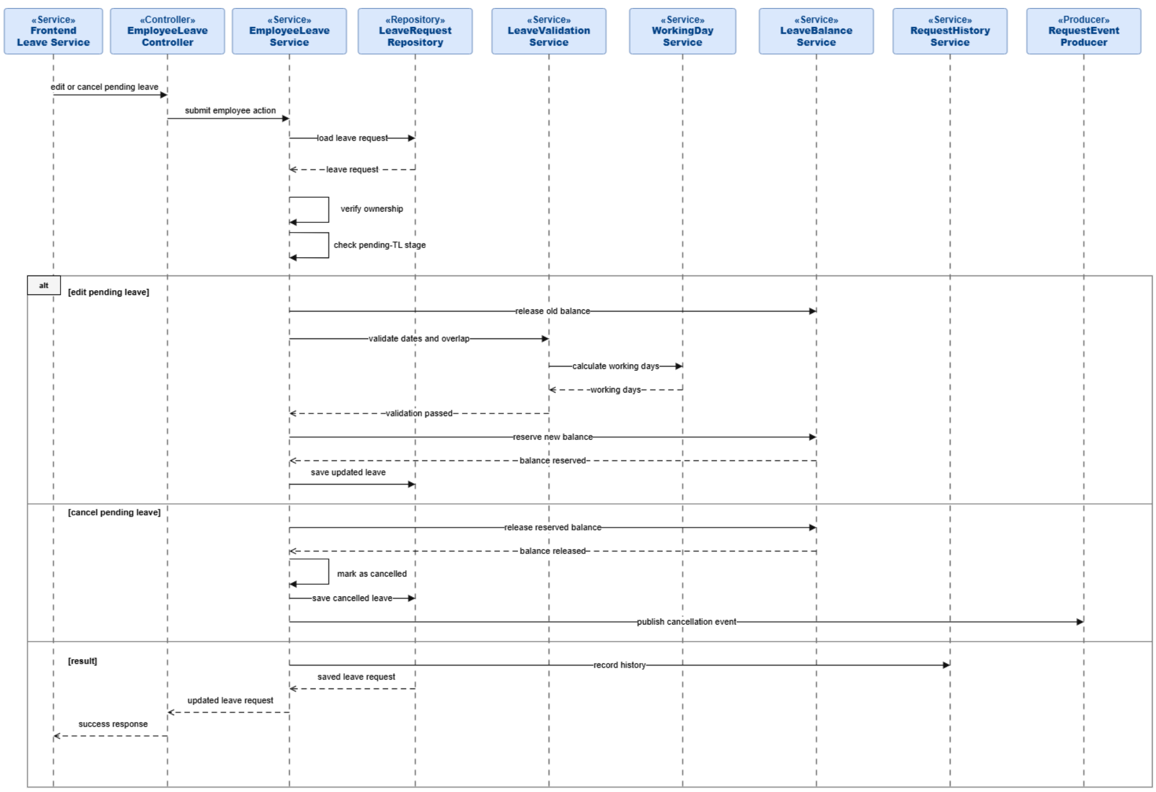


Figure 65: Conception Sequence Diagram – Edit or Cancel Pending Leave Backend

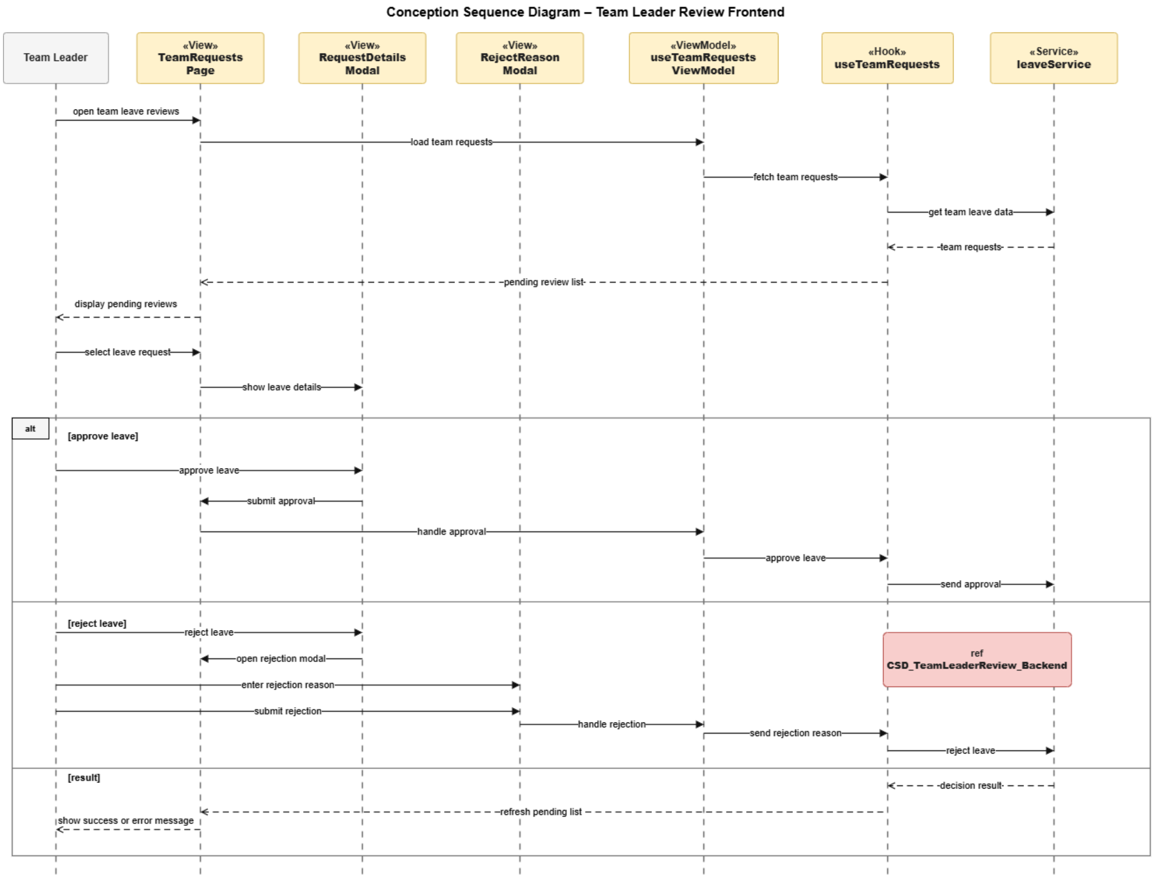


Figure 66: Conception Sequence Diagram – Team Leader Review Frontend

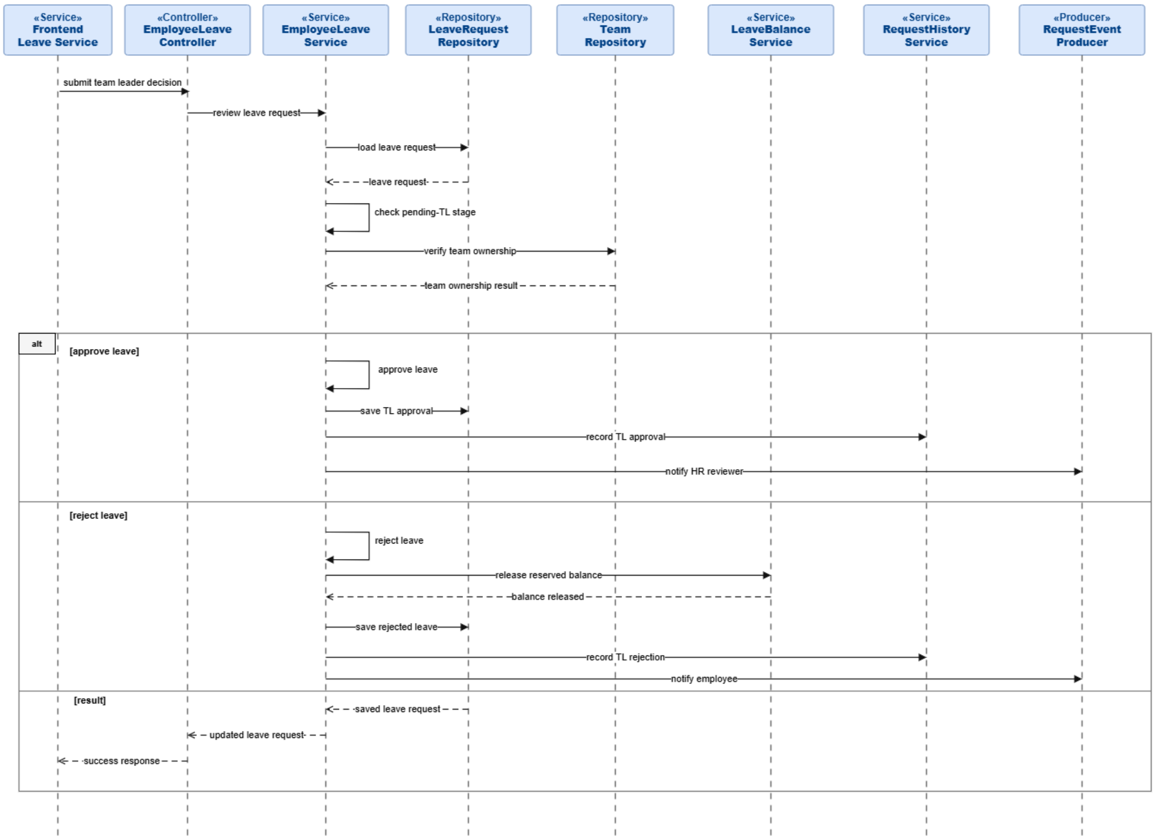


Figure 67: Conception Sequence Diagram – Team Leader Review Backend

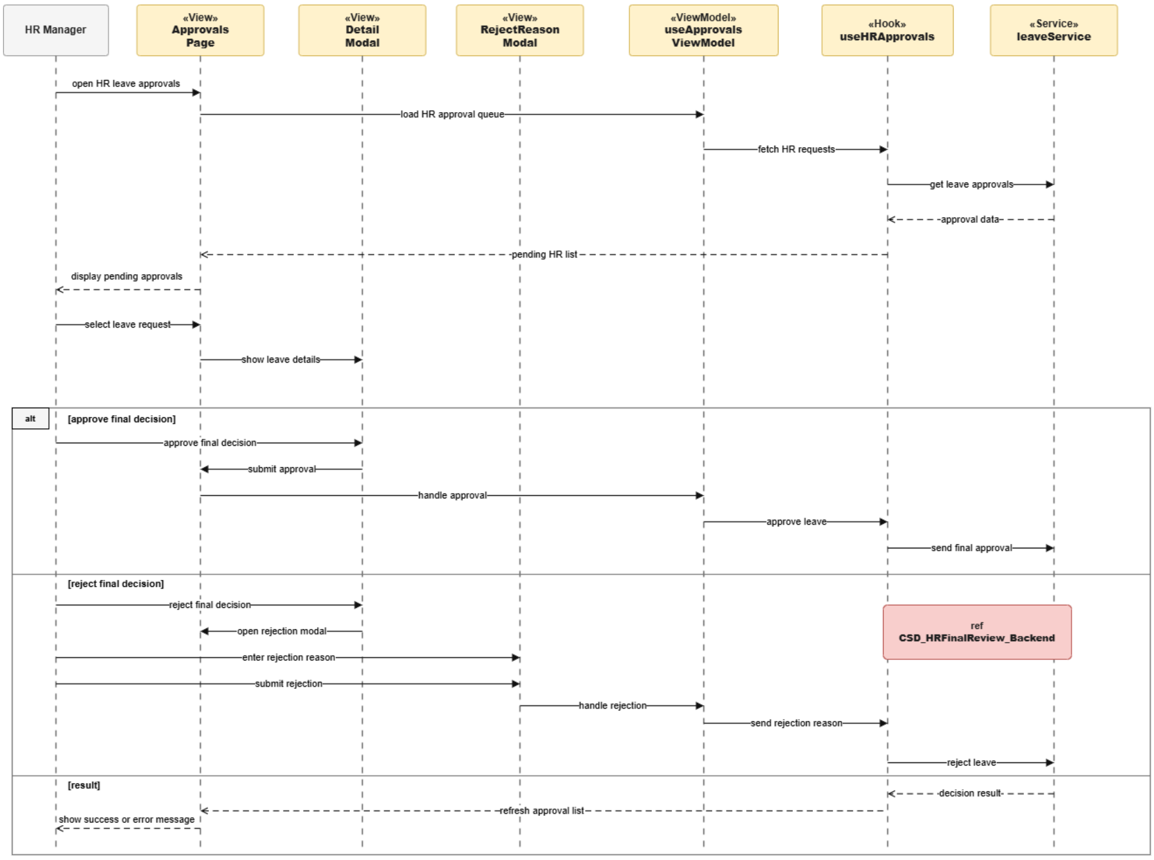


Figure 68: Conception Sequence Diagram – HR Final Review Frontend

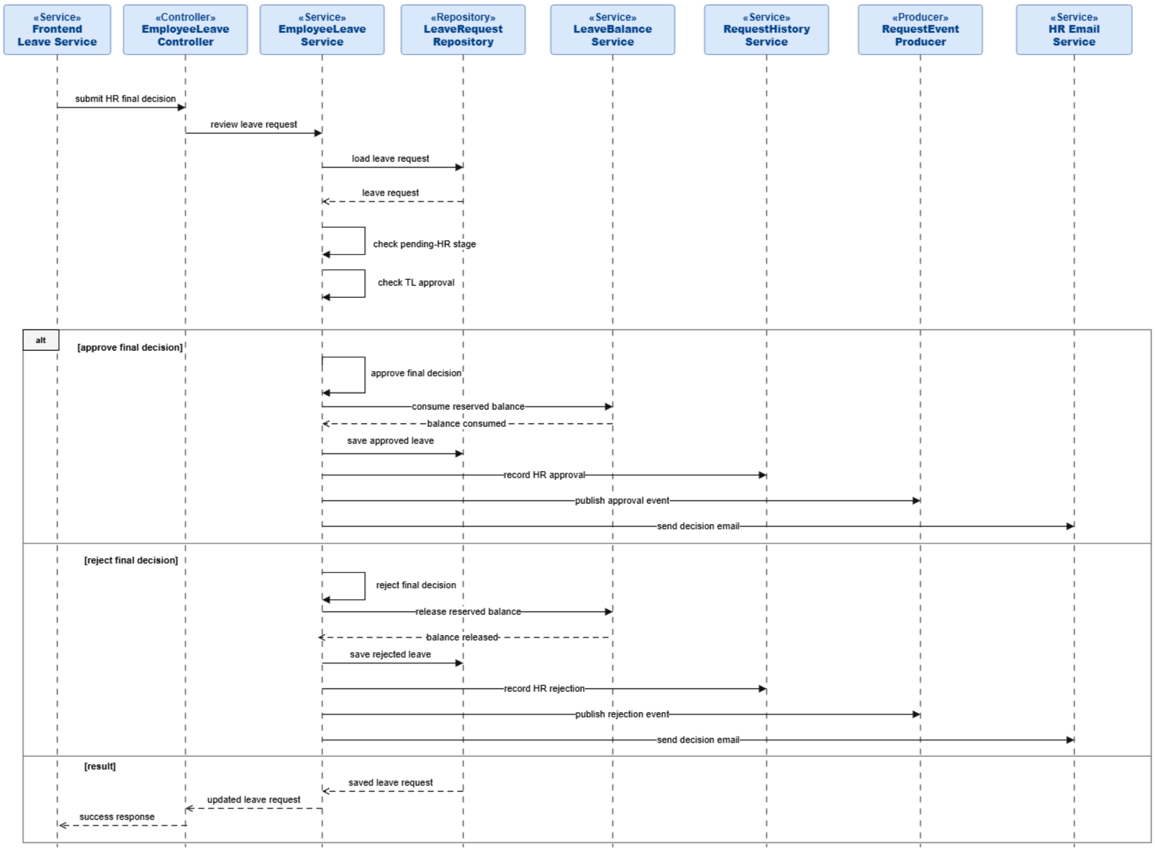


Figure 69: Conception Sequence Diagram – HR Final Review Backend

Realization

This section presents the main user interfaces reserved for Sprint 2 leave management. The final screenshots will be inserted after implementation.

Employee Leave Request Interface

sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_employee_leave_requests.png

Employee Leave Details Interface

sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_employee_leave_details.png

Team Leader Leave Review Interface

`sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_team_requests.png`

Team Leader Leave Calendar Interface

`sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_team_calendar.png`

HR Leave Approvals Interface

`sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_hr_approvals.png`

HR Leave Records and Filters Interface

`sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_hr_all_leaves.png`

Role-Aware Leave Calendar Interface

`sections/chapitre3/img/sprint2_leave_management/09_ui_screenshots/sprint2_company_calendar.png`

5. Sprint Review

At the end of Sprint 2, the leave-management workflow was reviewed as the first complete administrative process of the platform. The sprint enabled employees to submit, update, cancel, and track leave requests while applying working-day validation, public-holiday exclusion, overlap prevention, and leave-balance control. It also introduced the two-level approval workflow, where team leaders review requests first and HR managers make the final decision. In addition, role-aware calendars, filters, notifications, and approved leave PDF generation improved visibility and traceability across the workflow.

6. Sprint Retrospective

What went well	What needs improvement
The sprint delivered a complete leave workflow from employee submission to HR final decision.	**Working-day, holiday, balance, and overlap rules required careful coordination between frontend feedback and backend validation.**
The two-level approval process improved traceability through clear statuses, decision reasons, notifications, and history records.	**Calendar readability can still be improved when many leave events are displayed at the same time.**
The leave balance, history, filters, and role-aware calendars improved visibility for employees, team leaders, and HR managers.	**Future reporting can extend the approved leave PDF into broader analytics and export features.**

Table 5: Sprint Retrospective – Sprint 2

Conclusion

This chapter covered Release 1 of the ArabSoft internal platform, structured into two complementary sprints. Sprint 1 established the secure access foundation of the system through registration, authentication, password recovery, profile management, HR account setup, role assignment, and account activation control. Sprint 2 delivered the leave-management module, including leave submission, update and cancellation, working-day validation, leave balance control, team leader review, HR final decision, calendars, notifications, and approved leave reporting. These two sprints provided the first operational version of the platform, combining secure user access with a complete administrative workflow. The next chapter will present Release 2, which extends the platform with additional administrative requests and HR management features.